

**DIRECTIVES FOR PROJECT FORMULATION AND
MANAGEMENT IN DRDO
(DPFM 2026)
(PART-1)**



March 2026

**Directorate of Planning & Coordination
Defence Research and Development Organisation
New Delhi-110011**

MESSAGE FROM SECRETARY DDR&D

Self-reliance in defence technologies, systems, and platforms is fundamental to achieving 'Aatmanirbhar Bharat'. The Defence Research and Development Organisation (DRDO) is central to this mission, utilizing its vast technological expertise to meet India's diverse defence requirements. However, the rapidly evolving nature of warfare, propelled by disruptive and advanced technologies, necessitates continuous innovation, agility, and the development of sophisticated, future-ready systems to ensure a decisive advantage for India's Armed Forces.

Given the inherent complexity and High Risk nature of defence R&D, successful project execution demands meticulous planning, adaptable implementation, and robust monitoring. Comprehensive due diligence, proactive risk assessment, and the flexibility to navigate unforeseen technological challenges are paramount. The Directives for Project Formulation and Management (DPFM-2026) provides a streamlined and robust framework to address these demands, empowering DRDO Labs to formulate, execute, and monitor projects with precision and efficiency. By prioritizing simplicity, adequacy, and procedural clarity, DPFM-2026 serves as a strategic enabler of project success.

DPFM-2026 introduces innovative mechanisms, including quantifiable Technology Readiness Levels (TRLs) tailored to the DRDO context, proactive risk management with sunset milestone in the project. These tools establish a standardized, transparent, and accountable approach to tracking progress throughout the project lifecycle. By integrating these systems, DRDO becomes exceptionally well-equipped to deliver cutting-edge defence technologies, significantly advancing India's pursuit of Aatmanirbharta and global technological leadership within an increasingly dynamic defence ecosystem.

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग
नई दिल्ली 110011-
दिनांक 30 मार्च 2026

सेवा में,

अध्यक्ष,
रक्षा अनुसंधान एवं विकास संगठन,
डीआरडीओ भवन, राजाजी मार्ग
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विषय: DIRECTIVES FOR PROJECT FORMULATION AND
MANAGEMENT IN DRDO (DPFM-2026)

महोदय,

I am hereby directed to convey the sanction of the President of India for adoption of 'Directives for Project Formulation and Management in DRDO (DPFM) 2026' as enclosed.

2. Procedures laid in DPFM-2026 will be applicable for all projects of DRDO taken for sanction subsequent to date of this letter. This document will also be applicable for execution/ monitoring and closure of all ongoing projects of DRDO. DPFM-2026 is not applicable for Projects approved through Empowered Committee/ Political Committee (EC/PC) route. However, special provisions will be applicable wherever necessary.
3. Sanctions and corrigendum issued in respect of ongoing projects will remain valid. However, all such projects shall be aligned with DPFM-2026 with respect to project execution, reviews, monitoring and closure.
4. All future activities costing less than ₹10 Cr. shall be classified as 'R&D Activities', however ongoing projects costing ₹5 Cr. and above and less than ₹10 Cr. will continue as projects and follow DPFM-2026. These shall continue till their PDC w.e.f. release date of DPFM-2026.
5. All projects costing ₹10 Cr and above shall require Acceptance-of-Necessity (AoN) of DMC before sanction except those whose AoN is already accorded by DAC/DPB/SPB or Apex council of DRDO-MHA.



6. Amendment to DPFM-2026 will be issued with the concurrence of Addl. FA (R&D) and approval of Secretary DDR&D and Chairman DRDO.
7. Any clarification w.r.t. implementation or interpretation of the directives will be referred to DP&C, DRDO HQ for examination and necessary action.
8. This issues with the concurrence of the Ministry of Defence (Finance) vide their Dy. No. 166/MOD/Fin(R&D) dated 30th Mar 2026.

भवदीय,
30.3.26
(एवन कुमार शर्मा)
अवर सचिव, भारत सरकार

Encl: As above

Ink Signed Copy to:-

CGDA, New Delhi

PCDA (R&D), New Delhi

Principal Director of Audit
(Defence Services), Pune

CDA (R&D), Hyderabad

CDA (R&D), Bengaluru

JCDA (R&D), Pune

D(R&D), Ministry of Defence

File Copy

Copy to

All DG (Cluster)

All DG (HQ)

Addl. FA (R&D), New Delhi

All Lab/DYSL Directors

All Centre Heads

All Corporate Directors

All IFA (R&D)

PREFACE

A structured approach is followed by DRDO to bridge the knowledge gaps, enhance technological capabilities, develop contemporary test methodologies, establish advanced test facilities and infrastructure and develop complex systems based on the requirements of Services Head Quarters (SHQs) or on its own assessment of requirements for the future. These activities are carried out in the form of Projects/Programmes.

DPFM-2026 aims to streamline the project sanction process and effectively implement the standardized monitoring and review structure in DRDO projects.

Salient features of DPFM-2026 are given below:-

- ❖ Re-defining of PDC extensions due to scope change/non-availability of platforms as “Re-fixing of PDC” for TD(S) and MM projects.
- ❖ Time Estimation - Realistic Time (PDC), Pessimistic Time and Sunset Milestone.
- ❖ Channel of Submission by maximum 4 Levels.
- ❖ Responsibility & Accountability Framework (RACI Matrix).
- ❖ Seed funding by Build-up for pre-project activities.
- ❖ Provision for projects to be categorised as ‘High Risk’ in Futuristic and Advanced Technologies.
- ❖ Mandatory PERT chart and WBS for a project proposal - 4 levels WBS for MM & TD(S) and 3 levels WBS for TD(T) and S&T.
- ❖ ‘Roadmap of DRDO’ to be used as a Single Consolidated Plan Document for projects.
- ❖ Ensuring functional Lab Technology Council for TRL assessment.
- ❖ Creation of “Database of Systems & Technologies”.
- ❖ Effective date of sanction- quarterly.
- ❖ ‘Six-Monthly milestones’ and ‘PERT Chart’ part of Govt. Sanction Letter.
- ❖ Fluidity of cost for S&T and TD(T) projects, firming up after CDR.
- ❖ Nomination of Deputy Project Director (DPD) for projects in addition to Project Director (PD) for every project.
- ❖ Experts database suggested – Experts to be drawn from this pool for PRC/IRC/PDR, TiEC, DDR/CDR etc.

- ❖ Independent PD & DPD for every project and sub-project, project management training and performance review every 2 years.
- ❖ Preparation of Technical Completion Report within project PDC.
- ❖ ‘Valuation of Technologies’ and ‘Economy Impact Analysis’ introduced.
- ❖ Doing away with project categories – Basic S&T, UT, PS & LSP.

Any amendment to DPFM-2026 shall be issued with the concurrence of Addl. FA (R&D) and approval of Secretary DDR&D and Chairman DRDO. Exceptions, where DPFM-2026 directives are not applicable, should be submitted for approval of Secretary DDR&D and Chairman DRDO along with Financial Concurrence. Clarification in the form of Advisories will be issued by Directorate of Planning & Coordination (DP&C) as and when directed by Secretary DDR&D and Chairman DRDO / Deptt. of Defence R&D Management Council (DMC).

The applicable formats and checklists which are to be referred in conjunction with this guideline document are being promulgated as ‘**Handbook of Formats & Checklists of Directives for Project Formulation and Management (DPFM-2026) in DRDO**’. For ease of cross-reference, Handbook contents are denoted as DRDO.DPFM.FT.SN for Formats; DRDO.DPFM.CL.SN for Checklists and DRDO.DPFM.AL.SN for Abbreviation lists. These contents are indicative and may be revised and issued by Directorate of Planning and Coordination (DP&C) with the approval of DG (R&M) on requirement basis.

ACKNOWLEDGEMENT

The **Directives for Project Formulation and Management in DRDO (DPFM-2026)** have been finalized after several deliberations and discussions.

Valuable inputs and comments provided by Project Directors, Lab Directors, Directors (PM), Cluster DGs, HQ Directorates, Integrated Financial Advisors (R&D) and Additional Financial Advisor (R&D) have been incorporated.

Recommendations of internal and external review committees have been deliberated for implementation. Based on discussions held with ‘Committee of DGs’ including Addl. FA (R&D) and their suitability to Project Management in DRDO these have been implemented in DPFM.

Continuous guidance, support and inspiration by DG (R&M) has helped in organizing the chapters, incorporating the recommendations by various offices & committees and introduction of new provisions aligning with standard project management practices to address immediate concerns of Project Directors and Lab Directors.

The team at DP&C, led by Shri Kamlesh Kumar, Scientist ‘G’ and comprising members Shri Shashank Sharma, Scientist ‘E’, Shri Lalit Mohan, Scientist ‘E’ and Shri Pawan Sharma, Technical Officer ‘C’ has worked hard to ensure the methodological compilation of the directives. Contribution made by the team is exemplary.

Acknowledgement is also extended to Director, DESIDOC and team for printing this document.

New Delhi

Date : Mar 2026

Director Planning & Coordination

DRDO HQ, New Delhi

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

1.1.1 A Programme / Project essentially consist of well-defined objectives which are planned to be achieved under given time, cost and other resource constraints. Research & Development (R&D) requires exploration of existing knowledge and skills, along with developing new knowledge in related technical domains, to bridge gaps required to develop a system, technology or process not developed hitherto indigenously or otherwise. Accordingly, uncertainties in the final outcome become an integral part of the R&D processes and its management. A programme or a project would plan to achieve pre-defined objectives, with an anticipation of bridging the knowledge gaps within the set period, by defining the milestones and also identifying the risks involved and its planned mitigation.

1.1.2 Project management has evolved, perfected and formalized in Defence Research and Development Organisation (DRDO) with time. Since the inception of DRDO up to the year 2005, there were no consolidated guidelines issued for executing the projects in the Labs. However, the programmes and Labs had their management plan and could be traced way back even in 1980s. In the year 2006, Procedures of Projects Formulation and Management (PPFM-2006) was released after wide consultation across R&D spectrum and was aimed to provide necessary guidelines and clarity to Programme Directors and Project Directors for timely and successful completion of projects. These guidelines were revised in the years 2014 and 2016 and released as PPFM-2014 and PPFM-2016 respectively. In the year 2021, these guidelines were revised with due emphasis on further simplification, adequacy and effectiveness of the procedures for achieving self-reliance and technological excellence in defence technologies towards 'Aatmanirbhar Bharat'. These were issued in April 2021 as "Directives for Project Formulation & Management (DPFM-2021)".

1.1.3 R&D project management is a typical process and should not be compared with conventional projects where uncertainties are minimal. Certain flexibilities are accorded to R&D projects where the possibility of meeting all the objectives with its associated risks. The knowledge and experience gained during any R&D project is a significant asset and is useful in many other subsequent projects and activities. Hence, not meeting the objectives verbatim of such projects is not considered as a failure of the project.

1.1.4 The basis of undertaking programmes and projects by DRDO is the problem statement or the requirements mentioned in plan documents of Indian Armed Forces or DRDO. 'Roadmap of DRDO' (D-RdMAP) is being considered as single consolidated plan document for taking up new projects. Projects taken based on 'Roadmap of DRDO' shall be classified as '**Listed Projects**'. Projects taken based on Acceptance of Necessity (AoN) given by Defence Acquisition Council (DAC), Defence Procurement Board (DPB) or Services Procurement Board (SPB) will also be considered planned projects. In specific cases, programmes or projects shall be undertaken either to meet immediate requirements of Armed Forces or to develop indigenous capabilities in emerging technology areas not fully captured in D-RdMAP. These will be considered as 'Non-Listed' projects. These will be added in D-RdMAP which will be updated on six monthly basis.

1.1.5 The 'Roadmap of DRDO: 2025-2034' is expected to be reviewed and updated after three years. The updated Roadmap will map systems and technologies aligning with 'Integrated Capability Development Plan (ICDP)' of the Armed Forces, which is under compilation. Therefore, DPFM-2026 may be reviewed at least every five years to merge the corrigendum/advisories, consider the lessons learnt in the period, requirement of any additional/modified guidelines to align with the requirements of revised 'Roadmap of DRDO' and other policy documents to achieve 'Viksit Bharat'.

1.2 CRITERIA FOR CLASSIFICATION OF PROGRAMME & PROJECT

1.2.1 Programmes and projects are similar in nature with respect to their scope of activities, but differing on the scale of execution due to difference in cost. Following guidelines will be adopted for classifying programmes and projects for execution purposes in DRDO:-

1.2.1.1 **Programme:** All proposals in the delegated financial power of Cabinet Committee on Security (CCS) shall be classified as programmes. However, existing ongoing programmes will be continuing till their Probable Date of Completion (PDC). **Programmes will not have any category.**

1.2.1.2 **Project:** All proposals which have financial outlay up to the Delegated Financial Powers (DFP) of Finance Minister (FM) and not less than ₹10 Cr or as directed by Deptt. of Defence R&D Management Council (DMC) from time to time shall be classified as projects.

1.2.2 Programme and Project (including Programme Director and Project Director) shall be used interchangeably in the document with the meaning restricted to as defined above.

1.2.3 Labs may work on forward looking technologies. Activities whose cumulative value is less **than ₹10 Cr or as directed by DMC** from time-to-time shall be referred to as **‘R&D Activities’**. Such activities shall be decided by **Cluster Council**. **These ‘R&D Activities’ will be reviewed periodically by Lab Director and Cluster Council. Project number and unit code will not be assigned in such cases. Expenditure will be borne from Lab under Build-up head. Low Cost-Long Term Technology Development & Prototyping in the Labs may also be undertaken through Build-up funds as ‘R&D Activities’ if deemed necessary.**

1.3 TECHNOLOGY READINESS ASSESSMENT (TRA)

1.3.1 Historical Perspective: A methodical evaluation of R&D project is essential for systematic analysis of its technological content and mapping it with the expertise and facilities available in the country. Assessing the technology readiness of DRDO/Lab to undertake a project has been the practice. However, the process to do so has been changing as the technology readiness level of the country has advanced and the lessons learnt in the technology readiness assessment process have been incorporated.

Growth of competence with respect to any specific project in the Lab was mapped for the projects in the Project Management Plan since 1970s. Thereafter, Decision Aid for Technology Evaluation (DATE) was introduced in the year 2006 as a decision support tool for technology evaluation of R&D projects. The methodology incorporated in DATE specifically addressed the system development projects of DRDO. It facilitated the systematic analysis of

project for its technology content and evaluation of feasibility in the context of technological expertise and facilities available in the country. The methodology enabled the project team to assess the importance of different technologies for each of the subsystems of the system, the team's confidence in exploiting the national technological resources and finally determined the potential for successful completion of the project. The methodology also helped the project team and the Lab to identify and quantify possible technology gaps affecting successful system development. 'National Resource Index' (NRI) for the technologies and sub-technologies was evaluated by specially constituted expert groups taking into consideration the resources (expertise and facilities) available at DRDO Labs, national research institutions, academia and industries in the country. This was in use till year 2015.

Project Evaluation Assessment and Readiness Level (PEARL), highly modified version of Decision Aid for Technology Evaluation (DATE), was introduced during year 2016 at the time of release of Procedures for Project Formulation & Management (PPFM - 2016). The main objective of PEARL was to assess difficulty/ease of development and also to make realistic time assessment for development of a system/sub-system. In the process, Module Readiness Level (MRL) was derived for each module/sub-system followed by denoting a Project Readiness Index (PRI) for a system which was the minimum value of MRL of the various sub-systems. MRL was scaled from a level of 1-10, wherein 1 denoted ab-initio research to start and 10 denoted modules in production/Commercial-off-the-Shelf (COTS). MRLs would be extrapolated to PRI of the programme / project which helped to classify the category under which the project could be taken {PRI between 1-4: Science & Technology (S&T) projects, 3-7: Technology Demonstration (TD) projects and 6-10: Mission Mode (MM) projects}. PEARL analysis was not only done at the start of the project but also at the end of the project so as to provide a better picture of the technological progress made during the project. PEARL was in use till the year 2020, prior to introduction of DPFM-2021.

1.3.2 The aim of a project is to increase the technological capabilities at its completion. Technology Readiness Levels (TRLs)* are the most common measure for systematically communicating the readiness of new technologies or new application of legacy or existing technologies to be incorporated into a system or programme. They serve as a compendium of characteristics that describe increasing levels of technical maturity based on demonstrated capabilities. TRL is traditionally measured on a scale of 1-9, with 9 being the highest. However, TRL 10 has been introduced in DRDO to indicate Commercial Off-The-Shelf (COTS) items

that are readily available in the open market or already developed by DRDO. The expected TRLs, in the context of DRDO are given in Table 1.

1.3.3 TRL should be monitored as a project is progressed through the process called Technology Readiness Assessment (TRA). It is a systematic, evidence-based process which evaluates maturity of critical technologies vital to the performance of a larger system. It is also a measure to identify risks and formulate risk mitigation strategy.

1.3.4 In DRDO projects, a technology assessed and proved in the previous projects can be introduced at TRL 3 for current project. Same TRL of a legacy system will not be carried forward if there is change in any of the aspects such as changes in design (including material and manufacturing process) or changes in operating environment conditions etc.

**Used by NASA, US DoD and European Agency reference 'Technology Readiness Assessment Guide' on Best Practices for Evaluating the Readiness of Technology for Use in Acquisition Programmes and Project issued by the U.S. Government Accountability Office in January 2020*

Table 1: Technology Readiness Levels Adapted for DRDO			
TRL INDICES	DRDO DEFINITION	QUALIFIERS	SUPPORTING DOCUMENTS
TRL -1	Basic principle and research	Scientific publication with probable concept / application.	Scientific Publication/ Research Reports
TRL -2	Basic technologies assessed and technical concept formulated	- Scientific/Technical report with feasibility and potential application. - Literature study of world-wide trends, identification of proposed development model, testing and validation plan and possible technical risk areas.	- Feasibility study report - Concept design report

TRL - 3	Experimental proof of concept and preliminary design	• Lab results showing validation of critical properties, studies and Lab measurements to validate analytical predictions. • Preliminary Design Review for components/sub-systems completed.	• Preliminary Design Document (PDD) and Preliminary Layout Drawings • Report on comparable system developed with test details of specifications achieved. • System Definition Document (SDD)
TRL - 4	Technology validated in Lab environment	• Configuration finalisation for system / product specifications. • Generation of System/Product Tree. • Detailed Design Review completed. • Lab prototype realised.	• Identification of Development cum Production Partner (DcPP) for a TD (System) Project. • Detailed Design Document and Layout Drawings. • Integration Layouts.
TRL – 5	Technology validated in simulated environment	• Components / Sub-systems manufactured as per Acceptance Test Procedures (ATP). • Design validation for performance at component / sub-system level. • Components/sub-systems integrated in Lab environment for functional assessment.	• ATP document. • Approved sub-system/component design and drawings for prototype manufacture. • Approved integration plan. • Component/Sub-system Functional Test Reports. • Quality Assurance Plan for sub-system / component.

TRL - 6	System/ Sub-system model or prototype readiness for trials in simulated environment	• Sub-systems/Systems manufactured as per Quality Assurance Plan (QAP) & accepted as per ATP. • Completion of Qualification tests for components/sub-systems from DcPP. • Critical Design Review and implementation of recommendations. • Readiness for Prototype Trials in simulated environment.	• Qualification Test Certificates for components/ sub-systems. • Approved Interface Control Documents • Critical Design Review report of components/ sub-system. • Manufacturing Drawings of QAP and ATP cleared sub-systems/ components. • Prototype Trial Plans.
TRL - 7	System prototype demonstration in relevant / simulated environment	• Realisation of prototype from DcPP. • Completion of Prototype Trials. • Prototype cleared for Preliminary Staff Qualitative Requirement (PSQR) Validation Trials. • Completion of Technical Documentation.	• Prototype Trial Reports. • PSQR Validation Trial definition document at system/ sub-system level. • Approved Technical Documents. • Approval by Executive Body.
TRL - 8	Actual system completion and qualified through PSQR validation trials for performance requirement against Trial Directive.	• Completion of PSQR Validation Trials on systems manufactured by DcPP-based on QA/ ATP, for performance requirements. • Completion of Technical Documentation.	• PSQR Validation Trial (for performance evaluation) reports duly vetted by the Trials Team.

TRL - 9	Actual System operated over the full range of operational conditions as per PSQR parameters against Trial Directive.	• Completion of PSQR validation/ User Trials / Final Acceptance Trials.	• Jointly vetted PSQR Validation Trial reports. • Approval by the highest Review Committee.
TRL - 10	System in production/ COTS or already developed by DRDO	• System/sub-system/ components under production meeting the performance requirements.	• Production order.

1.3.5 Project Readiness Indices (PRIs): Lowest of the TRLs of sub-systems, sub-sub-systems or technologies envisaged under a project will be considered as the Project Readiness Index (*sub-systems, sub-sub-systems and components may be used interchangeably in the entire document*). It is a single index which represents readiness level of the whole project at any stage. It should be used as a reference during selection of new projects under a category. **However, for a programme TRL Assessment will be done to assess the risk & uncertainties involved, and its impact on PDC for preparation of risk mitigation strategies.** As Programmes will not have any category, these will be denoted as 'PG' in Govt.

Letters for Programme Numbering as well as for Unique Sanction Code (USC). PRIs for selection of projects under different categories are given in **Table 2**.

Table 2: Project Readiness Indices (PRIs) and Project Categories			
Sl. No.	Project Category	Minimum PRI at Project initiation	Minimum desirable PRI at Project completion
1	S&T Projects	1 [#]	3
2	Technology level Demonstration Projects i.e., TD(T) Projects	2	4
3	Systems level Technology Demonstration i.e., TD(S) Projects	4	7
4	Mission Mode (MM) Projects	6	9
Note: # Basic research should not be taken by Lab other than by DYSLs. Basic research based projects may be undertaken through Academia/DIA-CoEs/R&D Boards. - TRLs/PRIs will be not be applicable for IF projects. - Details of project categories are mentioned at Para 1.5.			

1.3.6 Technology Readiness Assessment (TRA) by Lab Technology Council: Each Lab should ensure functional Lab Technology Council. Necessity of a member from academia and/or industry may be explored for the project, while doing TRA. The role of external members may be intimated at least five working days prior to convening the meeting and their comments be recorded in TRA report. The report will be presented before Peer Review Committee (PRC)/Preliminary Design Review (PDR) and will be enclosed with project proposal.

1.3.7 The Project Director (PD) and Deputy Project Director (DPD) designate have the responsibility of carrying out self-assessment of TRLs for each sub-system in the project based on Work Breakdown Structure (WBS) and prepare six-monthly TRL progression for entire project duration as TRL roadmap. PERT diagram and detailed WBS should be mandatory part of TRA. WBS should essentially cover the activities up to minimum four levels for MM and

TD(S) and minimum three levels for TD(T) and S&T category project. The project team will measure technology maturity on the TRL scale based on qualifiers and supporting documents stated at **Table 1**. Thereafter, TRA should be reviewed by Lab Technology Council and TRA report will be informed to PRC or PDR committee.

1.3.8 TRL progression during project execution phases will be reviewed in every Executive Board (EB) meeting {Project Monitoring & Review Board (PMRB) in case of projects where CFA is Director} and update TRL level of each of the sub-systems, sub-sub-systems etc. upto level 4 of WBS in the project in centralized Project Management Information System (PMIS) at DRDO HQ (or ERP, when implemented).

1.4 METHODOLOGY FOR TRLs AND PRI ASSESSMENT

The methodology for assessment of TRLs, PRI and TRA Roadmap during project sanction and execution will be as follows:

1.4.1 TRL Assessment at project sanction stage

The designated PD has to take the following steps for establishing TRLs of the sub-systems, components or technologies during project sanction stage: -

- (a) Draw the sub-system block diagrams detailed to component level. The sub-systems can be further deconstructed into processes. Minimum 4 levels of product/work breakdown for MM & TD(S) and 3 levels of product/work breakdown for other category of projects should be undertaken for clarity on the present level of expertise/capability and the quantum of work involved in the project execution. Further levels of breakdown are dependent on the confidence of the team to provide requisite documents as proof.
- (b) The deconstructed structure may not necessarily lead to a hierarchical breakdown structure, but could create a mesh structure when this is linked between the sub-systems and/or components across different modules.
- (c) The sub-systems and components may also be grouped into relevant modules for understanding the amount of integration required in the system which would require suitable skills, processes and infrastructure.
- (d) Identify readiness level index in accordance with the criteria defined for TRL indices. Once TRL of each sub-system, component and process has been determined,

the necessary documentation has to be collated with supporting documents, as stated at Table 1.

(e) Evolve PRI as lowest value of all sub systems TRLs i.e.

***PRI** = min TRLs of Sub-Systems (SS), Sub-Sub-Systems (SSS) etc.*

(f) Draw the anticipated roadmap of TRL and PRI with project timeline. A project timeline and cost estimation commensurate with the TRA Roadmap has to be drawn by the project team.

(g) The TRL report comprising of supporting documents and TRA roadmap should be presented to the reviewing authority, i.e. Lab Technology Council.

(h) Illustration of roadmap of TRA & PRI for a typical system is given in Para 1.4.6.

1.4.2 TRL and PRI Recommending Authority

Designated PD and the DPD has the responsibility of carrying out self-assessment of TRLs. However, there is a need for an independent verification of requisite documents to authenticate the professed PRI. These will be carried out as follows:

(a) Chairman, Lab Technology Council may seek valuable inputs/opinion by subject experts external to the project team. The Lab Technology Council will carry out the review of the proposed PRI based on submitted document and recommend the final PRI and TRL progression.

(b) The PRC or PDR Committee will record the PRI, category of the project and give recommendation for indigenous development of imported subsystems (if applicable) as per TRL roadmap presented by PD designate & Chairman, Lab Technology Council.

(c) Director PM, O/o DG (Cluster) will ensure presentation of TRL/PRI during PRC/PDR meeting by designated PD to inform PRC/PDR committee. Chairman PRC/PDR may be explained about the process of TRL/PRI in advance.

1.4.3 Filled Format based on TRA Roadmap will be reviewed by Lab Technology Council and will be informed to PRC/PDR committee as given at DRDO.DPFM.FT.02.

1.4.4 TRL Assessment during Project Execution Stage

The TRL will serve as a benchmarking tool. It will be beneficial to track progress of development of a specific technology/sub-system during each stage of the development. It is essential to define TRA values at intervals. This TRA Roadmap will aid in systematic assessment of how far technology/sub-system development has progressed during the execution of a project. It is recommended that assessment of TRLs should be done during Critical Design Review (CDR) also, in addition to EB/Project Management Board (PMB).

The EB should ensure that the technology/sub-system have advanced to the targeted maturity before start of integration, testing and operational readiness review's phase. The PD is responsible for presenting the current status of TRLs and ensuring planned transitions to next higher stage as proposed during the PRC/PDR and thereafter during the periodic review meetings such as PMRBs and PMBs/EBs.

1.4.5 Technology Readiness Assessment at Project Completion

The TRLs and PRI assessment at project completion stage should be used as a certificate of having achieved project scope by analyzing each of the modules/sub-systems readiness at the completion of project activities. An assessment of the final TRLs of each sub-system and the resulting PRI along with requisite documentary proof should be recorded in the TRA report. PD will present the TRL progression as discussed in monitoring and review committee meetings (EB/PMRB) at project closure stage to Lab Technology Council. There may be instances of not achieving planned TRL at completion for which justification/explanation can be recorded in the report.

1.4.6 **Illustration of Roadmap for TRA and PRI for a Typical System is given below (2 Level Break-down):**

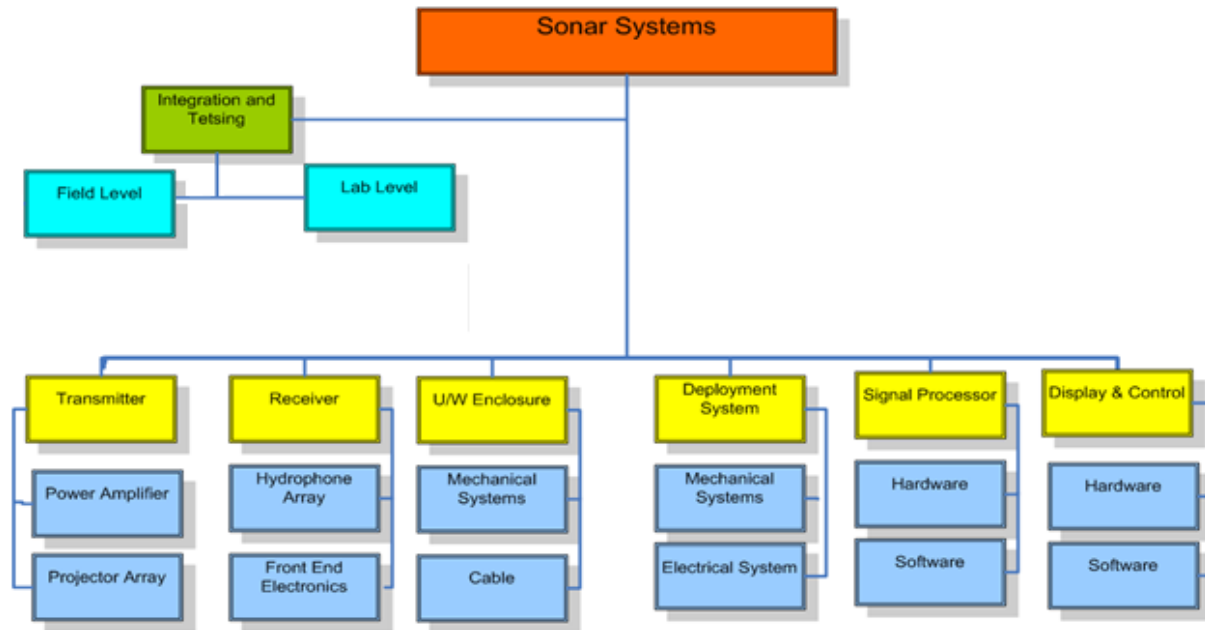


Illustration of WBS- 2 Levels for Sonar System

Sub-Sub-System (SSS)	Sub-System (SS)	PRC/PDR (sanction)		After 06 months		After 12 months / CDR		After 18 months		After 24 months	
		TRLs									
		SSS	SS	SSS	SS	SSS	SS	SSS	SS	SSS	SS
Power amplifier	Transmitter	3	3	3	3	5	5	6	6	7	7
Projector Array		4		4		5		7		7	
Hydrophone Array	Receiver	3	3	3	3	5	5	6	6	8	7
Front End Electronics		4		4		5		7		7	
Mechanical Systems	U/W Enclosure	4	3	4	3	5	5	7	7	7	7
Cable		3		4		5		7		7	
Mechanical Systems	Deployment System	3	3	4	4	6	5	7	7	7	7

Electrical Systems		3		4		5		7		7	
Hardware	Signal Processor	3	3	3	3	5	5	7	7	7	7
Software		3		3		5		7		7	
Hardware	Display and Control	3	3	3	3	5	5	7	7	7	7
Software		3		3		5		7		7	
Min TRL of Sub-System			3		3		5		6		7
PRI of Project	(3)										

Illustration of Roadmap for TRL and PRI progression for Sonar System

1.4.7 PRI for Programme

A Programme involves many systems and each system may be integrated to some other. Each individual system should be independently assigned a PRI based on TRL of its constituent sub-systems. Since each system may be complex and starts apart, assigning a PRI to the Programme may not be justified. However, for a Programme consisting of various sub-systems the feasibility may be checked to indicate an overall PRI.

1.5 PROJECT CATEGORIES

DRDO undertakes projects under different categories aligned to the requirements. The categories give an indication about the broad nature of activities and also help Lab to allocate resources in a manner meeting the organisational goals. Different categories of projects, their selection criteria, broad scope for each category of project and expected outcome have been given in succeeding paragraphs. Pre-requisites and expected outcome of various project categories have been stated in the pre-requisites and expected outcome diagrams given at subsequent sections. A single or combination of pre-requisite/s can be considered during project selection. A single or combination of outcome(s) is feasible during project completion.

1.5.1 Science and Technology (S&T) Project:

1.5.1.1 S&T projects are undertaken for research activities in alignment with long term technology requirements or Technology Forecast by Directorate of Futuristic Technology Management. These projects are for competence build-up in emerging, complex and futuristic scientific & technological areas. The scope could include ab-initio research, design studies and

analysis, design and development of working models or prototypes, modelling and simulation and tests in controlled environment. Bridging the knowledge gap in partnership with academia on work-share basis can be an important feature. The objectives of such projects are expected to be optimistic given the current knowledge base and the outcomes are also expected to have widely varying results. Partial achievement of the project objectives or not meeting few of the pre-defined objectives would not be considered failure, since uncertainties will be high under these projects. It is suggested that Labs should aim to have 10% of their projects by numbers in S&T category (Basic Research should not be taken by Labs). However, Labs doing basic research will have a much greater percentage of S&T projects in this category. **The reference diagram with pre-requisites and expected outcome for S&T category of projects is given in Fig. 1.**

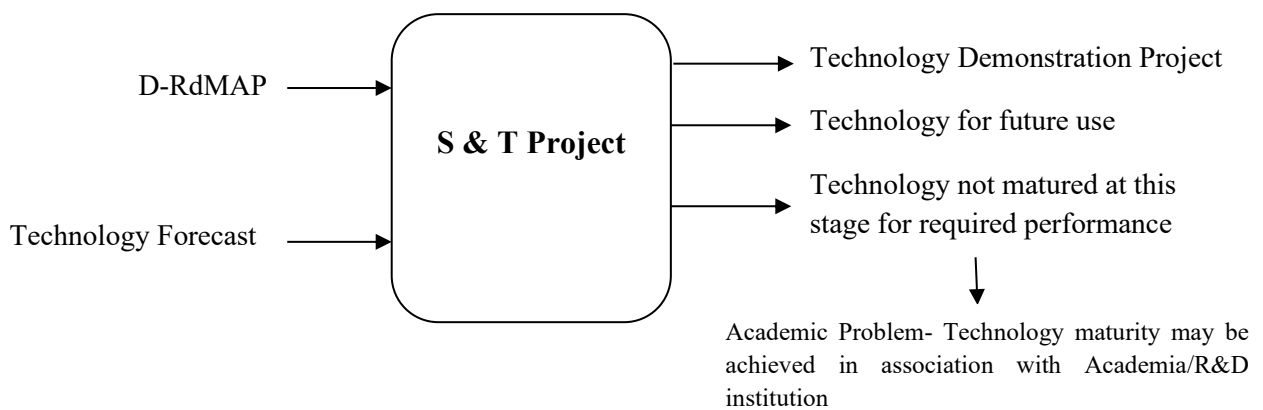


Fig.1: Pre-requisites and expected outcome of S&T Projects.

1.5.1.2 The expected outcome of an S&T project would be study report establishing the feasibility of converting some or all of its objectives in technologies, suggesting the probable applications of the knowledge or technology. The minimum TRL at the completion of S&T project should be 3.

1.5.1.3 In case, some of the objectives of the project have not been achieved or met partially, a follow-on study project in S&T category can be taken up by the Lab in collaboration with academia to bridge the knowledge base. Based on recommendation of project sanctioning authority, all such cases would not be considered a shortfall in performance or failure of the project to achieve desired objectives. Such projects will be completed as per procedures at Chapter 8. Some of the study results of these projects would be towards knowledge generation and competence build-up and may not have an immediate application.

1.5.2 Technology Demonstration (TD) Project

1.5.2.1 Technology demonstration and maturity is the key for self-reliance in defence systems. TD projects would be taken up to establish a technology or to evolve a process/test methodology or to develop a system for which the basic technologies are already developed or available through industry. TD projects could be initiated as a follow-up action of S&T projects and based on D-RdMAP. To develop competency in emerging technology areas, these projects may also be taken up to meet futuristic requirements of the Armed Forces. Academia could be involved to handle specific problems. The scope of a TD project should include design and development of prototype(s), testing and evaluation, analytically and experimentally establishing a system, process or test methodology.

1.5.2.2 TD projects can be further sub-categorised as Technology level TD projects i.e. TD(T) and as System level TD projects i.e. TD(S). **The reference diagram for pre-requisites and expected outcome of TD category of projects is given in Fig. 2.**

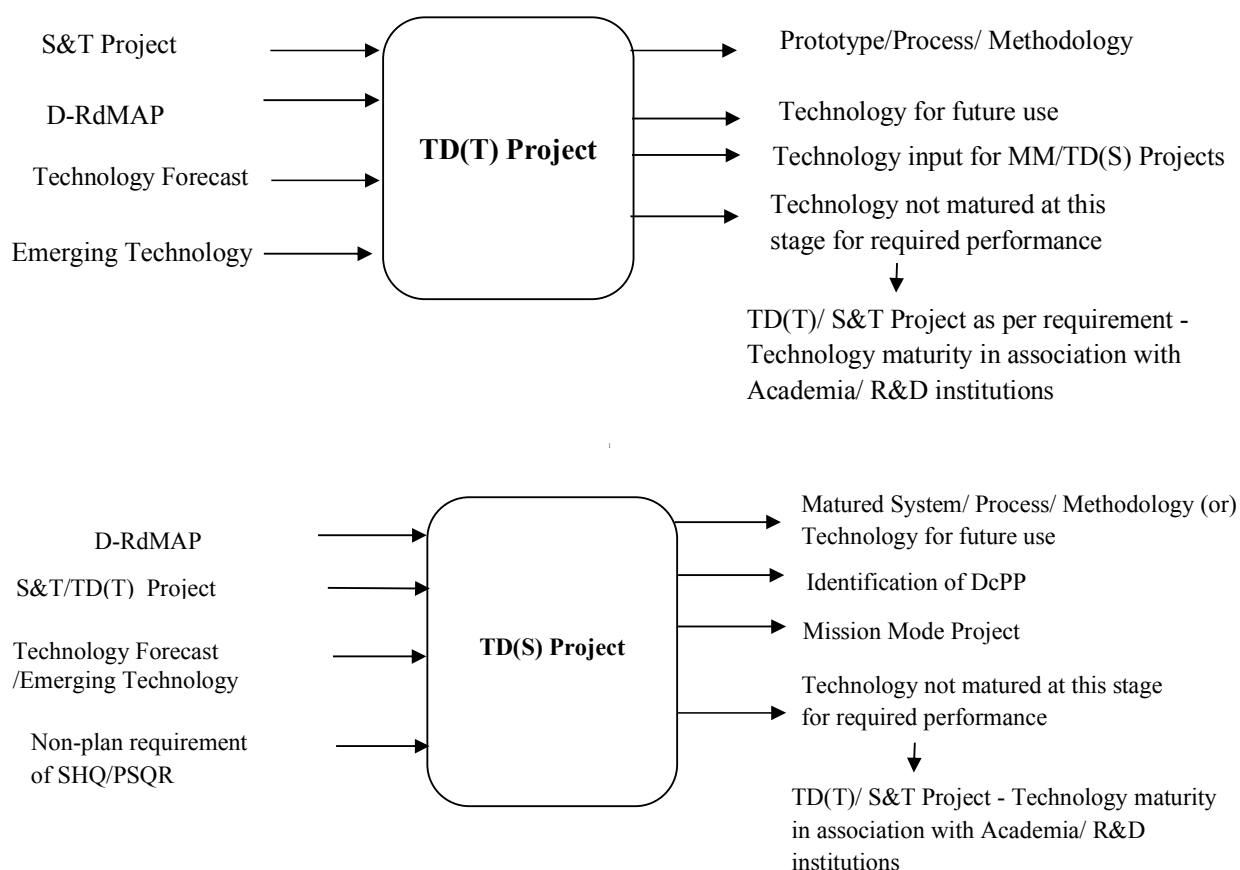


Fig. 2 : Pre-requisites and expected outcome of TD(S) and TD(T) projects.

1.5.2.3 The TD(T) projects shall be undertaken for enhancement of technology levels for realisation of a component or a system, such that the technology can be customised for development of variants of the components/systems. Technology development for a process/test methodology under a TD(T) project is expected to have higher uncertainties. Therefore, shortfall in the pre-defined objectives or not meeting all the pre-defined objectives would not be considered total failure of the project. **At completion of TD(T) project, the TRL should increase from 2 to 4 or higher.**

1.5.2.4 Availability of draft PSQR is desirable to take up a system level TD project. One or more DcPP(s) may be identified in the TD(S) project along with other technical objectives. It is expected that all essential performance parameters shall be met. Participation of Services Head Quarters (SHQ) reps in all system level TD projects is desirable. **At completion of TD(S) project, the TRL should increase from 4 to 7 or higher.**

1.5.2.5 If a TD(S) project is taken by a Lab and it is not based on draft PSQR, then a communication should be made by the nodal Lab with SHQ to get it included in roadmap of Services/ICDP.

1.5.2.6 A TD(S) project may lead to Mission Mode (MM) project, whereas, the outcome of TD(T) projects, the products/technologies/processes may be used in subsequent system level TD projects or as an input for MM projects. If a TD(S) project doesn't complete its Scope (objectives and deliverables) within its sanctioned PDC, the outcome of the project will be intimated to DISB. In case, a TD(T) project does not meet all the objectives, it will be completed with Partial Success as per the procedures in Chapter 8. A follow-up TD or S&T project can be taken up to build the necessary technological capabilities.

1.5.3 Mission Mode (MM) Project

1.5.3.1 MM projects should be undertaken only after Acceptance of Necessity (AoN) from Defence Acquisition Council (DAC), Defence Procurement Board (DPB) or Services Procurement Board (SPB). Prior to submission of Feasibility Analysis to DAC/DPB/SPB, details of these (feasibility study) will be presented in DMC & accordingly Minutes of Meeting (MoM) indicating **go-ahead** by DMC for these will be issued prior to AoN by DAC, DPB or SPB. Cluster Council and DMC be apprised after DAC/DPB/SPB, but prior to sanction, that the project PDC and scope are in sync with DAC/DPB/SPB approval.

1.5.3.2 Spin-offs from existing projects may also be inducted in organisations within Ministry of Home Affairs (MHA). Modifications in existing systems to cater to the requirements of CAPF/State Police/MHA may be taken up as an R&D activity. Any new design & development

requirement from MHA pertaining to ‘National Security’ may be undertaken as MM project if AoN is granted by Apex Council of DRDO-MHA and the same mechanism may be followed as explained in para 1.5.3.1

1.5.3.3 Lab has to ensure that any project based on AoN accorded by DAC/DPB/SPB and Apex council of DRDO-MHA should not be dropped at later stage. Sanction and completion of these projects should be accorded priority.

1.5.3.4 DRDO not being a commercial organisation, no indent/supply order from the Services should be accepted by Labs for taking up projects based on AoN by DAC/DPB/SPB.

1.5.3.5 PDR should be completed before sanction of MM project. Trial methodology should be a part of PSQR or made available separately. Lab and Directorate for Interaction with Services for Business (DISB) should aim to get the trial methodology, platform availability and efforts to be made to get CONOPS from the Services before PDR. For all design and development related activities in MM projects, cost will be borne by DRDO and generated Intellectual Property (IP) will belong to DRDO only. However, funding from Services will be explored for prototypes, platform modification & trials and will be ensured for field deployable deliverables.

Basic or applied research for any sub-system or component should not be undertaken in MM projects. It is expected that major technology blocks are already available and demonstrated in similar systems. **The reference diagram for pre-requisites and expected outcome of MM category of projects is given in Fig. 3.**

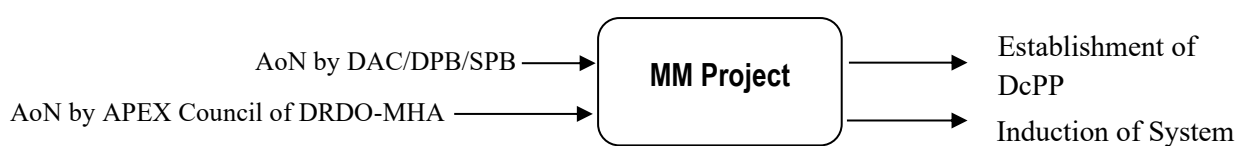


Fig.3: Pre-requisites and expected outcome of MM projects

1.5.3.6 MM projects should lead to readiness of systems as per the PSQR parameters so that it can be converted to SQR leading to induction and production. MM projects would envisage manufacture of specific number of deliverables required for demonstration of system capability vis-à-vis PSQRs during validation trials. The systems used for PSQR validation trials should be produced by the DcPP as this will reduce the trial cycle as per provisions of DAP-2020. Since MM projects are SHQ intensive, their participation at different stages of project is mandatory. A Joint Project Monitoring Team (JPMT) of DRDO and SHQ would be constituted on case-to-case basis.

1.5.3.7 MM projects are to be taken up to demonstrate system level capability against AoNs given by DAC/DPB/SPB. It is expected that minimum TRL should be 6 at the start of the project. The TRL at the completion of MM project is expected to be 9. In case, an imported sub-system/component is being considered for an MM project, indigenous development of the sub-system/component may be undertaken by the Labs through separate projects or through Technology Development Fund (TDF) or DRDO-Industry-Academia Centres of Excellence (DIA-CoE).

1.5.3.8 System developed by DRDO as an outcome of MM projects is expected to have export potential. These systems should be designed and developed in such a way that whenever there is a requirement for export of the particular system, it can be done without any additional work on these systems to convert it from indigenous version to export version. Therefore, necessary hardware & software provisions may be kept in place in the system to enable export with marginal variation in specifications.

1.5.3.9 Availability of PSQR and trial directives/methodology, Acceptance Test Procedure (ATP), minimum quantity for induction, selection of DcPP(s) and successful completion of development trials will lead to user trials. Trial report jointly vetted by DRDO and SHQ with trial sentencing by the SHQ, and acceptance of the system by the user shall be the expected outcome of the project.

1.5.3.10 PSQR compliance and adherence to trials directives is essential for successful completion of trials. Interpretations and assumptions by the Lab in understanding of these should be clarified and preferably taken in formal way by the Lab with SHQ to avoid any ambiguity.

1.5.4 Infrastructure and Facilities (IF) Project

1.5.4.1 A typical and advanced infrastructure, technical and test facilities are created for validating the technology/system developed or proposed to be developed under ongoing or future projects and for evolving test methodologies and protocols. These may also be taken for establishing a specialized skill development facility or complex manufacturing facility. These may be a mix of construction (civil works) and installation of equipment to be procured and installed therein. The infrastructure and test facilities required for multiple projects or

development of technologies, processes, products or for applied research may be taken up through Lab Build-up funds.

1.5.4.2 IF category projects should be used **preferably** for large facilities which require acquisition of land or land development along with setting up of test equipment and facilities.

1.5.4.3 Commissioning of test equipment and facility is the expected outcome of the project. PRC should be conducted for technological assessment of scope and its implementation. **TRL assessment is not applicable for IF Projects. Applicability of PRC may be assessed separately by Lab Director based on the scope of the project.** However, proposal of these projects will be reviewed by **Infrastructure Review Committee (IRC)** constituted in line with PRC of TD(T) & S&T projects, comprising Director Civil Works & Estates (DCW&E) / Chief Construction Engineer (CCE) as Infrastructure expert.

1.5.4.4 IF projects should clearly demarcate the ‘essential’ construction (civil works) as well as equipment to be procured and installed therein.

1.6 EXPECTED OUTCOME IN PROJECT DUE TO UNCERTAINTIES

1.6.1 Completion of project objectives is dependent on test facilities and corresponding infrastructure. Creating the test facilities in S&T, TD or MM project adds to the technological uncertainty as availability of facility in time becomes pre-requisite for the success of the project. Such facilities should be delinked from the R&D project to avoid its failure and should be taken up through Build-up route before sanction of project.

1.6.2 Based on low TRLs, risk and uncertainties a project proposal may be identified by Lab as a ‘**High Risk**’ project. The project will be marked as High Risk based on recommendations by Lab Technology Council and vetting by PRC. While seeking AoN by DMC for such projects which will be either S&T or TD(T) category, designated PD will apprise DMC, the High Risk areas and its impact on successful completion of the project. Based on deliberations, DMC may concur its status as a High Risk project and it will be recorded in MoM. **If a project has been identified by DMC as High Risk project, even total failure is acceptable.** At any given time, total High Risk projects will not be more than 15% of total number of ongoing projects in a Lab. This status recognition will be mentioned in project sanction letter.

1.7 REVENUE DEBT REMITTANCE (RDR) WORK

1.7.1 RDR works can be undertaken at the request of external agencies, such as BrahMos, Department of Space (DoS), Department of Atomic Energy (DAE), Defence Public Sector Undertakings (DPSUs), Public Sector Undertakings (PSUs), Central/State Government agencies and private industries, on the basis of their requirements. The work could include consultancy services, development of prototypes for technology demonstration, manufacturing limited number of prototypes for a developed system in the absence of ToT to a production agency, and providing test support at DRDO Labs etc.

1.7.2 RDR works are to be exclusively funded by the external agency. Detailed guidelines for execution of RDR works issued by Directorate of Finance & Material Management (DF&MM) should be followed. Standard Operating Procedure (SOP) for fund allocation and expenditure management is available on DF&MM's portal in DRONA (DRDO Rapid Online Network Access, the Intranet of DRDO).

1.8 COLLABORATIONS UNDER PROJECTS

1.8.1 Labs may collaborate within and outside DRDO to tap expertise for fulfilling project objectives through following provisions.

1.8.2 **Sub-Projects** are essential for managing large, complex projects by breaking them into smaller more manageable activities, which improves focus, resource allocation, progress tracking and delegation. Successful and timely execution of these sub-projects plays a significant role in achieving the overall objectives of main project. Success of a complex project is dependent on a concerted and co-ordinated efforts by participating Labs with their domain specific expertise and services to the nodal Lab. Sub-projects can be awarded by nodal Lab/centre to participating DRDO Labs. Sub-projects should be submitted along with the project proposal while processing the project for sanction. Financial outlay for a sub-project will be same which is for a project.

1.8.3 **Work packages** can be awarded through MoU by the nodal/participating Lab to constituents of other Government of India (GoI) Departments such as DAE, DoS, Department of Science and Technology (DST) etc.

1.8.4 **CARS and CAPSI:** Labs may avail professional expertise through (i) Contract for Acquisition of Research Services (CARS) from academic institutions (ii) Contract for Acquisition of Professional Services with IDST (CAPSI). Institute of Defence Scientists and

Technologists (IDST) utilises the vast pool of knowledge, experience and technical know-how of the retired scientists of DRDO and is recognised as a 'Deemed Academic Institution' for the purpose of acquisition of professional services. CARS may be avoided in MM projects. Details of the CARS i.e. scope, institute, Project Investigator (PI), title, cost, PDC & research fellows engaged and outcome of CARS projects should be shared with Directorate of ER&IPR (Extramural Research & Intellectual Property Rights) to avoid duplicity and creation of centralised database at DRDO level. Any technology developed through CARS or CAPSI should be entered in the central 'Database of Systems & Technologies' linked with PMIS to support the reusability.

1.8.5 **Consultancy contracts** will be awarded for seeking professional expertise which cannot be met through CARS and CAPSI.

1.8.6 Guidelines for sanction and execution of sub-projects are given at relevant chapters whereas the guidelines issued by DF&MM, DRDO HQ shall be followed for sanctioning and execution of CARS, CAPSI and consultancy contracts.

CHAPTER 2

PRE-PROJECT ACTIVITIES

Pre-project activities are the foundation for the project during its sanction and execution process. These activities use minimum resources but have maximum impact on the outcome of the project. Activities pertaining to Feasibility Studies, Peer Review, Preliminary Design Review, Infrastructure Review, Time Estimation, Cost Estimation, preparation of Draft Project Proposal including Plans for Procurement, Risk Management, Tests & Trials, Qualification & Certification etc. should be carried as a part of pre-project activities. These activities may require six months to one year (based on complexity) and will be undertaken with seed funding (within Director's Power) and team allocation with concurrence of Cluster Council Meeting (CCM). During execution of pre-project activities, detailed review will be held and the proposed PDC will be ascertained. After successful completion of these activities, a presentation will be given in DMC to seek AoN and further processing the proposal for approval by CFA. These activities may be carried out by Participating Labs also. The participating lab will assist the nodal lab in finalizing the pre-project activities.

2.1 SELECTION OF PROJECTS

- Roadmap of DRDO (D-RdMAP) to be considered as DRDO Technology & Systems Perspective Plan. This document will serve as a Single Consolidated Plan Document for projects to be undertaken by the Labs. This document will be reviewed almost every 03 years based on technology requirement at that time.
- Labs will select a project from D-RdMAP to be taken up in current financial year which will be presented in DMC individually after completing pre-project (PRC/PDR) activities for Acceptance-of-Necessity (AoN) including detailed scope, cost and duration. Based on deliberations held in DMC, AoN will be accorded with a validity period as applicable.
- D-RdMAP to have short-term and long-term plan for initiating development of technologies, systems and infrastructure & test facilities.
- Projects should not to be undertaken in the areas mentioned in the already published list of 108 & 75 systems which may be appended (in future) or which industries can do.
- There may also be situations when a project needs to be undertaken based on immediate requirements of Services or technology forecast by Directorate of Futuristic Technology Management (DFTM). Such projects may be taken up with the

recommendation of Cluster Council & AoN by DMC as special case independent of its listing in D-RdMAP.

- Whenever Labs are reporting or presenting list of projects in any forum/report, sub-projects (irrespective of its cost) and activities below ₹10 Cr (or as decided by DMC) should not be reported as these are not categorized as Projects.

2.2 DRAFT PROJECT PROPOSAL

A draft project proposal should be prepared, incorporating details of feasibility studies, project scope including objectives and deliverables, goals and milestones, operating environment/conditions, boundaries & constraints, preliminary work done, comprehensive technology scan in related areas, plan for development and tests, risk identification technologies received from foreign countries and its percentage, imported components and its indigenisation plan etc. Additionally, it should also include mitigation plan for identified risks and possible effect on completion schedule, requirement of additional specialised infrastructure and test facilities, resource requirement in terms of scientific, technical & contractual manpower and funds, estimated time and the current & expected TRLs.

2.3 RESPONSIBILITY AND ACCOUNTABILITY FRAMEWORK: RACI MATRIX / CHART:

2.3.1 Team identification and proper distribution of activities is the key to success of every project. The RACI matrix (also known as RACI diagram or RACI chart) is created to ensure that all stakeholders are on the same page, and working together in unison to achieve the undertaken objectives. This chart identifies and defines the roles and responsibilities of team members in relation to the activities in a project. A RACI matrix uses the letters R, A, C, and I to categorize team responsibilities. **RACI stands for Responsible, Accountable, Consulted, and Informed.**

2.3.2 The Work Breakdown Structure (WBS) will lead to identification of various activities resulting in an activity chart of a project. Responsibility and accountability can be assigned properly if WBS has been prepared in such a way wherein bigger activities are broken down to multiple levels of activities. To prepare RACI matrix for a project, WBS needs to be broken down to minimum 4 levels of Activity Chart for MM and TD(S) and minimum 3 levels of Activity Chart for TD(T), S&T and IF category having well defined timelines and required

resources. For a project having multi sub-projects, separate RACI matrix may be prepared for each of its sub-projects.

2.3.2.1 Responsible (R)

(R) is the one responsible for getting the work done. When labelled ‘responsible’ in a RACI matrix, it indicates that the person(s) expected to be executing the activity. Necessary administrative support should be provided by the Lab to (R).

2.3.2.2 Accountable (A)

This is the person who is responsible for overseeing the activity and making sure these are done properly. They (A) may not execute the activity instead are managing and ensuring the completion of the work. This person is responsible for signing-off the activity. Only one person will be accountable for the activity. Necessary administrative support and resource control should be provisioned to (A).

2.3.2.3 Consulted (C)

(C) will assist by providing information and support about a particular activity or deliverable. They are not directly responsible for an activity instead provide necessary information that will help the ‘R’ to get assigned activity completed.

2.3.2.4 Informed (I)

(I) Individual or group of individuals are the one who are to be kept updated about the progress of an activity or deliverable. They are commonly higher or top management or stakeholders who need to be informed about the progress of the project but does not have direct role in execution.

2.3.3 The RACI Matrix can be considered as the guiding document which every team member may refer as the project progresses. It keeps the project on track by clearly defining who is responsible for what. When every team member knows exactly what their responsibility is, a team can avoid miscommunication, wasted time and confusion, resulting in higher rate of completion / success.

2.3.4 Structure of a RACI Chart:

1. **Activity list:** The chart which is compatible with WBS, lists all the activities that need to be completed to achieve project goals. Each activity is typically listed in the first column of the chart.

2. **Roles:** The roles (Responsible, Accountable, Consulted, and Informed) are listed across the top row of the chart. Each role corresponds to a specific responsibility within the activity.
3. **Assignments:** Within the matrix, each activity has a corresponding cell where the roles are assigned for that particular activity.

2.3.5 Application of RACI:

- **Clarity:** Provides clarity on who is responsible for each activity (Responsible role).
- **Accountability:** Clearly identifies who is ultimately accountable for the activity's completion (Accountable role).
- **Consultation and Communication:** Indicates who needs to be consulted (Consulted role) and who needs to be informed (Informed role) throughout the activity's lifecycle.
- **Alignment:** Ensures alignment of responsibilities with project goals, reducing misunderstanding and conflicts.

RACI Matrix Template *(filled for illustration purpose only)*

Activity (N)	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Activity 1	Dy. PD	PD	Expert	Apex Body Team Members
Activity 2				
Activity 3				
Activity 4				
...				
...				
Activity N				

2.4 RISK ANALYSIS AND RISK MANAGEMENT PLAN

2.4.1 Risk Mitigation Plan should be a top-down approach. The risk identification and its mitigation plan together with the Product Breakdown Structure (PBS)/Work Breakdown Structure (WBS) should be included in the draft project proposal. It should be reviewed and vetted by the Peer Review Committee (PRC) or Preliminary Design Review Committee (PDR). It should also be presented as part of Detailed Design Review (DDR), Critical Design Review (CDR) and other project reviews.

2.4.2 Risks in an R&D project occur primarily in areas given below (the list is illustrative and not exhaustive):-

- (a) **Technical Risk:** Design iterations to meet the performance and operational requirements, change of components due to obsolescence or due to design requirements, changes in operating environment due to changed concept of operations, limitations of manufacturing processes & like.
- (b) **Resource Risk:** Delays due to procedural issue related to procurement, development contracts, non-performance of vendors, technology/export/import denial and human resource.
- (c) **Test & Evaluation Risk:** Test requirements vis-à-vis availability of test facilities and necessary infrastructure, availability of test methodology, additional test requirements due to changed design/ operating conditions & others.

2.4.3 Risk can also be identified by the factors which are primarily responsible for delay. These can be categorized as External Factor and User Factor. Process, Administrative, Management, Logistical, Operational, Technical or Engineering challenges, which may contribute directly or indirectly for delay may also be identified for resolution. Identification of these factors may be done in accordance with **Annexure 'I'**.

2.4.4 Risk Mitigation Plan broadly should involve the following steps: -

- (a) Identify the risk - to be drawn from the PBS/ WBS

The risks can be identified through one or more of the following: Reviews & analysis, SWOT analysis, brainstorming among experts, Delphi techniques, assumption analysis, cause-and effect & influence diagrams, Monte Carlo simulations etc.

- (b) Assess the probability of occurrence of the risk and impact on project duration, cost and performance. This assessment should be based on past experiences and qualitative judgement, ABC analysis etc.

- (c) Prioritize risks - Based on Impact/Probability.

2.4.5. Plan for implementation of risk mitigation strategy.

2.4.6 Track, monitor and update the risks throughout project cycle.

2.4.7. The projects may adopt any other suitable method for specific requirements.

2.5 PROCUREMENT PLAN & DEMAND APPROVALS

2.5.1 An overall procurement plan should be prepared, detailing all sub-systems & components to be procured with their estimated timelines for different procedural actions, cost and source of supply. Standard procurement time should be assumed for planning based on average time taken in the Lab and number of iterations in tendering.

2.5.2 Labs may initiate procurement actions for projects to be submitted for approvals/PDC extension/cost enhancement as per Procurement Manual-2025 as amended from time to time.

2.5.3 Execution of Procurement Plan will be reviewed by PMRB and EB and necessary corrective actions, if any, may be recorded.

2.5.4 Due diligence to be done at the time of identification of long lead items and most critical component/sub-component/raw materials/devices and equipment for maintaining a minimum stock being imported from other countries and their approval should be taken at the initiation of the project to avoid delay caused by disturbed/interrupted supply chain. Lab may prepare indigenization plan for such items during start of the project & initiate action so that indigenization effort goes separately parallel to the project.

2.6 HUMAN RESOURCE ACCOUNTING

2.6.1 **Accounting of Manpower:** The requirement of Scientists year-on-year basis for the ongoing and new projects has been indicated in the DRDO Roadmap. Accordingly, manpower availability and additional/dedicated requirements should be indicated with reason, if any. Manpower should be catered in/as design team, documentation team, prototype manufacturing and testing team, procurement in-charge, quality officer etc.

2.6.2 The man-days of Scientists, DRTC and Admin & Allied Cadre personnel utilized in the project should be accounted. Quarterly record of manpower utilization to be reviewed in every PMRB (till Technical Completion) and consolidated in the Technical Completion Report.

2.7 TESTING PLAN /TEST INFRASTRUCTURE/ TEST METHODOLOGY

2.7.1 A project proposal will have estimation of the number of trials, redoing of the trials in case of failure, inadequacy or improper results which may be needed in a project. It will also have the details of the trials which can be simulated. The simulation capabilities should be enhanced and the number of simulation trials should be done for reducing the trials in an actual environment.

2.7.2 A first level of test plan including time and cost estimations should be evolved. It should contain assessment of existing facilities and its suitability to meet the test requirements proposed in the project. The project proposal must include tests required, test equipment/facilities required and their availability, test methodologies to meet the performance as envisaged in the project. The project proposal should document whether the required test facilities are available or being created. A project may be taken only if the required test facilities are either available or being created as part of the current project/build up. In case such facilities are not available with DRDO, a long-term contract/Inter Govt. Agreement (IGA)/Memorandum of Understanding (MoU) with agencies or a plan/way ahead mutually agreed by DRDO and User for testing will be considered for sanction of project.

2.8 SELECTION OF DEVELOPMENT CUM PRODUCTION PARTNER (DcPP)

Selection of DcPP is required to develop the industry partner for prototype manufacturing, testing and life-cycle management of systems. Preparatory activities for selection of DcPP may be initiated during pre-project sanction stages wherever the commensurate details are available. The process for selection of DcPP shall be governed by DISB guidelines DISB/DPP/27656/P-1/Para-72 dated 15th Mar 2019 as amended from time to time.

2.9 PROJECT PEER REVIEW AND INFRASTRUCTURE REVIEW

2.9.1 **All TD(T) and S&T projects shall be mandatorily peer reviewed** by an expert committee i.e. PRC for assessment of viability and technical adequacy of the proposal. **For MM & TD(S) projects, PDR should be conducted in lieu of PRC.** TRA report will be prepared by Project team and shall be presented to Lab Technology Council for review. The reviewed TRA report will be presented to PRC/PDR Committee as the case may be.

Specific recommendations of PRC/PDR should be recorded for consideration or implementation. These recommendations, duly approved by the Chairman should be included in the project proposal. PRC/PDR recommendations which are not agreed will need to be highlighted during presentation to DMC for seeking AoN.

2.9.2 **Infrastructure & Facilities (IF) category projects shall be presented to Infrastructure Review Committee (IRC).** IRC constituted in line with Peer Review Committee (PRC) essentially comprising Director, DCW&E or rep as Infrastructure expert will assess the viability and technical adequacy of the proposal. For projects other than IF category, having civil works a combined meeting of PRC/PDR (as applicable) and IRC may be convened.

2.9.3 Competent Authorities for constitution of PRC/IRC is given below in **Table 3**:

Table 3: Competent Authority for Approving the Constitution	
PROJECT CFA	COMPETENT AUTHORITY
Lab Director	*Lab Director
DG (Cluster)	DG (Cluster)
Secretary, DDR&D and beyond	Secretary, DDR&D
* Note: (a) While seeking approval of Competent Authority for constitution of Committees, the basic details of project, viz. Name, Cost, PDC, Category, CCM minutes, alignment with 'Roadmap of DRDO' and bio-data of external members (outside DRDO) may be provided. (b) Composition of the Committee may be intimated to Director (PM), O/o DG (Cluster).	

2.9.4 **Broad Constitution of PRC/ IRC should be as under: -**

- (a) Chairman (Eminent person preferably from outside DRDO)
- (b) Two external experts (eminent domain experts from academic institutions/ R&D institutions/ Industry) (not mandatory for IRC)
- (c) Two DRDO experts mandatorily from other Labs (not mandatory for IRC)
- (d) Director DCW&E or rep (for IRC)
- (e) Chairman, Lab Technology Council to address TRL & PRI values/analysis (not applicable for IRC)
- (f) Director DP&C, DRDO HQ (every PRC and for IF in IRC)
- (g) Director (PM) of DG (Cluster)
- (h) Representatives of participating Labs (if applicable)
- (i) Nominated Scientist 'G' by DG (TM) (not applicable for IRC)
- (j) Counsellor (Defence Technology) Abroad / DIC – For projects with significant import content or under Govt-to-Govt. (not mandatory for IRC)
- (k) Project Director designate (Member Secretary)

Note:

- *Competent Authority or Chairman PRC/IRC may also co-opt any eminent person from Academia/Industry to participate in review process.*
- *Exemption of a Member from the suggestive constitution of the Committee (PRC/IRC) has to be approved by the Competent Authority.*

2.10 PRELIMINARY DESIGN REVIEW (PDR), DETAILED DESIGN REVIEW (DDR) AND CRITICAL DESIGN REVIEW (CDR)

2.10.1 The committee of **PDR** should comprise of the following members. However, members listed at **(d) to (q)** may be invited for **DDR and CDR** on requirement basis as proposed by PD.

- (a) Chairman (eminent technology expert preferably external to DRDO)
- (b) Two external experts (eminent domain experts from academic institutions/ R&D institutions/Industry)
- (c) Two DRDO experts mandatorily from other Labs
- (d) Director (PM) of DG (Cluster)
- (e) Rep ISSA [for MM & TD(S) projects]
- (f) Rep SHQ (for QR based/SHQ AoN cases)
- (g) Rep DISB, DRDO HQ (for QR based/SHQ AoN cases)
- (h) Rep CAPF (for APEX Council of DRDO-MHA AoN cases)
- (i) Rep DLIC (for APEX Council of DRDO-MHA AoN cases)
- (j) Director DP&C or Rep. for MM & TD(S) projects
- (k) Nominated Scientist 'G' by DG (TM)
- (l) Director DQR&S or Rep {if CFA is beyond DG (Cluster)}/ Director SQR (Cluster) {if CFA is DG (Cluster)} / Head R&QA {if CFA is Lab Director}
- (m) Director DCW&E or Rep (in case civil work)
- (n) Director DIPAS or Rep (for MM projects)
- (o) Director CAIR or Rep (for MM projects)
- (p) Chairman, Lab Technology Council to address TRL & PRI values/analysis
- (q) Counsellor (Defence Technology) Abroad/DIC – For projects with significant import content or under Govt-to-Govt. (if applicable)
- (r) Project Directors of sub-projects
- (s) Project Director designate (Member Secretary)

Note:

1. *Competent Authority or Chairman PDR/DDR/CDR may also co-opt any eminent person from Academia/Industry to participate in review process.*

2. *Exemption of a Member from the suggestive constitution of the Committee (PDR/DDR/CDR) has to be approved by the Competent Authority.*

3. *Competent Authority should nominate eminent person or experts in MM category from DIPAS and CAIR for Human Factor Integration (HFI) & Human Reliability Analysis (HRA), applicability of Artificial Intelligence respectively to participate in review process.*

2.10.2.1 The proposal for constitution of PRC/PDR should be linked with 'Roadmap of DRDO' and relevant details (viz. Category, Cost, PDC) may be provided in the proposal before approval of CFA. In case it is not linked with roadmap, necessity of taking up the project may be provided.

2.10.2.2 **The competent authority for Constitution of PDR, DDR and CDR Committee is same as per Table 3.** Detailed guidelines pertaining to these design reviews is given in Chapter 5.

2.10.3 Planning Peer Review/Preliminary Design Review

2.10.3.1 **Documents for PRC or PDR.** The following documents have to be prepared by designated PD and will be forwarded to the PRC or PDR Chairman, Director (PM), O/o of DG (Cluster) and members at least 05 working days in advance: -

- (a) Draft Project Proposal along with its Executive Summary
- (b) Draft PDR document (for MM and TD(S) projects)
- (c) TRL assessment report & PRI reviewed by Lab Technology Council
- (d) Composition of PRC or PDR committee
- (e) Roles of external constituents of PRC or PDR along with bio-data highlighting the relevant expertise/experience.

2.10.3.2 Director (PM), O/o DG (Cluster) shall examine the Draft Project Proposal and Executive Summary submitted by Lab prior to the conduct if the composition has been approved by Competent Authority and the details have been shared with all mandatory members.

2.10.3.3 **Broad Agenda of PRC or PDR.** To deliberate upon following aspects and give recommendations:

- (a) Deliberate on the scope comprising objectives, goals and milestones, deliverables, operating environment/conditions, boundaries & constraints, need of the project vis-à-vis charter of the Lab. Ensure that it is not in the list of systems that DRDO will not do.
- (b) Adequacy of pre-project activities.
- (c) Comparison w.r.t global scenario and contemporary systems/technologies along with related specifications and year of research/induction with giving references to authentic sources. Export potential based on qualified data of Friendly Foreign Countries (FFC) from international databases such as Janes and SIPRI may also be deliberated.
- (d) **Design Aspects**
 - Available design alternatives.
 - Selection of a suitable design option.
 - Reasons for adopting a specific approach.
 - Likely number of design iterations.
 - Analysis of requirement vs. design
- (e) PDR Committee shall review design methodology, test and validation plan of systems and subsystems, Preliminary QAP, hardware and software development plan, qualification plan, system interface requirements and system realisation plan.
- (f) Technologies re-use from other DRDO projects and Labs may be presented. Import content, percentage indigenization and plan to indigenize import content may also be deliberated.
- (g) **Realisation methodology:** Identification of optimum method of product realisation/technology maturity.
- (h) **Risk Factors/Grey Areas:** Risk factors and grey areas which are likely to pose challenges for successful execution of the project and their mitigation strategy.
- (i) **Responsibility & Accountability matrix (RACI).**
- (j) **Time Estimation:** Estimated time of the project should be finalized as per the recommendation of Time Estimation Committee (TiEC).

- (k) **Project Schedule:** To examine project schedule i.e. whether realistic and achievable timelines are projected considering possible contingencies. Broad schedule w.r.t six monthly milestones may be deliberated.
- (l) **Category of the Project:** Category under which proposed project should be undertaken be deliberated and recommended.
- (m) **Realisability/Achievability** of major milestones linking financial outlay and timelines of the project as per schedule indicated in the PERT Chart.
- (n) **QA Plan:** Quality assurance (QA) plan should be deliberated with nominated QA member of the Lab and should cover all aspects viz. design, manufacturing and testing. It shall examine Quality & Reliability requirements, Quality Function Deployment (QFD) analysis, Failure Mode Effect Analysis (FMEA)/Failure Mode Effect & Critical Analyses (FMECA) analysis and plans to achieve QR goals.
- (o) **Infrastructure Plan:** Available/ to be created infrastructure for system/sub-system realisation and adequacy of Infrastructure facility for planned test.
- (p) **Testing Plan:** Testing Plan of Lab model or prototype should be deliberated together with acceptance criteria.
- (q) **Documentation Plan**
- (r) **Cost Estimation:** Rough order of cost estimates may be provided based on budgetary quote/e-mail quotes etc. to assess the approximate cost of the project. However, estimated cost of the project should be finalized as per the recommendation of project Cost Estimation Committee (CEC).
- (s) **TRL levels/ PRI indices** at the beginning of project (at sub-system level and overall project level) and expected increase in these indices at the time of project closure should be discussed in detail and final values expected to be achieved shall be documented.
- (t) **Deliberations on CONOPS (Concept of Operations) with User Rep/SHQ.**

- (u) Committee to assess the detailed system analysis of MM and TD(S) projects using standard practices and guidelines formulated by DRDO as amended from time to time.
- (v) Deliberate on applicability of Human Factor Integration (HFI) & Human Reliability Analysis (HRA) and Artificial Intelligence.
- (w) Preliminary Identification of DcPP.
- (x) The PDR Committee shall review the Preliminary Design Document (PDD) to ensure the correctness and completeness of the requirements/specifications and assess the technical feasibility and adequacy of the design to meet the requirements/specifications. Based on the above review, PDR committee shall recommend improvements (if any). The approved PDD will be the baseline technical document for the proposed project.
- (y) Procurement plan, safety plan, resource allocation plan and any other points deemed necessary.

2.10.3.4 Role of Constituents of PDR/PRC/IRC Committee

➤ **Project Director**

- (a) Prepare project documents for PRC or PDR as per DPFM-2026 guidelines and circulate to all members at least 05 working days before the meeting.
- (b) Record and apprise Chairman, PRC or PDR about the availability of required development expertise and infrastructure for development and production partners for targeted system and sub-systems.
- (c) Technology re-use from other DRDO projects and Labs may be presented with supporting data from centralized technology database linked PMIS.
- (d) Provide the resource plan, responsibility and accountability matrix (RACI), project schedule, project category, documentation plan and reuse of already developed technologies under other projects.
- (e) Import content, percentage indigenization and plan to indigenize import content may also be deliberated.

- (f) Ensure that comments from members are solicited prior to the meeting and forwarded to PRC/PDR Chairman for taking up during discussion.
- (g) Propose the TRLs and PRI of the project and TRL progression plan for project duration.
- (g) Ensure that comments from participating Labs are obtained and documented regarding identified sub-projects and resources allocated.
- (h) Issue the Minutes of Meeting (MoM) within 10 working days from the scheduled meeting date and to be included in final PRC or PDR Report.
- (i) Written inputs should be taken from every external member for inclusion in report.

➤ **DISB, DRDO HQ**

Provide inputs about the SHQs requirement/perspective plans which may have direct impact on the proposed project activities.

➤ **SHQ Representative**

- (a) Clarify issues related to General Staff Qualitative Requirement (GSQR)/Air Staff Qualitative Requirement (ASQR)/Naval Staff Qualitative Requirement (NSQR)/Joint Staff Qualitative Requirement (JSQR), etc.
- (b) Provide necessary inputs in absence of any formal Qualitative Requirements (QRs).
- (c) During PDR, discussion will be held on CONOPS for better understanding of system's operational scenario and expected evaluation methodology which will help in optimal design of the same.
- (d) To comment on the suitability and availability of service test platforms for the system under development.
- (e) To comment on suitability of deployment of the system on the targeted platform, complexity, dependence for platform modification and its availability.
- (f) Details about the existing systems in use with services and elsewhere in the world and other details about the operation, maintenance and use of the product.
- (g) Views regarding scope and time estimates etc.

➤ **ISSA**

To provide input or undertake system analysis studies in all MM and TD(S) projects.

➤ **DCW&E**

To provide input on civil work and infrastructure related aspect of the project.

➤ **DIPAS**

To provide input on Human Factor Integration (HFI) and Human Reliability Analysis (HRA) aspect of system being developed in project.

➤ **CAIR**

To provide input on applicability of Artificial Intelligence (AI) in systems and software.

➤ **Counsellor (Defence Technology)**

To provide input on import content or Govt. to Govt. collaboration relevant to the project.

➤ **Chairman, Lab Technology Council**

Assess the TRLs and PRI of the project and TRL progression plan for project duration. Identify technological gaps, grey areas, critical technologies and strategy to mitigate the same. Comment on the suitability of the project category based on TRA.

➤ **Nominated Scientist ‘G’ by DG (TM), DRDO HQ**

- (a) To provide input on identification of emerging technologies that are likely to have high impact in the short and mid-term on Defence and Security and about the appropriateness of the project vis-à-vis the technology trends elsewhere.
- (b) Provide details of technology developed through Academia/DIA-CoE that can be re-used.
- (c) Advise if the project can/cannot be undertaken by Academia.
- (d) Identify if any indigenization effort can go to Academic institute.
- (e) Offer comments on Technology Readiness Levels, technological gaps, grey areas, critical technologies and strategy to mitigate the same.

➤ **DP&C, DRDO HQ**

- (a) Check if the project being undertaken is as per charter of Lab, listed in ‘Roadmap of DRDO’ or in Technology Forecast by DFTM.

- (b) Suggest allocation of work share to other Labs for modules requiring specific technical expertise.
- (c) Ensure compliance and adequacy of the draft project proposal vis-à-vis prevailing policies and guidelines.

➤ **Director (PM), O/o DG (Cluster)**

- (a) Ensure that the project undertaken is as per charter of Lab and it is not in the list of systems that DRDO will not do. Assure CCM recommendation of Nodal and participating Labs have been incorporated.
- (b) Assist Lab, DG (Cluster) and DRDO HQ in finalizing PRC or PDR constitution. Chairman & Members be from database of experts, once in place & feedback about the experts is given to database after review.
- (c) Ensure compliance and adequacy of the draft project proposal vis-à-vis prevailing policies and guidelines.
- (d) Ensure re-use of technology or system or sub-system already developed as available in 'Database of Systems & Technologies' linked with PMIS.
- (e) Ensure that Chairman and all members receive necessary documents (draft project proposal, executive summary, TRLs assessment report duly verified by Lab Technology Council including details of 'what is expected from the PRC or PDR' etc.) 5 working days before scheduled meeting date.
- (f) Ensure that minutes incorporating important comments of the members are approved by the Chairman and issued in 10 working days.
- (g) Update database of experts about utilization of services of Chairman and Members and their performance.
- (h) Comment on the resource plan, responsibility and accountability matrix (RACI), documentation plan and reuse of already developed technologies under other projects.

➤ **External Technical Experts**

- (a) Closely examine system configuration, technologies and sub-technologies, contemporary technologies worldwide and import options, if any.

- (b) Confirm that projects and its major/critical sub-systems cannot be done by the Industry or Academia.
- (c) Ensure that wherever necessary, alternative technologies have been examined and given due consideration.
- (d) Look for the technological gaps/grey-areas and to examine how these are proposed to be overcome and to suggest alternate/suitable approaches for the same. Check that resources projected are adequate to meet the scope comprising objectives and deliverables of the project.
- (e) Ensure complete coverage by Risk Mitigation Plan.
- (f) Written inputs may be shared with Project Director for inclusion in report.
- (g) Review of proposed design approach/methodology and preliminary design and offer pertinent comments.

➤ **Chairman**

- (a) Examination of the documents submitted to the PRC or PDR and to call for additional information, if required. Meeting to be conducted only if the documents are received by all members at least five working days in advance. No deviation is to be accepted.
- (b) Ensure that design, realisations and testing aspects are adequately addressed in the documents.
- (c) Ensure presentation from Project Director on TRLs and Final PRI of proposed project in PRC.
- (d) Ensure that the Peer Review is conducted as per the document made available to the PRC or PDR and all issues highlighted in the Executive Summary be addressed.
- (e) Take views of each of the member giving them sufficient time to comment on the project proposal during the review and ensure that all the observations and suggestions of each of the member is recorded in the report.
- (f) Comment on project viability in view of TRL levels, adequacy of pre-project work, realisation plan and Test & Evaluation capacity.

- (g) Any other issue not discussed but considered necessary by the Chairman in the interest of the project can be addressed and documented.
- (h) Approve the minutes of the meeting of PRC or PDR after ensuring that the important comments of all the members have been incorporated.
- (i) Comment on project schedule proposed in the project.

Chairman may have these meetings can be held in multiple rounds comprising more than a day.

2.10.3.5 Recommendations of the PRC or PDR Committee

Relevant deliberations and recommendations of the PRC or PDR on topics stated in the agenda need to be minuted for consideration/implementation, and should be a part of the final project proposal. The draft minutes of PRC or PDR may be circulated to all the members to seek their comments to be recorded in the final minutes. Recommendations (Minutes of Meeting) of the Committee specifying project category, PDC & indicative cost duly approved by the Chairman should be appended with project proposal. The final project proposal may be suitably modified to incorporate planned actions/modifications in view of the recommendations of PRC/PDR else suitable justification should be given to DMC for non-inclusion of specific recommendations of PRC/PDR by the project team for consideration and final decision by DMC.

2.10.4 Conduct of PRC, PDR, DDR and CDR

2.10.4.1 PRC/PDR will assess project feasibility by taking into consideration the design aspect, realisation methodology, projected timelines, realizability/achievability and assessment of validation of TRLs.

- (a) The Chairman of the PRC, PDR, DDR and CDR would be nominated preferably from the central pool of experts (as & when available), otherwise should be an eminent person (preferably from outside DRDO) having domain expertise related to the project. Chairman may either be from a premier research organization, academic institution or from industry. In exceptional cases, if from within DRDO, Chairman should be from a different Lab working in similar area. It is desirable that the Chairman should have experience of managing/coordinating/directing projects of similar costs and technological complexities.

- (b) The PRC, PDR, DDR and CDR should have a mix of members from within and outside DRDO. External Members would also be nominated preferably from the central pool of experts (as & when available), otherwise should be eminent personalities from R&D institutions, academic institutions and industry. The expert should not be nominated from the academia/industry, which is likely to be contracted for CARS or development contract in the same project who have been associated previously or likely to be engaged in upcoming projects.
- (c) While processing the cases for conducting PRC or PDR, the designated Project Director will send a copy of draft project proposal and Technology Readiness Assessment (TRA) report, reviewed by Lab Technology Council, to all committee members. Recommendations made by Lab Technology Council regarding PRI & TRL values should be presented in the Committee.
- (d) Relevant deliberations and recommendations of the PRC, PDR, DDR and CDR Committee will be minuted and issued after due approval of the Chairman within 10 working days of the last meeting. These meetings can be held in multiple rounds comprising more than a day.
- (e) The Minutes of the PRC or PDR will be appended to the project proposal.

2.11 TIME ESTIMATION OF A PROJECT

2.11.1 Project schedule and overall project duration are estimated for the identified scope including objectives and deliverables. It should be drawn using a standard project scheduling tool or software. Experience of time taken in similar activities, time taken in procurement or import as per standard process, iterations in design/ procurement and professional judgement should be factored to estimate the project duration.

2.11.2 It is advised to refer lesson learnt in the previous projects (available in PMIS) or 'Reasons for Delay' reported in Audit/ review reports available on DP&C portal on DRONA.

2.11.3 Low Technology Readiness Levels and capacity assessment will lead to design iterations for arriving the final configuration. These shall be identified and deliberated in PRC or PDR and TiEC for analysing the impact on project timelines. In an R&D project, design iterations should be in-built into the process. Design iteration cycle can vary from case-to-case. Design iterations can lead to repeat of procurement cycle. Number of design iterations considered to arrive at the project duration should be documented. Complex technology areas,

development of which may have a bearing on the project duration, should be highlighted in the report.

2.11.4 **‘Pre-requisite’ for conduct of TiEC meeting** is completion of the following activities:

- a) Draft Project Proposal.
- b) Preliminary time estimates based on similar projects, risk envisaged, risk availability etc.
- c) Input from any Standard Time Estimation tool or DRDO developed Time Estimation & Assessment Module (TEAM) software tool may be presented during TiEC meeting, highlighting the ‘Realistic’ and ‘Pessimistic’ time estimates. The handholding for using the TEAM software is provided by ITM, with support from ISSA.
- d) Project team identification including documentation, QA & MMG Reps with clearly specified responsibilities duly approved by Lab Director.
- e) PSQRs available (For MM projects) / requirements captured {For TD(S) projects}.
- f) PRC or PDR report along with approved Minutes of Meeting (MoM).
- g) Interaction with probable Industry partners for realisation time & cost estimation.
- h) Project Execution Plan defining execution structure, status reporting, risk management plan, quality/certification plan, documentation plan, realisation plan, records management, approximate costing & schedules.
- i) Validation methodology with Integration, Testing and Evaluation Plan ensuring availability of Resources.

2.11.5 **Terms of Reference of Time Estimation Committee (TiEC):**

- (i) The proposed committee will deliberate on the aspects of ‘pre-requisites’ and the draft project proposal. Committee should propose the **realistic and pessimistic timelines** considering the following phases (as applicable):
 - Design Phase
 - Requirement finalisation
 - Specification finalisation
 - Procurement plan finalisation and initialisation (if applicable)
 - Execution Phase
 - Design Reviews
 - Placement of supply order and receipt of items (if applicable)

- Project Documentation (for Hardware/Software) including: -
Certification/Qualification, ToT, User Manual, Maintenance Manual
 - Realisation of sub-systems and Integration
 - Certification/Qualification (Hardware/Software)
 - Platform availability & Modification (if applicable)
 - Test & Evaluation Phase (as applicable)
 - In Lab
 - Rig
 - Platform/Field
- (ii) Time Estimation Committee may also deliberate on: -
 - a) The human resource requirement and available resources to have a holistic assessment of time required.
 - b) The PDC extensions taken by Lab and reasons for delay in the ongoing projects and completed projects in **last 10 years** while bringing out the report.
- (iii) The committee may be apprised of the procurement process, average procurement time in the Lab and the number of procurement in the project and type of tendering in each and the associated risk/repeat cycles.
- (iv) Time required for the project based on the design approach approved by PRC/PDR should be estimated considering: -
 - a) Estimated timelines of similar technologies developed in the past should also be considered and the number of design iterations factored due to functional, certification and manufacturing/quality requirements and also the extent to which the changes affect the project timelines with iteration.
 - b) Industry capability to manufacture components/products to the design specifications should also be factored and documented.
- (v) Availability of test facilities, extent to which tests required for projects are feasible in existing facility and expected shortfall in tests for complete scope of the project, if any, should be documented in TiEC report. Time required to establish new test facilities or alternate route to meet the overall project objectives should be detailed for time estimates.

- (vi) Committee should also deliberate and finalise on the ‘**Sunset milestone**’ i.e. a significant milestone or stage in a project, if not achieved, will lead to discontinuation of the project. R&D thrives on experimentation and there might be failures – but better to fail fast, fail smart and move on. In complex projects or High Risk technology development, more than one sunset milestone may be defined, but may be limited to maximum 3 sunset milestones. In case of single sunset milestone, it should not be too early or towards end of the project and should essentially be after DDR or mid of the project in terms of number of activities or expenditure. For sunset clause in MM project, MoM of the DAC and DPB may also be referred in case of any such indication.
- (vii) Inputs from each of the external member should be included in the report.

2.11.6 Time Estimation Committee (TiEC) will submit the recommendations of following within 10 working days from the date of meeting: -

- a) Realistic Time (PDC) (in months)
- b) Pessimistic Time (in months)
- c) Detailed PERT Chart with 3 or 4 level of WBS based on Category of the project.
- d) Sunset Milestone(s) (marked in the PERT Chart)
- e) Inputs of every external member may essentially be recorded in the final reports.

2.11.7 Time Estimation Committee (TiEC) Composition and Approving Authority is as under:-

Table 4 : TiEC Composition and Approving Authority					
Project / Prog. CFA	Approving Authority	Chairman	Internal Experts	External Experts	Member Secretary
Lab Director	Lab Director	Nodal Lab Scientist 'G' (non-member of project)	• One DRDO subject experts • Dy. IFA (R&D) • Head Project Management Office/ Head Tech Coord • Head MMG • Head QA	• 2 experts from Academia (At least One from IIM/ Reputed Management Institute or Management Deptt. of IIT/ Reputed Universities) • 1/2 experts from Industry (Pvt. Industry/MSME/ Startups) • For MM/ TD(S) Projects One Rep from SHQs/HQ-IDS*	Deputy Project Director
DG (Cluster)	DG (Cluster)	Nodal Lab Scientist 'G' (non-member of project)	• One DRDO subject experts • Rep of IFA (R&D) • Dir PM • Head MMG • Head QA	• 2 experts from Academia (At least One from IIM/ Reputed Management Institute or Management Deptt. of IIT/ Reputed Universities) • 1/2 experts from Industry (Pvt. Industry/MSME/ Startups) • For MM/TD(S) Projects One Rep from SHQs/HQ-IDS*	Deputy Project Director

Secretary DDR&D and beyond	DG (Cluster)	Scientist 'H' (non-member of project) <i>(If Sc 'H' is not available, Sc 'G' may also be considered)</i>	• One DRDO subject experts • Rep Addl. FA (R&D)/ IFA (R&D) • Head MMG • Head QA • Dir PM • Dir P&C (HQ) or Rep • Dir DFMM (HQ) or Rep • Dir DQRS (HQ) or Rep (for MM and TD(S) only)	• 2 experts from Academia (At least One from IIM/ Reputed Management Institute or Management Deptt. of IIT/ Reputed Universities) • 1/2 experts from Industry (Pvt. Industry/MSME/ Start-ups) • For MM/TD(S) Projects One Rep from SHQs/HQ-IDS*	Project Director
Note:- ➤ Approving Authority/ Chairman may also co-opt members from Academia/ Industry as necessary. ➤ * Service Officers posted at Labs should not be considered as External Experts ➤ Experts from Project Consulting firms could be considered for incorporation in the Committee in addition to industry experts. ➤ One of the academia Rep may be from Operations Management (OM) background. ➤ Committee may also consider 10 years track record of Labs for timely completion of projects. This will enable inclusion of local factors despite common policies across DRDO. ➤ ITM and ISSA to maintain the TEAM software and provide support to the TiEC. However, ITM will hold the overall responsibility. ➤ Chairman TiEC and internal members should also be part of Highest Review Committee (EB or PMRB) Meeting in-case available within the lab, if PDC extension is sought by the Lab.					

2.12 COST ESTIMATION

2.12.1 An R&D project is different from a traditional project on account of uncertainties of conversion of knowledge into products/systems. Many components/sub-systems/ technologies would be attempted for the first time in the project. Cost of the bought-out or Commercially-Off-The-Shelf (COTS) items can be realistically assessed; however, the assessment of development cost would be a challenge. Hence, professional judgement of cost based on past experience of project team would be an integral part of the Cost Estimation.

2.12.2 Cost estimation should be supported by a clear statement of items or services to be procured, quantities and scope of the same and basis for estimating the cost of those items/services. The said basis could be the Last Purchase Price (LPP) or Budgetary Quote (BQ) or professional judgement. Scope of the project including objectives and deliverables should

be translated into modules/systems/activities, associated test equipment, software, works, etc. and each further decomposed to its sub-system/component level, which should be costed. Assumptions on which costing is based should be defined e.g.

- (a) Basis of calculation of numbers/spares of modules/systems/sub-systems taken for costing.
- (b) Specifications frozen or accepted by the appropriate authority, peer-reviewed.
- (c) Methodology of implementing the project as accepted/proposed.

2.12.3 The general principle of product costing should apply to the costing of project too. Every estimated cost should have a basis or supporting data in the form of one of the following:

- (a) **Annual Escalation and Foreign Exchange Rate Variation (FERV)** may be applied on LPP. Percentage of escalation on the above per year should be taken into account. For example, Previous Purchase Price/Supply Order (SO)/Purchase Order (PO) either from the same Lab/ Cluster/ any other Govt. agency normalized to the present day estimated cost by including a 2% escalation per year for imported items and 3% escalation per year for indigenous items.
- (b) And/or **Internal Cost Estimation based on Experience/ Professional Officers Vetting (POV)**. Especially in case of sub-systems for which components/Line Replaceable Units (LRUs) are known to the Project/Lab, internal estimation is relevant in cases of customization, development of hardware and software. POV involves a team of experts reviewing and verifying project's estimated cost to ensure accuracy, viability and risk mitigation, often by analyzing project documents like Bills of Quantities (BOQ) and contracts. This process helps establish realistic financial requirement, identify potential issues like cost escalation and inform decision making for High Risk projects where no such development has taken previously.
- (c) If a previous cost is available (old PO for same or similar item) and modification is required, then percentage of work to be done should be indicated to give an idea of proportionate additional/reduced cost over the **Non-Recurring Engineering (NRE) cost** of the referral order.
- (d) If **new development** is required, then **Bill of Material (BOM)** and Development/NRE Cost break-up is required. That may be assessed based on LRUs/components with approximate numbers, cost and estimated work (skilled

members + unskilled members with prevailing man hour rates taken for cost estimation).

(e) Completely new activity/jobs can be supported with Budgetary Quote (BQ)/or Expert Inputs. Only one BQ shall be considered adequate at this stage. BQs have to be discounted by minimum 10% while including in the estimates.

(f) **Reserve stocks** of critical components for system level TD(S) projects and MM projects may be created to support production during initial phases. It may be considered as a risk mitigation strategy and should be highlighted with detailed justification and costing. However, the cumulative cost of reserve stocks should not exceed 5% of the total project cost. As provided in PM-2025, such reserve stock pile may be procured after clearly indicating to the CFA that it is a stock pile.

(g) In the case of major equipment and high value stores the justification for procuring them and not opting for hiring or outsourcing as a services, should be clearly brought out.

(h) In certain cases where estimation of quantity may not be practically feasible, a benchmark percentage of project cost/revenue expenditure could be considered.

2.12.4 Overall estimated cost including taxes and duties shall be considered for determining project CFA. AoN of DMC will be on the basis of cost estimated by CEC which will be considered for project sanction. However, if DMC directs to change the scope of work and/or rework the cost significantly then CEC will revisit the cost estimation based on DMC directions and the cost suggested by this committee will be considered for project sanction. In case deliverables/prototypes are reduced or increased by DMC such as number of missiles etc. and wherein change in cost is not exceeding 20% of the proposed cost in heads of Equipment/Stores and trial expenses, CEC may not be reconvened in such cases. This Committee will also factor impact of the infrastructure linked with the project, if any, while arriving at final cost and will be represented by an expert member from DCW&E. A CEC shall be constituted by Competent Authorities as given in **Table 5**.

Table 5: CEC Composition and Approving Authority						
Project CFA	Approving Authority	Chairman	Domain Expert	*Infrastructure Expert	Finance Member	Member Secretary
Lab Director	Lab Director	Nodal Lab Scientist 'G' / 'H' (non-member of project)	- Designated PDs of all Sub-projects from Participating Labs - Lab Technical Expert - Head MMG #Director Admin	Director DCW&E or Rep	Dir (Finance) / Sr. Dy. IFA / Dy. IFA (R&D)	Project Director Designate
DG (Cluster)	DG (Cluster)	Nodal Lab Scientist 'G' / 'H' (non-member of project)	- Designated PDs of all Sub-projects from Participating Labs - Lab Technical Expert - Head MMG #Director Admin (Cluster)			
Secretary DDR&D and beyond	DG (Cluster)	Scientist 'H' (non-member of project) (If Sc 'H' is not available, Sc 'G' may also be considered)	- Designated PDs of all Sub-projects from Participating Labs - Lab Technical Expert - Director DFMM #Director Admin	Director DCW&E	Director (Cost) or Rep Addl. FA (R&D)	
Note : - Constituting authority/ Chairman may co-opt members as necessary * Infrastructure expert will be essential member if civil works / infrastructure is a part of the project. # Director Admin shall ensure that cost break-up details are filled in Cost Estimation Module conceptualized and monitored by Office of Addl. FA (R&D).						

2.12.5 A detailed cost estimation report shall be prepared by the Lab including the detailed cost estimation sheets supported by documents like previous Purchase Orders (POs)/Supply Orders (SOs), internal estimations, POs from other Labs, BQs etc., professional judgment indexed as

per the module/activity/sub-system matrix for review by the CEC. Cost break-up details as per 'Cost Estimation Module' (available in DRONA) should be filled mandatorily with administrative support of DF&MM. In cases, where it is proposed to seek demand approvals along with the project sanction; the same may specifically be indicated to the CEC. The list of proposed procurements/demands have to be appended to the project cost estimates for due scrutiny by the CEC. The Spread Sheet format for Cost Estimation to be used is given in DRDO.DPFM.FT.01.

2.12.6 The CEC report, duly incorporating recommendations of all members shall be a part of the Statement of Case of the Project. Overall cost estimate approved by CEC should be incorporated in the SoC under appropriate budget heads in the cost break-up tables for various project categories given at Table Nos. 5 and 6. The Cost Estimation sheets signed by Chairman and all members may be appended to the SOC of the project. However, in case of disagreement by a member same may be recorded in CEC report.

2.12.7 The Bill of Materials (BOM) and activities in the project cost estimation is as envisaged during project formulation. Given the inherent uncertainties of R&D projects, these are to be considered indicative and variation in BOM items and activities can arise during project execution and should not hold up procurement action.

2.12.8 Basis for considering a cost item under a particular budget head is the governing Controller General of Defence Accounts (CGDA) correction Slip No. 20/2018 issued vide CGDA Note No. A/B/I/13626/XXXI/32 dated 04 Jun 2018. The approved estimates and support documents should be uploaded by nodal and participating Labs on the Cost Estimation Module software available on DRONA.

2.12.9 For all SHQ initiated design and development projects as per provision of DAP 2020, as amended from time to time, which will be undertaken as MM projects, the SHQ would seek the estimated unit production cost through DISB. The product cost should be computed by the project team and vetted by the Director (Cost) in the office of Addl. FA(R&D). For past cases of Design & Development by DRDO, where product cost is sought by SHQ for AoN, the cost vetting would be done by Director (Cost).

2.12.10 A Head wise summary for cost estimation is given in **Table 6 & 7**. Regardless of CFA, as per classification code and correction slip No.20/2018; MM, TD & IF projects will be booked entirely to Capital.

Table 6: Cost Break-up Table for MM, TD & IF projects (₹ in Cr.)				
Minor Head	Major Head 4076 – Capital	Nodal Lab	Participating Lab(s), if any	Total (FE)
	Sub Major Head – 05			
	Heads of Expenditure	Total (FE)	Total (FE)	
052 (Code Head-929/25)*	Transportation (Movement of Stores)			
	Equipment/Stores			
	CARS/CAPSI			
	Consultancy Contracts			
	Job Work/Contracts/Technical Services/Work Package			
	Hiring of Transport			
	Fuel/Oil/Lubricants for Project Vehicles			
	Contingency & Miscellaneous			
	Plant & Machinery			
	Project related Vehicles			
111	Works			
Total				

*Total project expenditure related to MM, TD and IF projects including equipment, hardware, consultancy, project related contingency, purchase/hiring of transport, freight, contracts for “Acquisition of Research Services (CARS)” under the project etc. will be compiled to this head.

Table 7: Cost Break-up Table for S&T projects (₹ in Cr.)				
Minor Head	Major Head 2080 - Revenue	Nodal Lab	Participating Lab, if any	Total (FE)
	Heads of Expenditure	Total (FE)	Total (FE)	
105	Transportation (Movement of Stores)			
110	Equipment/Stores			
	CARS			
	CAPSI			
	Consultancy Contracts			

(Code Head-856/01)**	Job Work/Contracts/Hiring of Technical Services/Work Package			
	Hiring of Transport, Fuel/Oil/Lubricants for Project Vehicles			
	Contingency & Miscellaneous			
111	Works			
Total (Revenue)				
Major Head 4076 - Capital				
052 (Code Head-929/24)**	Plant & Machinery			
	Project related Vehicles			
	Works			
Total (Capital)				
Grand Total (Revenue & Capital)				

**Booking of Expenditure for Science & Technology (S&T) projects will be as per PM-2025 as amended from time to time.

2.13 SYSTEM ANALYSIS OF PROJECTS

System Analysis activity should be carried out by ISSA (or a team where ISSA is part of it) for MM & TD(S) projects. TD(T) and S&T projects may also employ the system analysis activities in accessing technology development requirements for the specific projects, if required. System analysis should be carried out to assess the requirement of the Products/Systems/Technologies in the simulated war scenarios. Detailed guidelines formulated by Directorate of Futuristic Technology Management (DFTM) as amended may be considered.

2.14 ACCEPTANCE-OF-NECESSITY (AoN) BY DMC

2.14.1 Every project proposal being processed for seeking sanction of Competent Authority needs to get AoN by DMC. For MM projects, **go-ahead** by DMC should be obtained prior to AoN by DAC/DPB/SPB.

2.14.2 Pre-project activity needs to be done mandatorily through an R&D activity prior to processing for project sanction. Therefore, intimation to CFA of the proposed project is essential before taking up this R&D activity. Lab will present brief details of proposed project

comprising title, cost, duration, category, classification, linking with DRDO Roadmap and participating Labs etc. to obtain the initial clearance and in-principle approval.

- a) For projects under delegated financial powers of Lab Director and DG (Cluster), Labs to obtain the initial clearance of its Cluster Council to carry out PRC/PDR, TiEC and CEC. Thereafter, this proposal needs recommendation from Cluster Council prior to be presented in DMC for seeking AoN. The recommendation by Cluster Council will be considered as in-principle approval for these projects.
- b) For projects under delegated financial powers of Secretary DDR&D and above, Lab will obtain the initial clearance of Cluster Council and thereafter present the brief details in DMC for in-principle approval prior to carrying out PRC, PDR, TiEC and CEC. Thereafter, Lab will obtain the recommendation from Cluster Council before presenting to DMC for AoN after completing the pre-project activities.

2.14.3 The recommendations of PRC/PDR, TiEC and CEC will also be deliberated in DMC for deciding the final scope including objectives & deliverables, cost and timeline. The DMC may consider 'Roadmap of DRDO-2025' as an in-principle approval for projects in a quarter/six-months.

2.14.4 Sunset milestone and two Time Estimates: "Realistic" and "Pessimistic", suggested by TiEC will be deliberated in DMC based on which final recommendations for PDC of a project will be decided. Sunset milestone(s) and pessimistic time for a project will be recorded in MoM along with AoN for a project. PDC extension will be taken as an exception and Pessimistic time will be used as a criterion for PDC extensions, beyond which normally no extensions will be accorded. Furthermore, any PDC extension beyond the Pessimistic time will be based only on the recommendations of an 'Expert Committee' having external members and constituted by Secretary DDR&D and Chairman DRDO considering geo-political situation, exports ban and similar situations beyond control as a rare case. PDC extensions would mandatorily lead to review/change of PgD/PD.

2.14.5 Matters pertaining to AoN of new projects and PDC extension of ongoing projects may be delegated by DMC to the "Committee of DGs for Key Performance Indicator (KPI) review of Labs" for detailed discussion. Thereafter DMC may ratify the outcome of the deliberation by the Committee of DGs. The Committee of DGs should also take into account the performance of the Lab for grant of new projects and the PDC extension cases for rating of the Lab. The Committee of DGs should also verify the timelines for sanction of the project as per

the Roadmap of DRDO and six-monthly, yearly, biennial targets given by the Director as their KPI.

2.14.6 For SHQ linked proposals and projects under Govt. to Govt. (International Collaboration), Statement of Case (SoC) will be presented to DMC for seeking AoN prior to forwarding outside DRDO.

2.14.7 In case of 'Immediate requirements' projected by SHQ and to meet the necessity of 'Emerging Technology', DMC would approve taking up projects based on the recommendations of the Cluster Council & accord AoN as special case independent of its listing in D-RdMAP. Projects taken based on immediate requirements after DMC approval will be listed in D-RdMAP which will be updated on six monthly basis.

2.14.8 AoN of DMC for a project will be **valid for 6, 4 & 2 months** respectively for Secretary DDR&D, DG (Cluster) and Lab Director as CFA for Projects (**i.e., project proposal should reach to DP&C for Secretary DDR&D as CFA and Director PM for DG as CFA within this timeline**). For MM category projects, the project proposal for approval should reach the office of CFA within 3, 2 & 1 month from AoN by DAC/DPB/SPB respectively for Secretary DDR&D, DG (Cluster) and Lab Director as CFA. DAC/DPB/SPB instructions regarding sanction of projects may be given priority. In case validity of AoN by DMC has expired it needs to be revalidated by DMC. However, this is not applicable for Government to Government projects as long as the inter government agreement is existing. **Non-adherence to the validity of AoN may adversely affect the KPI of Lab and Lab Director.**

2.14.9 **Proposal by DYSL:** Any new activity/technology/plan costing Rs. 10 Cr or above, to be taken up is to be presented in DMC prior to its discussion in DYSL's Apex Committee irrespective of its cost. This presentation is only for intimation to DMC members and not for according AoN.

2.14.10 Format for presentation in DMC is given at DRDO.DPFM.FT.03A & 03B.

CHAPTER 3

PREPARATION OF PROJECT PROPOSAL

3.1 OVERVIEW

The project proposal comprises the Note describing basic details (need, scope including objectives and deliverables, category, cost, PDC), competence level, work done so far, critical factors, type of risks and its mitigation plan, brief on project execution plan, six-monthly milestones along financial outlay and timelines, details of DcPP/PA/LSI, testing plan, procurement plan, cost benefit analysis, recommendations of various committees (CCM, PRC/PDR, IRC, TiEC, CEC & DMC) along with Statement of Case (DRDO.DPFM.FT.04) and Signed copies of committee(s) minutes/reports.

3.1.1 Statement of Case (SoC) should be prepared as per the format at DRDO.DPFM.FT.04 and ‘**General Advisories**’ as in **Annexure ‘II’** may be followed wherever applicable. Overall project cost should be provided under appropriate budget heads as per the cost break-up templates provided in the SoC format. Deliverables of the project should be specified. The proposal should also contain the number of design iterations required at the component/sub-system level in a TD(T) project and number of design iterations and repetition of test and trials required for overall system/project level in TD(S) and MM projects. Expected time penalties should be mentioned with each iteration.

3.1.2 Six-monthly technical milestones with linked financial outlay shall be worked out by the Project Team and integrated in the SoC in the format given at **Table 8**. Due care should be exercised in identifying the milestones as the project progress shall be monitored based on these. The **sunset milestones** shall be clearly indicated.

Table 8 : Six-monthly Technical Milestones Linked with Financial outlay Forecasts			
Sl. No.	Milestones	Time (Months)	Financial Outlay (₹ in Cr.)
	<i>{Include Sunset Milestone(s)}</i>		
	Total		

3.1.3 DRDO will work on advanced, critical technologies and complex systems. It will not undertake project that industry can do. The same will be certified by the Project Director and Lab Director while seeking project sanction.

3.1.4 The list of sub- systems/technologies, issued from time to time by DG(PC&SI)/DISB, to be developed exclusively by Industry would be considered as bought out/Build-to-Specification (BTS) items and shall be detailed in the project proposal. Project should not be taken for the design, development or upgradation of systems, sub-systems or components given in the list.

3.1.5 Sub-project and Work Packages Proposals (if any) should be concurrently processed for sanction by the Nodal Lab in coordination with the participating Labs/other agencies. Nodal Lab should ensure that all its participating Labs have **concurrence** of Lab Director and their respective CCM for undertaking sub-projects. Category of sub-project shall be the same as the main project. Probable Date of Completion (PDC) of sub-project should be at least 3 months prior to that of the main project. The sub-project/work package proposals will be made as per the specimen format for project SoC at DRDO.DPFM.FT.04. The participating Labs along with their sub-projects have to be associated from the pre-project stage so that they can assess the technical challenges and work load vis-à-vis their own project commitments, at the inception stage. Participating Labs will be fully responsible for sub-project from its inception to closure. Administrative Closure of sub-project by the participating Labs should be forwarded to the nodal Lab within a year PDC of sub-project.

3.1.6 Labs should identify the Project Team in the SoC and should also confirm that undertaking a new project would not affect any of the on-going activities directly or indirectly. Persons involved during pre-project activities shall be essentially part of project team.

3.1.7 Monthly Expenditure Report (MER) of sub-project should be presented to the main project. However, ensuring Annual Audit of the Statement would be the responsibility of respective participating Labs.

3.1.8 Inputs of Programme Director (PgD) and Project Director (PD) of main project should be taken regarding the performance of PD of sub-projects & vice-a-versa during KPI based review of the Labs.

3.2 PROJECT PROPOSAL

The designated PD will prepare a project proposal containing details of pre-project activities and project attributes. The proposal should comprise of feasibility studies, literature surveys,

alignment of the project with ‘Roadmap of DRDO’ (D-RdMAP), design analysis, realisation and test plans, availability of test facilities, TRA, Time Estimation by Standard as well as DRDO developed Software tool, Responsibility & Accountability matrix (RACI), resource assessment, risk management plan, milestones etc. The proposal should contain following sections –

- **Need of the project**
- **Scope of the project:** Scope (including objective and deliverables) should be measurable and achievable. It has to be framed carefully as they are driver for the Project.
- **Category of the Project**
- **Cost Estimate of Project**
- **PDC of the Project (Realistic & Pessimistic)**
- **Linkage of Project Goal with Thrust Areas** laid down in DRDO Roadmap.
- **Literature Survey:** Literature Survey should bring out the relationship between the project goal and the state-of-the-art technological accomplishment w.r.t global benchmark. Global benchmarking for technical features, application, output performance etc. with similar accessible products should be done to explore alternate options/technologies/process for similar applications. Suitable reference should be given to reputed journals/databases when comparing with contemporary systems world-wide.
- **Cost Benefit Analysis:** Cost benefit analysis of the proposed system/technology should be supported with comparative study of indigenous system development vis-à-vis contemporary technologies and products available worldwide, their development period and cost. It should also amplify on economic benefit, capacity building, capability building enhancement of technical knowledge and potential for job creation.
- **Targeted indigenous content:** For MM and TD(S) projects, proposal should include likely indigenization content of first and last prototype and First of Production Model (FoPM). Based on the necessity TDF projects may be planned. The detailed indigenization plan of the sub-system/ component may be mentioned. This should include the

indigenization effort within the project in PDC as well as outside the project through Technology Development Fund (TDF) or DRDO-Industry-Academia Centres of Excellence (DIA-CoE) which may go beyond this project PDC.

- **‘Operation Research & System Analysis (ORSA)’ Studies**

ORSA studies should be carried out for all MM & TD(S) projects. Other categories of the projects can also employ the system analysis activities if desirable. Labs may carry out detailed analysis using standard practices in coordination with ISSA. Detailed guidelines formulated by the DFTM and input from ISSA as amended from time to time may be considered.

- **‘Artificial Intelligence (AI) Enablement’**

Proposal shall also cover AI capability in the system and AI used over past data as a part of AI enablement of systems in co-ordination with CAIR. A Trustworthy AI may be ensured in coordination with SAG.

- Human Factor Integration (HFI) & Human Reliability Analysis (HRA) of DIPAS
- **Competence level:** Technological competence in terms of Infrastructure & Human Resource available with the Lab and preliminary work done by the Lab.
- **Critical factors/technologies involved** in the project and pre-emptive measures to minimize the criticality.
- **Confidence Level** in accomplishing the Project through chosen design approach and identification of project as High Risk category.
- **Past project details:** Whenever Lab proposes new projects it needs to submit the list of projects completed by the Lab in past 10 years (Name, DoS, Cat, Original & Revised PDC, Original & Revised Cost, name of PD at project completion, Administrative Closure date etc.). Linkage of proposed project with past projects if any may also be provided.
- **Details of pre-project work** already undertaken.
- **TRL Assessment:** Methodology for arriving at TRLs is given at Para 1.3 & Table 1 & Format at DRDO.DPFM.FT.02 should be filled for assessment of final category of the project.
- Time Estimation by Standard as well as DRDO developed Software tool and ‘TiEC Committee’.

- PERT chart.
- Responsibility & Accountability Matrix (**RACI matrix**).
- **Risks Management Plan:** A risk management document which will identify risks and possible strategies to manage the risks should be prepared and attached in accordance with Chapter-2, DPFM-2026.
- **Preliminary Design** should be mandatorily completed before project sanction of MM and TD (s) project.

- **Specification / Scope of Work (SoW)**

In case of MM Projects, specifications (essential and desirable) must be discussed with User which will culminate into Qualitative Requirements (QRs). Firm QRs should be expedited / sought by the Labs in consultation with DISB, DRDO HQ as early as possible.

- **Project Execution Plan:** Project execution plan should specify major milestones of the project including Program Evaluation and Review Technique (PERT) and detailed Work Breakdown Structure (WBS).
- **Six-monthly technical milestones with linked financial outlay:** A macro level realisation plan with six monthly technical milestones along with financial outlay should be formulated. As further details evolve during project execution stages, the milestones should be updated. These milestones will be reviewed by Project Monitoring & Review Committees during project execution stages.
- **Sub-Projects/Work-Packages/CARS/CAPSI/Consultancy Contracts etc.**
- **Development cum Production Partner (DcPP):** Selection of DcPP shall be a mandatory milestone for MM projects which envisage deliverable systems/products. DcPP shall not be issued Free Issue Materials (FIM). Project Director has to certify that Industry/DcPP is not in the negative list of vendors as promulgated by the DRDO HQ from time to time.
- **Testing Plan:** It should contain assessment of existing facilities and its suitability to meet the test requirements proposed in the projects. The project proposal must include tests required, test equipment/facilities, test methodologies to meet the performance as envisaged in the project. The project proposal should document whether the required test facilities are available or being created. A project may be taken only if the required test

facilities are either available or being created as part of the current project/build-up. In case such facilities are not available with DRDO, a long term Contract/Inter Govt Agreement (IGA) / Memorandum of Understanding (MoU) with agencies or a plan/way ahead mutually agreed by DRDO and User will be considered for sanction of project. If test facilities of foreign country are proposed to be used, the possibilities of government permission /availability need to be reviewed and indicated after vetting of Counsellor (Defence Technology) abroad.

- **Required resources**

- Manpower Requirement
- Infrastructure requirements (if applicable) as per prevailing guidelines will be vetted by DCW&E/CCE(R&Ds) .

- **Other relevant issues** (If any).

CHAPTER 4

PROJECT SANCTION

4.1 OVERVIEW

4.1.1 Project sanction is a process of consolidating all approvals-in-principle dovetailed with the procedural requirements authorizing the expenditure of public fund for given objectives. Necessity of prudence and clarity of purpose supported by necessary documents should be followed. The project proposal should be prepared as per the DPFM-2026 guidelines and other statutory and regulatory documents issued from time to time.

4.2 COMPETENT FINANCIAL AUTHORITIES (CFAs) FOR SANCTION OF PROJECTS

4.2.1 The Delegation of Financial Powers (DFP) within Dept. of Defence R&D are governed by 'Schedule of Financial Powers' issued by DF&MM as amended from time to time.

4.2.2 Financial powers to CFAs beyond Secretary DDR&D i.e. Raksha Mantri, Finance Minister and Cabinet Committee on Security (CCS) are as per MoD ID No. 14(23)/2003/IFDP II dated 30 July 2010 as amended from time to time.

4.3 SUBMISSION OF PROJECT PROPOSALS

4.3.1 Project proposal file with title and project code name (if decided) and security classification[#] (Secret/Confidential/Restricted/Unclassified) with following essential documents should be processed for project sanction: -

- (a) Project proposal comprising of SoC containing necessary details.
- (b) PRC/PDR and TiEC reports
- (c) Cost Estimation Committee (CEC) report
- (d) AoN accorded by DMC (copy of MoM / extracts).
- (e) AoN by DAC/DPB/SPB and PDR document for MM Projects.
- (f) Draft PSQR/a formal communication by SHQ, if applicable for TD(S) projects.
- (g) Documents related to identification of DcPP (for MM projects).
- (h) Procurement plan and actions initiated (mandatory for MM projects).

- (i) List for procurement items for demand approval by respective CFA with concurrence of designated Finance Authority.
- (j) Sub-Project SoCs.
- (k) Draft Govt. Letter (DGL) conveying project sanction (DRDO.DPFM.FT.06) with Project No. as in Para 9 of this chapter. Lab name to be incorporated in project numbers as listed in Abbreviation section. Unique Sanction Code should be provided in the DGL as per DF&MM guidelines.

4.3.2 A project proposal file can only be classified as ‘**Top Secret**’ if the requirements are conveyed with same security classification from National Security Council Secretariat (NSCS) or Services.

4.3.3 While preparing project proposal, designated Project Director will ensure that proposal has been prepared in compliance with project guidelines providing information in designated formats. It is mandatory to **attach the necessary reports in bound forms signed by authorized signatory** & Minutes of Meetings (MoM). In case, proposal is incomplete then the desired information/report will be sought from Lab which will lead to delay[#] in sanction of the project. All papers in project proposal file should be numbered as “*X of Y*” w.r.t. total No. of pages in the file.

4.3.4 A **File Routing Chart** shall be pasted in project file (inside front cover of project file) wherein file clearance from various stages, i.e., go-ahead from CCM, completion of PRC/PDR & IRC (if applicable), TiEC, CEC, in-principle approval by CCM, agreement by DMC (AoN), initiation of the proposal:- Designated Project Director, Lab Director, Director PM, IFA & DG is recorded. The entries for clearance in HQ till issue of Govt. letter will be made at DAK tracking system (supported by DIT&CS) to track and record the overall time taken for sanction of the project. This file routing/tracking chart at Lab and Cluster will be created in PMIS / upcoming ERP system.

[#] *Debottlenecking R&D projects execution across DRDO- Study Report: - Available in DRDO at DP&C Portal*

Sl. No.	Stage/Office	Entry Date	Exit Date	Time Taken (Days)
1	CCM	----	DD/MM/YYYY	
2	PRC/PDR	DD/MM/YYYY	DD/MM/YYYY	N1
3	TiEC	DD/MM/YYYY	DD/MM/YYYY	N2
4	CEC	DD/MM/YYYY	DD/MM/YYYY	N3

.....				
.....	O/o Addl FA (R&D)	DD/MM/YYYY	DD/MM/YYYY	N5
.....				

4.3.5 Routing of case file for approval of CFA for ‘*Project sanction, PDC & Cost revision, Change of Scope, Change of Lab & Category and Project completion etc.*’ by the designated offices shall be as per **Table 9**. Checklist available at DRDO.DPFM.CL.02 may be referred before submitting the proposal.

Table 9 : Project File Routing and Issue of Sanction Letter (As per presentation given in DMC meeting held on 03rd Jan 2025: On 4 Levels of Channel of Submission- for Projects and Procurements)- within Department of Defence R&D		
CFA	Concurrence Level as stated in latest DFP	Proposed Routing of Files
Director of Labs	Dy. IFA	Initiating Officer (L-1) → Project Director (L-2) → Dy. IFA (L-3) → CFA (Lab Director for Sanction) (L-4) → Officer designated by Lab Director for issue of Sanction letter. <i>Note: -</i> (i) Noting for project proposal and Draft Govt. Letter will be initiated by Officer at L-1. (ii) Vetting Authority and UO Number Allotment by Director (Finance) / Sr. Dy. IFA / Dy. IFA. (iii) Letter Issuing Authority- Officer nominated by Lab Director.
DG (Cluster)	IFA (R&D)	Project Director (L-1) → Lab Director (L-2) → IFA (R&D) (L-3) → CFA [DG (Cluster) for Sanction] (L-4) → Director (PM) O/o DG (Cluster) for issue of Sanction letter. <i>Note: -</i> (i) Noting for project proposal and Draft Govt. Letter will be initiated by Officer at L-1. Inputs

		<i>required at L-3 & above shall be provided by Director (PM).</i> <i>(ii) Vetting Authority and UO Number Allotment by IFA (R&D) of concerned cluster.</i> <i>(iii) Letter Issuing Authority- Director (PM), O/o DG (Cluster).</i>
Secretary DDR&D	Addl. FA (R&D)	Lab Director (L-1) → DG (Cluster) (L-2) → Addl. FA (R&D) (L-3) → CFA [Secretary DDR&D for Sanction] (L-4) → US(R&D) for issue of Sanction letter. <i>Note: -</i> <i>(i) Noting for project proposal and Draft Govt. Letter will be initiated by Officer at L-1. Inputs required at L-3 & above shall be provided by Director P&C, Director DF&MM & Corporate DG wherever required in co-ordination with Director (PM) and Designated PD.</i> <i>(ii) Vetting Authority and UO Number Allotment by O/o Addl. FA (R&D).</i> <i>(iii) Letter Issuing Authority- US (R&D).</i>
Hon'ble Raksha Mantri (RM)	FA(DS)	Lab Director (L-1) → DG (Cluster) (L-2) → Addl. FA (R&D) (L-3) → Secretary DDR&D (L-4) → FA(DS) → Hon'ble RM for Sanction (Final Level) → US(R&D) for issue of Sanction letter. <i>Note: -</i> <i>(i) Noting for project proposal and Draft Govt. Letter will be initiated by Officer at L-1. Inputs required at L-3 & above shall be provided by Director P&C, Director DF&MM & Corporate DG wherever required in co-ordination with Director (PM) and Designated PD.</i> <i>(ii) Vetting Authority and UO Number Allotment by O/o Addl. FA (R&D).</i>

		(iii) <i>Letter Issuing Authority- US (R&D).</i>
Hon'ble Finance Minister (FM)		Lab Director (L-1) → DG (Cluster) (L-2) → Addl. FA (R&D) (L-3) → Secretary DDR&D (L-4) → FA(DS) → Hon'ble RM → Hon'ble FM for Sanction (Final Level) → US(R&D) for issue of Sanction letter. <u>Note:</u> (i) <i>Noting for Project Proposal and Draft Govt. Letter will be initiated by Officer at L-1. Inputs required at L-3 & above shall be provided by Director P&C, Director DF&MM & Corporate DG wherever required in co-ordination with Director (PM) and Designated PD.</i> (ii) <i>Vetting Authority and UO Number Allotment by O/o Addl. FA (R&D).</i> (iii) <i>Letter Issuing Authority- US (R&D).</i>

4.3.6 For projects beyond delegated Financial Powers of DG (Cluster), DP&C will co-ordinate for issue of sanction letter. Office of DG (Cluster) will maintain Lab-wise 'Projects Master Control Register' for their records regarding Project Number. The project proposal file and other documents will be sent back to DG (Cluster) office.

4.3.7 Digital searchable version (keywords) of project proposal will be created by Lab after issue of Govt. letter for sanction and subsequent revisions and upload on PMIS/ERP. Lab will be sole custodian of project files.

4.4 EXAMINATION OF PROJECT PROPOSAL: TIMELINES

Before sanctioning any DRDO project by the Competent Financial Authority (CFA), it has to be examined at various levels. Indicated timeline of sanctioning authority is given in Table No. 10:

Table 10: Indicative Timelines for processing Project sanction files (time starts from AoN by DMC)				
CFA	L-1	L-2	L-3	L-4
Director of Labs	04 weeks	01 week	02 weeks	01 week

DG (Cluster)	08 weeks	01 week	02 weeks	01 week
Secretary DDR&D	16 weeks	03 week	04 weeks	01 week

4.5 SPECIAL PROJECTS BY STANDING COMMITTEE

Sanction of certain classified/sensitive ‘Special Projects’ (cost upto Secretary DDR&D as CFA of project), which should be executed on “need-to-know basis” will be processed on file. The Standing Committee comprising of DG (Cluster), Addl. FA (R&D) and Secretary DDR&D would grant the status of ‘Special Project’ based on the proposal by designated Project Director and recommendation of Director of Nodal Lab. DP&C will be the Standing Committee Secretariat for putting up the case to Addl. FA (R&D) and conveying status of ‘Special Project’.

After grant of status of ‘Special Project’ by Secretary DDR&D, TiEC and CEC needs to be convened with invited members on “need-to-know basis”. However, conduct of PRC/PDR is not mandatory for such projects.

Sanction of projects which have been given the status of ‘Special Project’ by Secretary DDR&D, will be accorded by respective CFA as per DFP. Such project will be processed by the Lab, routed as per Table No. 9 and sanctioned by the CFA within 2, 4 & 6 months for Director, DG (Cluster) & Secretary DDR&D as CFA. Basic details (Project No. instead of Name, Cost, PDC and Nodal Lab) of these projects will be entered in PMIS. Review mechanism will be applicable as per chapter 6 of DPFM-2026. However, Chairman of the Monitoring and Review Committee may restrict the members on ‘**need to know**’ basis.

4.6 PROJECT SANCTION LETTER AND UNIQUE SANCTION CODE

4.6.1 All project sanction letters must also have a ‘**Unique Sanction Code**’, the details of which have been issued by DFMM. Maintenance of a control register containing details of Unique Sanction Code for each financial sanction issued by the individual CFA will be mandatory. Project sanction letter will be considered incomplete/invalid, if Project Number and Unique Sanction Code have not been mentioned in the letter.

4.6.2 **Effective date of sanction:** All project sanction letters will have an effective date of sanction which will be **first date of starting month for every quarter** i.e., 1st Jan will be effective date of sanction for projects sanctioned in the last (4th) quarter of the calendar year (between 01st Oct to 31st Dec). Similarly, for projects sanctioned in 1st/2nd/3rd quarter, effective date of sanction will be 1st Apr/1st Jul/1st Oct respectively. The provision of having an effective

date of sanction for a project will bring uniformity in sanction dates and help in effective planning of reviews. Effectiveness of this provision will be reviewed over a period of time.

4.6.3 By the time file is submitted for sanction by CFA for a project proposal, PD designate may ensure that for MM category projects at least 50% procurement cases are initiated. Similarly, at least 25% procurement cases may be initiated for projects in other categories.

4.6.4 During the period when a project has been sanctioned (Govt. Letter issued), but the effective project start date is in the coming quarter, supply orders can be placed. However, delivery of stores and payment to industry cannot happen during this period.

4.7 SUBMISSION OF PROJECT PROPOSALS FOR CCS SANCTION

Format as specified by Cabinet Secretariat (DRDO.DPFM.FT.05), as amended from time to time, shall be followed for preparation of Draft Cabinet Note (DCN) along with Appendices & Annexures to be put in project proposal file in addition to the documents mentioned in Section 4.3. The relevant guidelines for preparing DCN are available at website www.cabsec.nic.in.

4.8 SUB-PROJECT SANCTION

4.8.1 Sub-project to different participating Labs should be identified during the pre-project stage (Feasibility Studies/PRC/PDR) and informed to DMC during AoN stage. DMC may add or modify the list of participating Labs and their scope. The sub-project should have the Cluster Council (CC) approval before AoN to the project by DMC.

SoC for sub-project should be forwarded by the concerned participating Lab Director to the Nodal Lab. Cost Estimation for the sub-projects shall be done by the Nodal Lab along with the Reps from the Participating Lab(s) in CEC. Labs holding sub-project cannot allot further sub-projects, however CARS/CAPSI may be sanctioned. There is no distinction between 'Project' and 'Sub-Project' for project execution and PDs & Dy. PDs may be appointed for sub projects. These PDs of sub-projects from participating Labs will exercise delegated financial power assigned.

4.8.2 Sub-projects detailed in the main project SoC, while seeking original sanction, shall be sanctioned along with main project. The main project sanction letter will include sub projects and respective sub-project numbers.

4.8.3 The sub-projects will be executed in the same way as main project and should have Probable Date of Completion (PDC) at least 3 months before the PDC of the main project.

4.8.4 Participating Lab can take a sub-project only if its cost is above the minimum financial outlay which is needed for classification as project. In case of revision of Financial Powers of CFA, ongoing sub-projects linked with a main project of the nodal Lab will continue till its PDC. Low cost activities required to be undertaken by a participating Lab may be taken as an R&D activity from its Lab Build-up fund. In such cases, Participating Labs will ensure that needed technological support is provided to nodal Lab for which these activities have been undertaken and apprise them their current status in scheduled reviews. The names of these Participating Labs will also be mentioned in the Govt. Sanction Letter without any project number and cost.

4.8.5 As an exception, sub-projects not planned during the project sanction stage can be sanctioned with recommendation of highest review committee and will be sanctioned by the Competent Financial Authority (CFA) as per **Table 11**. The procedures for sanction of sub-projects shall be as specified in Para 4.8.1 above.

4.8.6 Copies of sub-project sanction letters and subsequent revisions should be forwarded to DP&C by the issuing office.

4.8.7 Labs holding sub-projects shall approach the Dte. of Finance and Material Management (DF&MM), DRDO HQ for allotment of Unit Code.

4.8.8 PMB (& equivalent boards) may sanction sub-projects to CCS Programmes as per Appendix B to Govt. Letter No. DRDO/DFMM/PL/83226/M/01/1976/D(R&D) dated 18th Dec 2019, as amended time to time.

Table 12: Competent Authority to Sanction Sub-project After Main Project Sanction*	
Cost of Project	Competent Authority to sanction of sub-project within the project cost
Lab Director as CFA	Nodal Lab Director
DG (Cluster) as CFA	DG (Cluster) of Nodal Lab
Secretary DDR&D or Hon'ble RM or Hon'ble FM as CFA	Secretary DDR&D
* Levels indicated based on GoI Letter No. DRDO/DFMM/PL/83226/M/01/1976/D(R&D) dated 18 th Dec 2019, as amended time to time.	

4.9 AWARD OF WORK PACKAGES, CONSULTANCY CONTRACTS

4.9.1 CFAs for work packages will be as stated for sub-projects in **Table 12**.

4.9.2 Funds for work packages and consultancy contracts will be released by Nodal/ participating Labs as per the payment terms and conditions laid in the MoU/ Contract.

4.10 SCHEME FOR ALLOTMENT OF PROGRAMME, PROJECT & SUB-PROJECT NUMBER

The project number has to be in the form of “XXXX/YY-ZZ/ ABCD-123”. Where, **XXXX** defines the category of the project, **YY-ZZ** financial year of the sanction of project and **ABCD** is the abbreviated name of the Lab to which the project is sanctioned (refer DRDO.DPFM.AL.01). The **(123)** number denotes serial No. of the project of a particular Lab.

XXXX Category		YY-ZZ Financial Year of Sanction
PG	Programmes approved by CCS	Only the last two digits of the financial year need to be given
MM (Service)	Mission Mode (Indian Army - IA), (Indian Air Force - IAF), (Indian Navy - IN), (Tri-Services - IDS) & (MHA- Ministry of Home Affairs)	
TD (S or T)	Technology Demonstration {TD (System) &TD (Technology)}	
S&T	Science & Technology	
IF	Infrastructure & Facilities	
ABCD Abbreviated full name of the Lab		123 – Digit (Sl. No. of the project to be obtained from the Lab)

Illustrative example of assigning Project Number for nodal Lab

MM(IAF)/26-27 /CABS-310	This reflects MM category of project sanctioned to CABS in the financial year 2026-27 and Sl. No. of the project sanctioned to the Lab is 310
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Illustrative example of assigning Project Number for nodal Lab

TD(S)/25-26/ARDE- 215	This reflects TD(S) category of project sanctioned to Armament Research & Development Establishment (ARDE) in the financial year 2025-26 and Sl. No. of the project sanctioned to the Lab is 215
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In case of multi-lab project, the participating Labs executing sub-projects will be assigned separate project numbers as per the scheme mentioned below:

Illustrative example of assigning sub-project numbers to participating Labs

TD(S)/25-26/ARDE-215.01 (TBRL-25)	➤ First sub-project given to TBRL (.01) by ARDE (Nodal Lab) which is 25 th running project of the Participating Lab, TBRL .
TD(S)/25-26 /ARDE-215.02 (HEMRL-36)	➤ Second sub-project given to HEMRL (.02) by ARDE (Nodal Lab) which is the 36 th running project of the Participating Lab, HEMRL.

4.11 APPOINTMENT OF PROGRAMME DIRECTOR (PgD), PROJECT DIRECTOR (PD), DEPUTY PROJECT DIRECTOR (DPD) AND PROJECT TEAM

4.11.1 Programme Director (PgD), Project Director (PD), Deputy Project Director (DPD) and the programme or project team shall be appointed to steer the programme or project. A Programme can have multiple Project Directors within a Lab or in participating Labs. Every project will have a PD and DPD who cannot be a PD or DPD for any other project or sub-project. DGs, Lab Directors, Centre Heads or Associate Directors will not hold the charge of PgD, PD or DPD. If the PgD or PD is appointed as PgD or PD of any other project or Lab Director or DG, (s)he shall relinquish charge of PgD or PD within 15 days of appointment to a new PgD or PD. Any existing PgD or PD cannot hold additional charge.

4.11.2 A PD and a DPD shall be appointed for the sub-project and this appointment has to be made in parallel to the sanction of the project.

4.11.3 Nominated PgD/PD/DPD should undergo minimum 2-Weeks of Project Management trainings course organized or co-ordinated by ITM/DIAT and pass the course. The course should be exclusively for Project Management and not just as a topic in any other course. The other QRs for PD, DPD may be decided by the Lab Director. In Labs, where Service officers are posted and they are associated with the project directly, their nomination for DPD may be considered provided they have sufficient tenure. The existing PDs with projects sanctioned in last 01 year should also be trained in Project Management, if not already.

4.11.4 The PD should exercise the financial Power delegated as per DFP-2019 as amended for sanction of expenditure for purchases required for sanctioned project. **Delegated Financial Powers of a Project Director should not be used by the Lab Director**, failing which the Lab Director will be held accountable for any delay in the project.

4.11.5 Appointment letter of a PgD, PD and DPD will be issued by DOP after due approval by Competent Authority based on nomination by Lab Director & recommendation by concerned DG (Cluster).

4.11.6 Guidelines for appointment of Programme Director (PgD), Project Director (PD) and Deputy Project Director (DPD) are as follows: -

- a) **Process for Appointment of PgD and PD for Programmes sanctioned by CCS:-**
PgD and PD in a Programme will be appointed for Programmes sanctioned by CCS. Labs should submit Nomination of PgD and PD in “Format for appointment of PgD, PD and DPD” along with bio-data of the PgD/PD designate duly recommended by DG (Cluster) to DP&C. This may be sent while submitting the proposal for approval by Competent Authority, i.e. Secretary DDR&D. Format is given as DRDO.DPFM.FT.07A. Order will be issued by Director DoP under intimation to DP&C.
- b) **Process for Appointment of PD for Projects sanctioned by Finance Minister (FM), Raksha Mantri (RM) and Secretary DDR&D:-** Labs should submit Nomination of PD in “Format for appointment of PgD, PD and DPD” along with bio-data of the PD designate duly recommended by DG (Cluster) to DP&C. This may be sent while submitting the proposal for approval by Competent Authority, i.e. Secretary DDR&D. Format is given as DRDO.DPFM.FT.07A. Order will be issued by Director, DoP under intimation to DP&C.
- c) **Process for Appointment of PD for Projects sanctioned by DG (Cluster) and Lab Director:-** Labs should submit Nomination of PD in “Format for appointment of PgD, PD and DPD” to Competent Authority for approval, i.e. DG (Cluster) along with bio-data of the PD designate. Format is given as DRDO.DPFM.FT.07A. Order will be issued by Director (Admin), O/o DG (Cluster) under intimation to DP&C.
- d) **Process for Appointment of Deputy Project Director (DPD) for Projects:-** Labs should submit Nomination of DPD in “Format for appointment of PgD, PD and DPD” duly recommended by Lab Director to Competent Authority for approval, i.e. DG

(Cluster) along with bio-data of the DPD designate. Format is given as DRDO.DPFM.FT.07A. Order will be issued by Director (Admin), O/o DG (Cluster).

e) Process for Change of PgD, PD and DPD for already sanctioned Projects/Programmes: -

If existing PgD/PD/DPD is going to superannuate/get transferred or the need arises for change in existing PgD/PD/DPD, Labs should submit nomination of PgD/PD/DPD within five working days from date of such change to Competent Authority in the Format given as DRDO.DPFM.FT.07B.

CHAPTER 5

PROJECT EXECUTION

5.1 PROJECT REALISATION PLAN AND RACI MATRIX

5.1.1 The project team should apprise detailed Work Breakdown Structure (WBS), major milestones, PERT, project execution plan, RACI matrix, Output/findings by Time Estimation Software tool, risk areas with mitigation plan and procurement cases as envisaged during project sanction stage to first review meeting of PMRB and EB. In case if there is any progress or deviation in milestone from approved SoC of project, it may be informed to the first review meeting i.e. PMRB & EB.

5.1.2 Clarity to participating Labs/work centres for their responsibilities is the key to success of the project. Thereafter, as further details/ changes evolve during project execution, RACI matrix should be updated and made available to the key stakeholders.

5.1.3 It is expected that during the execution of project, due to earlier identified risks and some unforeseen technical challenges, change in implementation and development strategies may become essential. Project Director shall brainstorm for new risk identification & their mitigation on regular basis with all stakeholders and resultant changes in the realisation plan, if any, and its implications on time and cost should be recorded and reviewed in all subsequent progress review meetings.

5.2 DETAILED DESIGN, DESIGN REVIEWS AND CONFIGURATION FINALISATION

5.2.1 System/Sub-system/Component level design exercise should be undertaken during the earliest phase of project. The project team may use in-house capabilities and external expert consultations. Configuration details, wherever required, should be finalised in the initial phase itself. Documentation of the design exercise is very important for records and continuity and special emphasis on design documents should be placed by the project team.

5.2.2 A documentation team shall be nominated in the SoC for all projects. This team shall be responsible for preparation of all the documents required as per the project timelines for different purposes.

5.2.3 Documentation team will ensure that documents prepared for a project cover all important aspects of complete project Life Cycle and will lead to full knowledge transfer to the industries

chosen as Industry/development partner which will later on be able to provide support on its own. Printed as well as soft copy of these documents should be retained by Lab and Director (PM).

5.2.4 The following design reviews are mandatory for all TD(S) and MM projects and should be with respect to identified milestones for the project. These reviews are optional for S&T category projects and not required for IF category of projects.

5.2.5 Preliminary Design Review (PDR): PDR should be done as a pre-project activity specifically for MM & TD(S) projects. PDR should focus on the design constraints, design methodology, test and validation plan of system and sub-systems, hardware and software development plan, system interface requirements and system realisation plan. The PDR Committee should review the Preliminary Design Document (PDD). The PDD approved by the Chairman of the Committee shall be the baseline technical document.

5.2.6 Detailed Design Review (DDR): The review would focus on sub-systems level detailed design. Acceptance of Detailed Design Document should be the milestone gateway for manufacturing clearance of first prototype.

5.2.7 Critical Design Review (CDR):

a) CDR should be carried out based on feedback of initial functional trials and performance validation trials including Environmental Tests. After CDR approval, the system with or without design changes can be cleared for manufacturing in numbers and full-fledged testing. CDR committee may suggest additional tests or retrials for performance validation.

b) At CDR stage, all the sub-systems should be clearly spelt out as Built to Print (BTP) / Built to Specifications (BTS). For all BTP systems and sub-systems, ToT is to be carried out. **ToT for complete system and sub-system to be carried out post development trials.**

5.2.8 System Design Review (SDR): In case of projects wherein development is related with 'Systems of Systems', Lab may also undertake System Design Review (SDR) in lines of PDR. Composition and Terms of Reference may be drawn accordingly.

5.3 DESIGN, DEVELOPMENT AND FABRICATION CONTRACTS

5.3.1 The core team should initiate procurement cases in accordance with prevailing DFP and Procurement Manual. Development contracts should be in conformance with Procurement Manual 2025 as amended from time to time. 'Standing Committee' provisions stated in

subsequent sections may be utilized by the programmes for urgent procurement needs (Monitoring and Review section in subsequent Chapter).

5.3.2 Interim design iterations and refinements is a continuous process in R&D projects. The consequent changes in procurement plans due to changes in design approach and realisation plan specified in project SoC and entailing changes in components/sub-systems/equipment within the overall cost of the project can be undertaken with the recommendation of PMRB. The same should be ratified by EB in subsequent review.

5.4 TRANSFER OF TECHNOLOGY (ToT) TO DcPP

Preparatory activities for nomination of the DcPP should be initiated during pre-project sanction stages. Selection of the DcPP should be in accordance with prevailing guidelines issued by DISB. ToT to DcPP and Production Agency (PA) will be in accordance with the prevailing 'DRDO Policy and Procedures for Transfer of Technology' promulgated by DIITM, DRDO HQ.

5.5 UPDATE OF "DATABASE OF SYSTEMS & TECHNOLOGIES"

"Database of Systems & Technologies" which is available on DRONA should be updated by the Project Director for every project after completion of Critical Design Review, achieving major milestones of projects and during preparation of Technical Completion Report.

5.6 REALISATION OF SUB-SYSTEMS AND INTEGRATION

Realisation of sub-systems/components, their testing, validation and integration should be progressed by the core team. Quality Assurance, Acceptance Test Plans and other Quality and Reliability documents should be prepared by Project Teams as per the Prevailing guidelines on 'DRDO Quality and Reliability Policy' and approved by Lab R&QA/MSQAA/other designated QA Agency.

5.7 TEST, TRIAL AND EVALUATION

5.7.1 Test Plan for internal trials would be prepared by the Project Team and approved by the Lab R&QA MSQAA/other designated QA Agency. Development trials on systems shall be done as per Trial Directives and Trial Methodology of SHQs, where applicable. JPMT/PMT shall be associated with trials, where applicable. DQRS, DRDO HQ shall be involved in test and evaluation activities. Lab will intimate the MM and TD(S) projects system level

Development/User trials schedule in advance to Chairman DRDO, DP&C, DISB in addition to participating Labs and trial range and thereafter convey the trial findings. DISB, DRDO HQ shall be the nodal Directorate to coordinate with the concerned SHQ Directorates for conducting User Trials as per trial directives or Development Trials where support from User is required.

5.7.2 Standard Operating Procedures (SOPs) on ‘Conduct of DRDO Internal Trials’ and ‘Project Teams Proceeding to High Altitude Areas for Conducting Trials’ issued by DISB, DRDO HQ as per details provided on DRONA should be adhered.

5.7.3 Trial results shall be reviewed by the JPMT (for AoN cases), PMRB & EB and way ahead shall be given.

5.7.4 Post promulgation of DAP-2020, the various trials have been subsumed under a single stage PSQR Validation Trial (PVT). The PVT consist of the following: -

- a) Functional/Performance Evaluation against PSQR parameters on the DcPP/PA produced systems.
- b) DGQA/MET/EMI/EMC trials, as applicable, for the systems.

5.7.5 DRDO Internal Trials Committee should assess the readiness of the systems for fielding for PSQR Validation Trials (PVT). This Committee will be formed by concerned DG (Cluster) under the Chairmanship of Senior Scientist (not below the rank of Scientist ‘G’), including Members, Chairman JPMT, internal domain experts, Reps from DQRS, DISB, CEMILAC for airborne systems, Lab QA Head and Project Director as Member Secretary. The Committee will review: -

- a) The readiness of the systems to be fielded for PSQR Validation Trials, including the analysis of Development Trials against the PSQR parameters, review of sub-system qualification, integration checks, confirmation to DGQA/MET/EMI/EMC parameters etc.
- b) Risk assessment and mitigation plan.
- c) Readiness of the documentation.
- d) The committee may recommend any additional test/shortcomings required to be addressed by the project with timelines before fielding the system for PSQR Validation Trials.
- e) The committee to certify that the equipment is ‘Fit for PSQR Validation Trials’.

5.7.6 Project Director will forward the final recommendation of the DRDO Internal Trials Committee, duly approved by DGs (Cluster), to DISB. Post receipt of the recommendations, DISB will inform Users on readiness for PVT.

5.8 ANNUAL AUDITED STATEMENT OF EXPENDITURE

5.8.1 Annual Audited Statement of Expenditure, duly vetted by the concerned AO(R&D), CDAs/PCDAs is required to be sought at the end of each Financial Year (FY) by the Nodal/Participating Labs as per the format at DRDO.DPFM.FT.09.

CHAPTER 6

PROJECT MONITORING AND REVIEW

6.1 CO-ORDINATING REVIEWS

6.1.1 Director (PM) of respective DG (Cluster) in consultation with Lab Director and Project Director should ensure conduct of reviews at intervals as stipulated in subsequent sections.

6.1.2 PD should workout project execution plan in sync with the Systems Engineering process and quantify progress % during review based on Milestone completion, financial progress, SOs placed, Documents submitted etc. w.r.t. original plan along with supporting artefacts.

6.2 PROJECT MONITORING AND REVIEW

6.2.1 In all **projects based on CCS approval or AoN accorded by DAC, DPB or SPB**, a 3-Tier periodic monitoring & review mechanism would be constituted i.e. 'Apex Board', 'Programme Management Board (PMB)/Executive Board (EB)' and the 'Project Management Board (PJMB)/Project Monitoring & Review Board (PMRB)'. Scheduling of Apex Board for projects accorded by DAC, DPB or SPB will be done separately for each of the Services. **However, for administrative purposes, the highest review board for these projects will be decided based on the cost of the project.** This Apex Board will also review the projects for which AoN has been accorded but the project is yet to be sanctioned. Standing Committee for Apex, PJB and PMB will be applicable only for CCS approved projects.

6.2.2 For all other **Projects under delegated financial power of Finance Minister (FM) and Raksha Mantri (RM) and Secretary DDR&D**, a 2-Tier monitoring mechanism would be constituted i.e. 'Executive Board (EB)' and 'Project Monitoring and Review Board (PMRB)'. However, projects under delegated financial power of Finance Minister (FM) and Raksha Mantri (RM) will be undergo '**Special Review**' under the Chairmanship of Secretary DDR&D on six-monthly basis. The constitution of the Special Review committee and the frequency of the review may vary with the project and be coordinated by DP&C in consultation with Project Director, Director PM, DISB/DLIC. DG (Cluster), will brief the progress separately to Secretary DDR&D regarding the projects under delegated financial powers of Secretary DDR&D on six-monthly basis.

6.2.3 For **Projects (except those under MM category) under delegated financial power of DG (Cluster)**, the 2-Tier monitoring mechanism would be constituted i.e. ‘Executive Board (EB)’ and ‘Project Monitoring and Review Board (PMRB)’.

6.2.4 Projects (except those in MM category) with Lab Director as CFA will be monitored by ‘Project Monitoring and Review Board (PMRB)’ and reviewed in Cluster Council once in six months.

6.2.5 In addition to the review mechanism applicable, ‘**Special Projects**’ will be reviewed by Secretary DDR&D with members as DG (Cluster) and Addl. FA (R&D). Chairman of the review committee, DG (Cluster) or Lab Director may invite members on need-to-know basis.

6.2.6 For all Projects, PMRB is the only ‘**Monitoring**’ committee in 3-Tier monitoring & review structure. All other higher Committees are for ‘**Review**’ of the project. Therefore, Lab Director may adopt additional internal mechanisms, as required, for monitoring the projects. Project based on **AoN accorded by DAC, DPB or SPB**, will also be monitored by JPMT (Joint Project Monitoring Team) with agenda specified in Chapter IV of DAP 2020 as amended.

6.2.7 The nomenclatures PMB and PJMB used for CCS Programmes are equivalent to EB and PMRB respectively in other cases except for delegated financial powers vested with PMB and PJMB. Constitution of the 3-Tier monitoring boards in the Programmes is given in the subsequent section.

6.2.8 Participating Labs will monitor and review sub-projects as stipulated for the main projects. The Project Director of the sub-project will forward status updates and technical review reports to the nodal Lab during the main project reviews and also in the interim as and when sought by the nodal Lab.

6.2.9 Each of the review meeting should be allocated sufficient time. Two separate reviews of the same project at different tier should not be held on the same day.

6.3 BRIEFING PAPERS FOR REVIEW MEETINGS AND MINUTES OF MEETINGS

6.3.1 Briefing papers for the monitoring and reviews should be provided by the Member Secretary to all concerned **at least 05 working days in advance** as per the format available at DRDO.DPFM.FT.10. This is to ensure informed participation by each of the member and contribution to the review. Director (PM) shall ensure that format for reviews and Minutes of Meeting (MoM) are being used by all concerned.

6.3.2 Minutes of Meeting (MoM) should be prepared by Member Secretary and seek approval of Chairman **within 10 working days**. These should also be uploaded in PMIS (ERP, when operational) subsequently along with the briefing papers and issued to all concerned. The minutes should essentially record agenda points (including issues to be deliberated), details of deliberation & discussion, proposed way-ahead/action-points/other decisions, expected date of completion of the action and responsible agencies. Delayed milestones should be recorded in minutes along with their implication on time, cost and mitigation plan. Additional requirements and changes in the design with its implication on cost and timeline should be specifically mentioned. Chairman of the Review Committee may ensure chance to each of the member to express their views and the same be recorded in the MoM. Specimen format for MoM is given at DRDO.DPFM.FT.11.

6.4 AGENDA OF 3-TIER REVIEW COMMITTEE MEETINGS

6.4.1 Common Essential Agenda Points for Review Committees

- Project overview.
- Action Taken Report for Action points of previous review.
- No. of Reviews by subordinate boards / committees since last review meeting.
- Programme/ Project activities since last meeting.
- Review of the six monthly technical & financial milestones.
- Review of Risk Management Plan and impact on overall project milestones with assessment of Responsibility & Accountability (as per RACI matrix).
- Review of procurement plan & its issues of respective CFA level, Expenditure status (Spent, commitments made and in pipeline).
- Time estimation by Standard and In-house developed Software tool for remaining duration.
- Assess reasons for delay and suggest corrective measures.
- Decision Points.
 - Programme/Project related managerial and technical issues
 - Issues raised by User or Industry Partner
 - Issues having timeline implications
 - Issues having financial implications

- For reviews after PDC of the project:
 - Reason for project not being closed on schedule
 - Details of pending payments
 - Status of admin completion

6.4.2 Apex Board (AB)

- Common essential Agenda points for monitoring committees
- Deliberation on key management issues which have cropped up during execution of Projects/ Programmes and solutions thereof.
- Discussion on unresolved techno-managerial recommended by EB and JPMT.
- Deliberations on SHQ and other inter-departmental requirements.
- Review and way ahead on the additional requirements by Services/ Changes as agreed by “PSQR Approving Body” in design with implication on cost and timeline.
- Ratification of recommendations made by lower-level committees i.e. Programme Management Board (PMB), Project Management Board (PJMB), Executive Board (EB) in respect of project cost revision and change in PDC (not beyond pessimistic PDC).
- Ratification of recommendations made by the lower committees on project completion in suitable category.
- Any other point with the approval of Chairman.

6.4.3 Executive Board (EB)/Programme Management Board (PMB)

- Common essential Agenda points for monitoring committees.
- Discussion on unresolved technical issues by PMRB and related managerial issues with their proposed solutions.
- Ratification of the TRL based on Qualifiers and Supporting Documents.
- Review sub-projects and work packages.
- Review of Sunset milestone w.r.t. its completion on time or its impact on PDC leading to management decision.
- Review the Recommendations of JPMT, wherever applicable.

- Recommendations on the additional requirements of SHQ/design changes as agreed by “PSQR Approving Body” with implication on time and cost.
- Ratification of PMRB recommendation for change in realisation plan including systems/equipment not envisaged in project proposal, project cost revision and change in PDC.
- Ratification of PMRB recommendation for closure of projects.
- Man-hour used in the project till date and infrastructure/ facilities utilized.
- Status of timely conduct of reviews i.e. PMRB, EB and Apex held vis-à-vis should had been done.
- Discuss deviation from roadmap and deviation from projected KPI.
- Status of Annual audited statement of expenditure.
- Any other point with the approval of Chairman.

6.4.4 Project Monitoring and Review Board (PMRB)/Project Management Board (PJMB)

- Common essential Agenda points for monitoring committees.
- Review PERT and any modification in it.
- Detailed WBS (indicating Bought out/BTP/BTS) indicating status of each module.
- Discussion on unresolved technical issues and their proposed solutions.
- Review of the Sunset milestone.
- Recommendations on the change of TRLs based on Qualifiers and Supporting Documents.
- Resolutions of issues internal to Lab.
- Initiatives by PD and DPD to ensure the project on track.
- Recommend alternate approach, whenever necessary.
- Recommending proposals for change in realisation plan including systems/equipment not envisaged in project proposal, cost revision and change in PDC.
- Man-hour used in the project till date and infrastructure/ facilities utilized.
- Recommending proposals for completion of projects in suitable category.

- Review and ratify that all reviews are on schedule as required. Ratify data updation in PMIS.
- Review ‘valuation of the technology’ and ‘economic impact data’.

6.5 COMPOSITION OF THE REVIEW COMMITTEES

6.5.1 Apex Board (AB)

Secretary, DDR&D	Chairman
Financial Advisor (Defence Services)	Member
Vice Chiefs / CISC / Deputy Chiefs or equivalent of Services	Member
DG (Acquisition) (for AoN accorded Projects)	Member
DG (Concerned CAPF) (for AoN accorded Projects by Apex Council of DRDO-MHA)	Member
DG (Cluster) of Nodal Lab	Member
DG (Cluster)s of Participating Labs	Member
DG (R&M)	Member
DG (PC&SI)	Member
Addl. FA (R&D)	Member
Director, Nodal Lab	Member
Chairman, JPMT (where nominated)	Member
CE, CEMILAC (as applicable)	Member
DGQA/AQA (as applicable)	Member
Director, P&C	Member*
Programme Director	Member Secretary

Chairman may co-opt/nominate other members as and when necessary.

* DP&C to function as Apex Board Sectt. for CCS Programmes.

Periodicity of meeting : Once in Six months

Note:-

(a) For Programmes whose PDC is over and Administrative Closure is under process, the necessity of Apex Board will be assessed by DG (Cluster) and review to be conducted based on direction of Chairman, Apex Board

(b) HQ DGs may nominate concerned Corporate Directors

Standing Committee for Apex Board for CCS approved Programmes

Secretary, DDR&D	Chairman
DG (Cluster) of Nodal Lab	Member
Addl. FA (R&D)	Member
Director, Nodal Lab	Member
Programme Director	Member Secretary

DP&C is the Apex Board Secretariat for Standing Committee to facilitate the implementation of decisions taken by Apex Board .

6.5.2 Executive Board (EB)/Programme Management Board (PMB)

DG (Cluster) of Nodal Lab	Chairman
ACNS/ACAS/ACOAS Plans (for AoN accorded projects)	Member
Rep DG (Acquisition) (for AoN accorded Projects)	Member
IG (Concerned CAPF) (for MHA Accorded AoN accorded projects only)	Member
Director, Nodal Lab	Member
Directors of Participating Labs	Member
Chairman, JPMT / Rep SHQ (for AoN based MM cases)	Member
Director DP&C, DRDO HQ	Member
Director DISB, DRDO HQ (for MM projects)	Member
Director DLIC, DRDO HQ (for MHA based AoN accorded projects)	Member
IFA (R&D), Nodal Cluster	Member
Director (PM) of DG (Cluster)	Member
Director DQR&S, DRDO HQ (for MM projects)	Member
SA to Service Chiefs (for MM projects)	Member
Rep of CEMILAC (as applicable)	Member
QA Agency (as required)	Member
Director DCW&E/CCE (for IF and other projects Civil works)	Member
Chairman, Lab Technology Council	Member
Programme/Project Director	Member Secretary

Chairman may co-opt/nominate other members as and when necessary.

Periodicity of meeting : Once in six months (till administrative closure)

- Note:-**
- 1) *Representatives of DcPP, if applicable, may be invited specifically to apprise the Board about the status of work allotted during the proceedings at appropriate time*
 - 2) *Director DF&MM, DRDO HQ, may be invited as member, if required, after project PDC is over and awaiting administrative closure, for pending issues with PCDA*
 - 3) *Chairman and Members of the initial TiEC should be invited for EBs and PMRBs ratifying PDC extension for the project for awareness on what went wrong and train them for future.*

Standing Committee for PMB

DG (Cluster) of Nodal Lab	Chairman
Director, Nodal Lab	Member
IFA (R&D)	Member
Programme Director	Member Secretary

6.5.3 Project Monitoring & Review Board (PMRB) / Project Management Board (PJMB)

Director, Nodal Lab	Chairman
Subject experts from R&D Centres/Academia/other DRDO Labs (<i>as required</i>)	Member
Director (PM) of DG (Cluster)	Member
DFA/Dy. IFA/Jt. IFA/Director (Fin)	Member
Project Directors of Sub Projects	Member
Programme Director/Project Director	Member
Head PMT/ Rep SHQ (for AoN / MM Projects) (as applicable)	Member
Director CAIR or Rep (for AoN accorded project: AI related)	Member
Director, DIPAS or Rep (for AoN accorded project: HFI & HRA related)	Member
Director ISSA or Rep (for AoN accorded project: ORSA Study)	Member
Rep of CEMILAC (as applicable)	Member
Rep of DGQA/AQA (as required)	Member
Lab QA/IV&V	Member

Deputy Project Director

Member Secretary

Chairman may co-opt/nominate other members as and when necessary.

Periodicity of meeting : Once in three months (till administrative closure)***Note:- 1) Representatives of DcPP, if applicable, may be invited specifically to apprise the Committee about the status of work allotted during the proceedings at appropriate time*****Standing Committee for PJB**

Nodal Lab Director/ Work Centre Director

Chairman

DFA/Dy. IFA/Jt. IFA/Director (Fin)/Rep of IFA (R&D)

Member

Project Director

Member Secretary

6.6 SPECIAL REVIEWS**Participants in Special Review by Secretary DDR&D***(Projects whose CFA is Secretary DDR&D, RM and FM)*

Secretary, DDR&D

Chairman

DG (Cluster) of Nodal Lab

Member

DG (Cluster)s of Participating Labs

Member

DG (R&M)

Member

DG (PC&SI) (for AoN projects)

Member

Addl. FA (R&D)

Member

Director, Nodal Lab

Member

Directors of Participating Labs

Member

Director, P&C

Member

Director, DISB (for AoN projects)

Member

CE, CEMILAC (as applicable)

Member

Director, DQRS/Director SQR

Member

Project Director

Member Secretary

Chairman may co-opt/nominate other members as and when necessary.

Periodicity of meeting: Once in six-months or as directed by Secretary DDR&D

6.7 ROLE AND FUNCTIONS OF STANDING COMMITTEES FOR AB, PMB & PJB FOR PROGRAMMES

The role of Standing Committee for demand approvals, financial sanctions, PDC extensions and other financial approvals will be as per DFP of DRDO and Procurement Manual-2025 as amended from time to time.

Standing Committee is not applicable for EB and PMRB.

6.8 ROLES & ACCOUNTABILITY IN PROJECT REVIEWS

Major Roles & Accountabilities (but not limited to) of Members of Review Committees/ Stakeholders is detailed below:

Dte/ Agency	Roles & Accountability
DP&C	- Participate in review meetings and assess overall project progress w.r.t. reference/original timeline (Milestone linked with financial outlay & timeline). - Share lessons learnt from other projects. - Convey any recent changes in DPFM or directives of DMC. - Arrange special reviews by Secretary, DDR&D (if required) to resolve major issues.
DISB	- Assist Lab to resolve QRs/AoN related issues with SHQs. - Resolve trial related issues with SHQs (trial directives, platform/test range availability etc). - Update and advise on procedures related to DcPP.
ISSA	Apprise about implementation of system analysis guidelines/ ORSA study for MM/TD(S) projects and update the review committee.
DQRS	- Suggestions/Inputs to Labs on project quality and safety related issues. - Implementation of DRDO Quality and Reliability Policy Guidelines. - Co-ordinate and resolve any issues with external QA agencies. - Inform the status of assessing and rating of manufacturing maturity of defence industry partners (SAMAR).

CAIR	▪ Identification and Implementation of AI applicability as per 'Artificial Intelligence Framework and Guidelines for projects and programmes' issued by CAIR and update to review committee.
DIPAS	▪ Implementation and update to review committee Human Factor Engineering (HFE) and Human Reliability Analysis (HRA) element in Mission Mode projects.
DCW&E	▪ Providing inputs on implementation of proposed and ongoing Civil Works.
Project Director (PD)	▪ Brief project progress as per the milestone linked financial outlay & timeline. ▪ Present current status/ progress as per review format during review of project. ▪ Highlight the critical issues / Risk associated & mitigation plan in project execution and impact on overall PDC. ▪ Review of RACI matrix. ▪ Monitor financial status of the project. ▪ Preparation & issue of Minutes of Meetings (MoM). ▪ Man-hour used in the project till date and infrastructure/ facilities utilized
Deputy Project Director	▪ Forwarding briefing papers to members, five working days in advance. ▪ Time estimation by standard and DRDO developed software tool for the remaining project duration. ▪ Record minutes with action points along with responsibility to concerned agency/Individual with timeline. ▪ Obtain audited Statement of Account Expenditure from concerned audit authority after project completion. ▪ Assure follow up of broad action points deliberated during previous reviews. ▪ Update online project status on PMIS/ERP.
Director (PM)	▪ Issue AB, EB & PMRB calendar in coordination with Lab in the beginning of the calendar year and ensure compliance. ▪ Confirm during EBs that project status has been updated online in PMIS/ERP on DRONA.

	▪ Ensuring that briefing papers are sent to members, five working days in advance. ▪ Liaise with DG (Cluster)s to track status of action points deliberated during review meetings.
Lab Director	▪ Assign scope based functional project team, allocate adequate manpower and other resources. ▪ Expedite allotment of sub projects to participating Lab (if any). ▪ Ensure timely conduct of reviews and implementation of action points. ▪ Seek any special review for resolving important issues, if required.
DG (Cluster)	▪ Monitor progress vis-à-vis milestones. Suggest corrective actions to mitigate milestones slippages. ▪ Resolve inter Lab related issues. ▪ Risk Mitigations. ▪ Update DMC regarding delays in projects on six-monthly basis during KPI based review.
Chairman, Lab Technology Council	▪ Assess the TRL progression during the execution of the project (envisaged vis-à-vis achieved). Offer comments on technological gaps, grey areas, critical technologies and strategy to mitigate the same.

CHAPTER 7

PROJECT SCOPE, DURATION AND COST REVISION

All efforts shall be made by the Project Team to complete the project within the allotted time and cost. However, due to the unforeseen technological challenges and other issues, duration of the project and/or its cost may require changes. During the execution of projects, re-allocation of funds under the same major head may be required. These corrigenda are to be considered as per merit and need to be justified for the reason for changes, proposed way-ahead for completion of balance activities, risk mitigation strategies etc.

The Note proposing the change should have the basic details of the projects, need for change of Scope with justification, work done so far, reasons for PDC extension, reasons for cost revision, revised six-monthly milestones with financial outlay, comparative PERT chart (**original vis-à-vis proposed**), revised cost break-up table and recommendations from relevant committees as applicable for Scope, Duration and Cost revision.

While sending any proposal to DRDO HQ for approval of CFA, viz. change in project scope, duration, cost, category or Lab, all volumes of project file with volume number, date of opening etc. mentioned on each file, need to be sent to HQ instead of a single part case.

7.1 SCOPE, COST AND PDC REVISION

7.1.1 Extension of the project duration or changes in the project cost or scope, for reasons specified, shall be recommended in not later than 02nd last meeting (i.e. at least 3+ months for projects with Director as CFA and 6+ months in other projects) of the highest review committee.

7.1.2 PDC delay should be reported in the review meetings and PMIS from its earliest occurrence/awareness. A delay of 3 months or more cannot be reported suddenly by the project in any review committee. It has to be recorded in PMIS, PMRB and EB.

7.1.3 Criteria for PDC Extension: -

- a) Large Quantum of proposed work has been completed and High Risk has been addressed. PDC extension sought for small quantity of work/ low risk/ further investigations.
- b) In case large quantum of proposed work is pending, sunset clause may be opted by the highest review committee.

- c) Sufficient achievements have been made and same is needed to be protected and continued, but significant risk still remained. If the risk occurs after this PDC extension then project can be completed with partial success or closed.
- d) Projects having High Risk still remaining unaddressed, PDC extension should be sought only if technology is Critical and no other “Plan B” available at present.

7.1.4 The explanation for the time extension may be mapped to one or more of the following, but not limited to: -

- a) Re-definition of objective
- b) Delay in achieving technical breakthrough
- c) Procedural delays in critical path activities
- d) Force Majeure conditions

7.1.5 All projects seeking PDC extensions, irrespective of time, cost and category, will be presented in DMC minimum 03 months prior to current PDC.

7.1.6 DMC may delegate the detailed review to the Committee of DGs to ensure linkage of delay with KPI of the Lab. The decision of the Committee of DGs may be presented in the DMC and ratified. Labs may plan suitably in advance as the KPI review of the Labs is held only twice a year.

In exceptional cases, the Committee of DGs may meet on quarterly basis to review the projects for PDC extension prior to DMC. However, all such PDC extensions will be counted during the subsequent KPI based review of the Lab.

7.1.7 All MM Projects with PDC extension shall be updated in PMIS on monthly basis for:-

- a) Slippage by months from existing PDC of project.
- b) Reasons for delay
- c) Steps that can be taken to mitigate the delay

7.1.8 For PDC extension by 50% or more in MM projects, the requirement for the extension should be peer reviewed by committee constituted by Secretary DDR&D, comprising of external experts and user representatives before presenting in DMC. Some members of this committee should preferably be same as in the initial Time Estimation Committee (TiEC) and PDR Committee.

7.1.9 Presentation should be given to DMC (or committee of DGs, if delegated) in the format provided in DPFM 2026 (DRDO.DPFM.FT.03B) with reasons for delay. Accountability for delay, if any, in addition to Project Director & Lab Director may be brought out.

7.1.10 Based on the recommendations of the DMC, the extension in the duration of the project and / or change in its cost shall be sanctioned by the corresponding CFA with the concurrence of respective Financial Adviser.

7.1.11 Any change in Scope in TD (S) and MM category of projects sanctioned based on PSQR/AoN by Services leading to change in project planning and timelines shall be agreed upon if these changes are received after approval from PSQR/AoN approving body/Indenting Authority. In such cases Lab should work out mutually time and cost penalty leading to revision in timeline and cost. Revision in timeline will not be considered as PDC extension however, it will be considered as '**Re-Fixing of PDC**' for that particular instance. However, the count of the PDC extension, if any, prior and post this instance will continue to be counted for KPI.

7.1.12 Non availability of platforms at pre scheduled times as indicated in the PERT and which has agreement/concurrence of User may also cause delay which is beyond the control of project team. Delay in project timelines due to this should not be called as revision of the PDC but it will be considered as '**Re-Fixing of PDC**'. However, if the project has not indicated the platform requirement in the PERT or not adhered to its indicated timeline for use of platform due to delay in readiness for the trial, it will not be considered for re-fixation of PDC.

7.1.13 **Re-Fixing of Cost** (for projects other than sanctioned by CCS):

- a) If cost of a project is proposed to be enhanced by less than 20% but not beyond initial recommendations of CEC, which has been reduced during the sanctioning process, without change of scope, will be considered as 'Re-Fixing of Cost' without re-assessment by CEC.
- b) As the activities in S&T, TD(T) and High Risk projects would have never been done before, firming up of cost in some of these projects at the time of project sanction may be difficult. The cost during sanction will be considered as Rough Order of Magnitude (ROM) having fluidity. In such cases, upto 20% escalation in the ROM will be considered as 'Re-Fixing of Cost' without re-assessment by CEC.
- c) For S&T, TD(T) and High Risk projects, after Critical Review/CDR, the suggested changes and re-assessment of risk may lead to change in scope. In such cases, the final cost may be decided by re-assessment of CEC and will be considered as 'Re-Fixing of Cost'.

7.1.14 Any revision in scope of project may delay the project. Any additional scope due to internal requirements should be discouraged.

7.1.15 In respect of projects sanctioned by Raksha Mantri, the case for extension in PDC shall be dealt with as per provisions of DFP, as amended from time to time.

7.1.16 For processing PDC extension cases, SoC has to be initiated by the nodal Lab as per DRDO.DPFM.FT.12. Checklist at DRDO.DPFM.CL.03 may be utilized by the Director PM, O/o DG (Cluster) .

7.1.17 The guidelines stipulated in DFP, as amended from time to time, will be followed for re-allocation/ enhancement/ reduction of funds of a sanctioned Project. Re-allocation of funds and cost revision proposals should be submitted as per DRDO.DPFM.FT.13 and the checklist at DRDO.DPFM.CL.04 may be utilized by the Director PM, office of DG (Cluster)s while processing the cases. Cost revision proposals resulting in approval of higher CFA should be supported by cost estimations duly vetted by CEC with some members preferably same (efforts should be made to have some members from previous CEC) as initial CEC.

7.1.18 **All PDC and Cost Revision sanctions**, issued by designated CFA, will be assigned fresh “**Unique Sanction Code**”. Specimen Govt. sanction letters for PDC extension and cost revisions are at DRDO.DPFM.FT.16 and DRDO.DPFM.FT.17 respectively.

7.2 SUB-PROJECTS REVISIONS

Scope revision, PDC extensions/cost revisions and re-allocation of funds under sub-projects will be sanctioned by the CFAs stated in **Table 12 (section 4.8)**, with the recommendations of highest review committee. Any revision in the cost break-up table related to sub-projects within CFA Powers of sub-projects may be undertaken directly.

7.3 TRANSFER OF A PROJECT FROM NODAL LAB TO ANOTHER LAB

7.3.1 The steps to be followed for transfer of a sanctioned project from one Lab to other are:

- (i) The case will be presented during CCM and EB review respectively. Recommendations should be recorded in the minutes.
- (ii) Proposals for **project transfer between Labs** shall be recommended by EB and presented to the DMC. Proposal should also mandatorily include transfer of the dedicated members with no linkage with other projects of the project team, at least till the PDC of the project.

(iii) Audited statement of expenditure should be sought from the concerned audit authorities, till the cut-off date.

(iv) A sanction letter would be issued by the CFA stating closure of old project number and old unit code and allotment of new project number and new unit code. Project number and unit code of the sub-projects will also be changed.

(v) Transfer of expenditures as per audited statement of expenditure will be done from old unit code to new unit code.

(vi) Project inventory (except capital equipment and facility) and inventory records will be transferred to new Lab along with outstanding audit observations.

(vii) All procurement case files should be transferred on as-is-where-is basis. New Lab will be authorized by the CFA to place supply orders for all cases initiated by the previous Lab.

7.3.2 Specimen formats for transfer of a project are given at DRDO.DPFM.FT.19.

7.4 CHANGE OF CATEGORY OF PROJECT

7.4.1 There may be instances where a project may need to be initiated pending the completion of process of AoN by DAC, DPB or SPB/PSQR finalization and quantity commitment from the SHQs. In all such cases, the project can be initiated in TD(S) category and can be changed to MM category subsequent to completion of procedural requirements.

7.4.2 Change of category of projects shall be done with the recommendations of the highest review committee, intimation to DMC & approval of CFA with concurrence of corresponding Financial Advisor.

7.4.3 Specimen formats for change of a project category during project currency is given at DRDO.DPFM.FT.20.

7.5 PROJECT COMPLETION PRIOR TO PDC

The PDC of the project need no revision for finishing the project before the PDC. It should be brought out in the Project Completion Letter. Projects completed prior to PDC may be given due credit during KPI review. Significant early completion may appropriately be considered for **Commendation**.

CHAPTER 8

PROJECT COMPLETION OR CLOSURE

Completion or Closure of project has to be initiated by the participating Lab and nodal Lab well within sanctioned PDC or on project meeting all its stated objectives, whichever is earlier. **Upon completion of the Project PDC, no new contract/supply order should be committed.** All items/stores must be received in the Lab before end of project PDC. However, payment of already committed stores can be released beyond PDC. Like project sanction, a case for project completion or closure has to be moved for approval of CFA. On the similar lines, completion of the sub-project is to be intimated (Technical Completion Report) by the participating Lab to nodal Lab, at least 3 months prior to the PDC of the main project.

The Note proposing the approval of Competent Authority for Administrative Closure should have the basic details of the projects viz. project scope, envisaged vis-à-vis achieved, achievement made under the project, way forward, lessons learnt, statement of expenditure vetted by CDA (R&D) and recommendation by DG (Cluster) and recommendations from relevant committees as applicable.

8.1 PROJECT COMPLETION AND CLOSURE CATEGORIES

8.1.1 A sanctioned DRDO project may be completed or closed under following categories. Detailed procedures for completion or closure of projects under these categories are given at subsequent sections.

- (a) **Completed:** Project meeting all specified objectives within the original or revised cost and duration approved by the CFA.
- (b) **Completed with Partial Success:** Projects with **defined set of objectives** may result in successful completion of certain objectives and partial accomplishment of balance objectives. Such projects, where part of objectives could not be completely met, would be considered as partial success. Projects completed in this category should ensure have achieved more than 75% of the 6-monthly milestones.
- (c) **Stage Closed:** A sanctioned project can be stage closed, where considerable effort and funds have been spent and considerable success has been achieved, yet concrete results could not be achieved. This closure is resorted to only in extra-ordinary circumstances e.g. when faced with insurmountable scientific or technical challenges, obsolescence, changes in specifications or requirements causing major changes in cost

and timelines. Labs should amply justify the advantage of **stage closing** the project rather than pursuing it further and also provide utilization plan of assets/ hardware and benefits accrued under the project. Henceforth, such closed projects may be interpreted as stage closed project in this document.

(d) **Cancelled Project.** A sanctioned project can be cancelled, where continuing the project is not considered viable due to changed requirements/technology or due to administrative reasons. In all such cases, expenditure of greater than 5% of the sanctioned project cost should not have been incurred.

8.1.2 Projects not achieving the ‘sunset milestone’ within time frame may also be closed or completed with partial success. It will depend on expenditure and accomplishments till ‘sunset milestone’ as recorded in DMC MoM at the time of according AoN for the project.

8.1.3 For all type of closure categories, SoC has to be prepared for Technical Completion for CFA’s approval as per format at DRDO.DPFM.FT.14 and for Administrative Closure as per format at DRDO.DPFM.FT.15.

Note:- Throughout this Chapter, “completion” may be interpreted as successful completion, completion with partial success, stage closure or cancellation as the case may be.

8.2 PROJECT SUCCESS EVALUATION

8.2.1 Once the last activity is completed, the project has to be evaluated for measuring achieved outcome vis-à-vis defined success criteria, i.e. the qualifiers and support documents defined in **Table 1**. Evaluation method is given below.

- (a) Success evaluation of DAC/DPB/SPB AoN based MM projects w.r.t. PSQR validation trials.
- (b) For other ongoing MM or UT projects, PSQR validation trial outcome will be regarded as success determinant by the highest review committee while recommending project completion.
- (c) For TD(S), TD(T), S&T, IF and ongoing PS projects, an Independent Technical Committee for Evaluation of Project Success drawn from outside project will evaluate project outcome and certify successful completion.

8.2.2 Independent Technical Committee for Evaluation of Project Success

8.2.2.1 Once all project activities are completed, the project has to be evaluated for measuring achieved outcome vis-à-vis defined success criteria. The project should also be analyzed to find out reasons for positive and negative deviations.

8.2.2.2 For TD(S), TD(T), S&T, IF and ongoing PS projects, an Independent Technical Committee for evaluation of Project Success, drawn from outside project will be constituted by Competent Authority in the **2nd last highest review board** and its report should be presented in the last meeting of EB or PMRB (whichever is higher) before PDC of the project. The Committee will evaluate project outcome and certify successful completion based on the success criteria/expected outcome defined for the project in terms of scope including objectives and deliverables etc. as defined in the sanction letter and the project proposal. The report should have the table indicating success criteria or expected outcome vis-à-vis achieved outcome.

8.2.3 Criteria for Evaluating Successful Completion of TD(S), TD(T), S&T and IF Projects

- **TD(S) and TD(T) Projects:** Successful completion of the project scope i.e. design & development of prototypes; tests/evaluations and demonstration in Lab/intended test environment; technical reports etc; listing accomplishments vis-a-vis project scope.
- **S&T Projects:** Successful completion of the scope i.e. research, prototype development, simulations, trials and findings, research reports/publications/patents, compilation of technical completion report listing accomplishments vis-a-vis project scope. Due consideration to be given while assessing the success as research under S&T projects at completion may not always exactly result in envisaged objectives. Each successful TD and S&T project should lead to addition of at least one system/technology in 'Database of Systems & Technologies'.
- **IF Projects:** Completion of installation & commissioning, acceptance test of the facility based on specifications. Acceptance test report listing successful completion of the scope i.e. completion of FATs, equipment delivery, completion of civil works, acceptance test approved/endorsed by end user.

8.3 COMPETENT AUTHORITY FOR CONSTITUTION OF INDEPENDENT COMMITTEE FOR EVALUATION OF PROJECT SUCCESS

Nodal Lab Directors will constitute the committee for projects within their delegated financial powers. For projects beyond delegated financial powers of Lab Director, DG (Cluster) will constitute the committee. The Competent Authority will also decide time frame for submission of the report in such a way that the final report is available to highest review committee of the Project during its meeting for recommendations of Project Completion.

Table 13 : Constitution of Independent Technical Committee for Evaluation of Project Success	
Chairman	Scientist 'H' / Scientist 'G' (Non-project member)
Members	Two Expert Member(s) with domain knowledge from other Labs/ outside project (nominated by DG Cluster)
	Member from nodal Lab
	Academia expert & Industry expert (as applicable)
	Director PM for Projects with DG (Cluster) as CFA
	Director DP&C for projects beyond DG (Cluster) as CFA
Member Secretary	Project Director
Any other members may be co-opted by Constituting Authority or Chairman	

Terms of Reference:

- The committee will independently assess and ratify project success as per criteria laid in the project proposal, SoC and not just the Govt. Sanction Letter.
- A report to that extent has to be prepared by the committee, duly ratifying compliance of the project achievements with the success criteria and submit the report to competent authority for approval and acceptance of the report within allotted time.
- The compliance should be ensured in accordance with the paragraphs in SoC/project proposal during sanction of project and as amended thereafter. It should have a table indicating success criteria/expected outcome viz-z-viz achieved milestones.
- To ensure preparation of Technical Completion Report for submission within PDC and updated PMIS database.
- The Committee should vet the online entries done by the project team for the lessons learnt, Technologies Developed, Valuation of Technology, Economic Impact and comparable MM/TD(S) Systems worldwide (in terms of development cost, timeline, specification).

Competent Authority for Approval and Acceptance of Evaluation Report for Project Success

Project CFA	Competent Authority
Lab Dir	Lab Director in consultation with Director (PM) of O/o DG (Cluster)
DG (Cluster) or beyond	DG (Cluster)

The Evaluation Report for Project Success has to be forwarded to DG (Cluster) through Lab Director for intimation.

***Note :** This report will be submitted to the EB/PMRB (whichever is higher), while reviewing the project for recommendation of project completion.*

8.4 RECOMMENDATION OF HIGHEST REVIEW COMMITTEE

The highest review committee shall recommend the completion of the project. **Agreement of DMC** shall be required for projects completed with partial success, closing or cancelling a project. Such projects should be put upto DMC prior to PDC of the project or hitting the roadblock milestone, whichever is earlier.

8.5 INTIMATION TO USER POST DEVELOPMENT TRIALS

For TD(S) projects in which a system/equipment is realized as a deliverable, the following to be carried out at the time of project completion:

- (i) Once a system/equipment has been developed and development trials are completed; Labs to intimate DISB that the development trials have been completed and system is ready for User Trials (development trial report and the QR compliance, if applicable, to be enclosed with the intimation).
- (ii) DISB, thereafter, to intimate the concerned SHQs about the completion of development and offer the system for User Trials requesting to SHQ for the following:
 - (a) If development is initiated as per agreed PSQRs, SHQ to initiate action for accord of AoN.
 - (b) Else, SHQ to initiate action for formulation / finalization of PSQRs.
- (iii) A copy of this letter by DISB to SHQ will be shared with concerned Lab and should be mandatorily enclosed as a part of the project completion report.

8.6 TECHNICAL COMPLETION REPORT

8.6.1 A Technical Completion Report for all projects should be prepared by the documentation team/project team within PDC of project. Participating Labs and agencies should submit its Technical Completion Report of sub projects and work packages to the nodal Lab **within its PDCs**. PD and DPD will ensure its submission within the PDC. The report should be prepared by the documentation team and the Deputy Project Director, recommended by Project Director and approved by Lab Director for all projects. The report will be finally submitted to DG (Cluster) for acceptance.

For projects with PDC prior to Dec 2026, the technical completion report may be submitted within 6 months of completion of PDC.

8.6.2 All Quality & Reliability (Q&R) audit observations (Non-Compliances) as per DRDO Q&R Policy Guidelines should be addressed before technical completion of the project.

8.6.3 Technical Completion Report is an in-house document. It should **essentially contain** the following –

- Complete project history.
- Details of objectives envisaged vis-à-vis achieved/shortfall.
- Achievements of the project.
- Detailed justification/reasons for shortfalls.
- References to design documents/technical reports evolved during the project execution.
- TRL assessment reports (initial, six-months periodic and final).
- Details of Systems and Technologies developed.
- ToT/Production Order details etc.
- Problems faced, lessons learnt & way forward.
- AI enablement, HFE & HRA implementation, ORSA Study on developed systems/product {for MM & TD(S) only}.
- Test results.
- Data for ‘Economic Impact’.
- Data for ‘Valuation of the Technology’ {Manpower utilization in terms of Man-months (employees/scientists including project-based manpower), infrastructure used (of the Lab, without making any payment)}.

- Publications: reports, papers, patents.
- Proposed utilization of developed technologies.
- List of BTP systems/sub-systems for which ToT is being / has been carried out.
- Declaration for submission of a copy of final version of software (including code) developed in the project to Centralised Database of Lab, DG Offices & Central repository of DRDO.
- Approach for the future.

8.6.4 Distribution List: A copy of the technical completion report should be retained by the Lab and the O/o DG (Cluster) for future reference. A soft copy of the report should be forwarded to DP&C, DRDO HQ and uploaded on PMIS for future retrieval. The achievements and shortfalls, references to technical reports (such as project proposal, TiEC, CEC, IRC/PRC/PDR, DDR, CDR etc. created during the project life cycle) evolved during the project along with its location of archival in repository should be stated in the Technical Completion Report. All Technical Completion Reports and supporting documentation should be stored in central repository of the (a) Nodal Lab, (b) DG (Cluster) and (c) DESIDOC for future reference. Technical completion report should also be shared with participating Labs. Guidelines for storage of classified documents may be followed in the repositories, wherever applicable.

8.7 PREPARATION OF TECHNICAL COMPLETION REPORT

8.7.1 Executive Summary: It should contain following:-

- Background - Scope of the project.
- Major achievements compared to Global Scenario & Nos. of ToTs generated, IPR generations, No. of papers published related to project, details of infrastructure created, details of DcPP.
- Percentage time variation in the project as compared to original PDC, % variation in cost as compared to original cost.
- No. of industries (Tier I, II & III), start-ups and institutes involved, % indigenization & imported components that have been indigenized.
- Major Trials/Experiments.
- List of countries those already have this system, when was it developed by that country, time required, development cost and team size.

- Export potential that has already been explored for the product till now, give status.
- Type of completion

- **Project Performance**

Summarize the actual performance of the project against the planned performance. All projects may vary to some extent from the original plan, these variations (if any) should be identified and the reasons for the variation be described.

- **Performance against Objectives/Outcome:-**

- (a) Describe the actual performance of the project vis-a-vis project objectives and targeted outcomes.

- (b) Describe the actual performance of the project in relation to the delivery of the outputs. Whether all planned outputs are delivered & are accepted? Did the quality of the outputs meet expectations?

- **Performance against Schedule:-** Describe the actual performance of the project against the project schedule and reasons for delay, if any.

- **Performance against Budget:-** Describe the actual performance of the project against the project budget and reason for cost variation, if any.

- **TRL Assessment Reports**

An assessment of final TRLs of each sub-system with resulting PRI may be enclosed in the final report along with relevant supporting documents.

- **Database of Systems and Technologies**

Closure report of the project should essentially contain the list of system(s) and technologies developed as an outcome of the project. These shall be updated in the central database- 'Database of Systems and Technologies' providing all relevant fields/details.

8.7.2 Lessons Learnt

Reasons for success/partial success	Lessons Learnt	Way Forward

TRL Matured during the Project: Mention TRLs calculated at the time of sanction of the project, 6-monthly TRL change in each of the sub-systems and the level of TRLs reached at the time of closure of the project with suitable justification.

8.7.3 Appendices to Technical Completion Report

- A copy of the signed declaration of acceptance by the relevant users for each of the deliverables.
- A copy of minutes of highest review committee recommending the technical completion of the project.
- References to Project technical documents (report/document No. & location) such as PRC, PDR, DDR, CDR and Minutes of technical review meetings, TRA report (indicating initial, 6-monthly and final TRL), Project definition documents, System definition documents, Technical / developmental / ATP documents, development trials etc.
- PSQR validation trials & its reports.

8.7.4 Digitization of Project File

As the Technical completion Report is prepared, the complete project file should be digitized with 'search option' and be uploaded in PMIS, till Enterprise Resource Planning (ERP) system is functional.

8.7.5 Way Forward

Plans for further use of the prototypes/technologies developed in the project may be stated. Product development plans for Mk-II by the Industry Partner, SHQs acceptance or any other related future roadmap for the product may be mentioned.

8.7.6 Accepting Authority

Technical Completion Report should be approved by Lab Director & accepting authority will be DG (Cluster).

8.8 ADMINISTRATIVE CLOSURE REPORT

8.8.1 Administrative closure report of the project should be prepared by the Deputy PD, recommended by the Project Director and approved by Lab Director. The report should be finally submitted to DG (Cluster) for acceptance and then forwarded for approval of CFA well within **12 months** after project PDC. The report serves the purpose of audit authorities. Participating Labs and agencies should submit the admin completion report to the nodal Lab for respective sub-projects and work packages within 06 month of expiry of respective PDCs. PD, DPD and project team will be responsible for replying any audit query raised during or after PDC of the project.

8.8.2 Audit observations, not settled, may continue to be active till settled, however, Administrative closure should not be delayed only on account of this.

8.8.3 Administrative Closure Report should **essentially contain:-**

- Project attributes.
- Brief summary of technical achievements (objective envisaged vis-à-vis achieved).
- No further new commitment should be done in the project after completion of the PDC. The items/stores must be received in the Lab before the PDC. Payment of committed stores can be released beyond PDC based on contractual obligations/warranty etc. Concerned CDA office will audit the Statements of Accounts of Expenditure. It will also bring out outstanding balance payment due to contractual obligation/warranty etc. after PDC of the project, in the Audit Statements (Audit report/Certificate). Administrative closure of the project will be carried out based on this audit report/certificate. Administrative closure of the project will not be held up because of balance payment after the PDC of the project which is due to contractual obligation/warranty etc.
- Minutes of the highest review committee recommending completion of the project.

8.9 COMPETENT AUTHORITY FOR ADMINISTRATIVE CLOSURE OF PROJECT

8.9.1 **Completed projects**, with all specified objectives achieved within approved cost and timelines, which have been recommended for completion by the highest review committee,

will be administratively closed with the approval of Secretary, DDR&D or CFA whichever is lower, with concurrence of designated Financial Advisor. Routing of completion case file will be the same as stated at **Chapter 4** (applicable upto Secretary DDR&D). Upon accord of approval for Project completion, a completion letter will be issued by the designated office (DRDO.DPFM.FT.18).

8.9.2 Projects Completed with partial success, Stage Closed & Cancelled projects, where project objectives are not met in its entirety, Lab should identify reasons for not achieving the pre-defined objectives. These should be reviewed by the **Committee for Evaluation of Project Success**. Findings of the committee should be submitted to the highest review committee, which will recommend completion (with suitable category) of the project with further way ahead. In case of MM and TD(S) projects, completion category should be based on trial outcome and recommended by the highest review committee. In-principle Agreement of DMC shall be obtained before processing such projects. Thereafter, case should be moved for approval of Secretary, DDR&D or CFA whichever is lower, with concurrence of designated Financial Advisor for completion. Projects in the MM category which could not be completed will be intimated to the User through DISB. Transaction of Business Rules (ToBR) may be followed in case of CCS projects completed in this category.

8.9.3 CFA for administrative closure of sub-projects will be as stated at **Table 12** with the recommendations of Project Director of Nodal Lab and concurrence of designated Financial Advisor. Participating Labs will submit the Administrative Closure Letter to Nodal Lab.

8.9.4 Completion of Work Packages. Statement of expenditure duly vetted by CFA of organization executing the work package should be annexed with the completion report. CFA for Work package completion will be as stated for sub-projects at **Table 12**.

8.9.5 All Categories of Project Completion, Stage Closure or Cancellation should have technical completion and administrative closure report.

8.9.6 In case of Administrative Closure of the project is delayed beyond the permissible period of 12 months, appropriate reasons for delay may be provided in tabular format highlighting the duration of delay in an office in the project while preparing the Note for seeking the approval of Competent Authority.

8.10 WAY FORWARD AFTER PROJECT COMPLETION

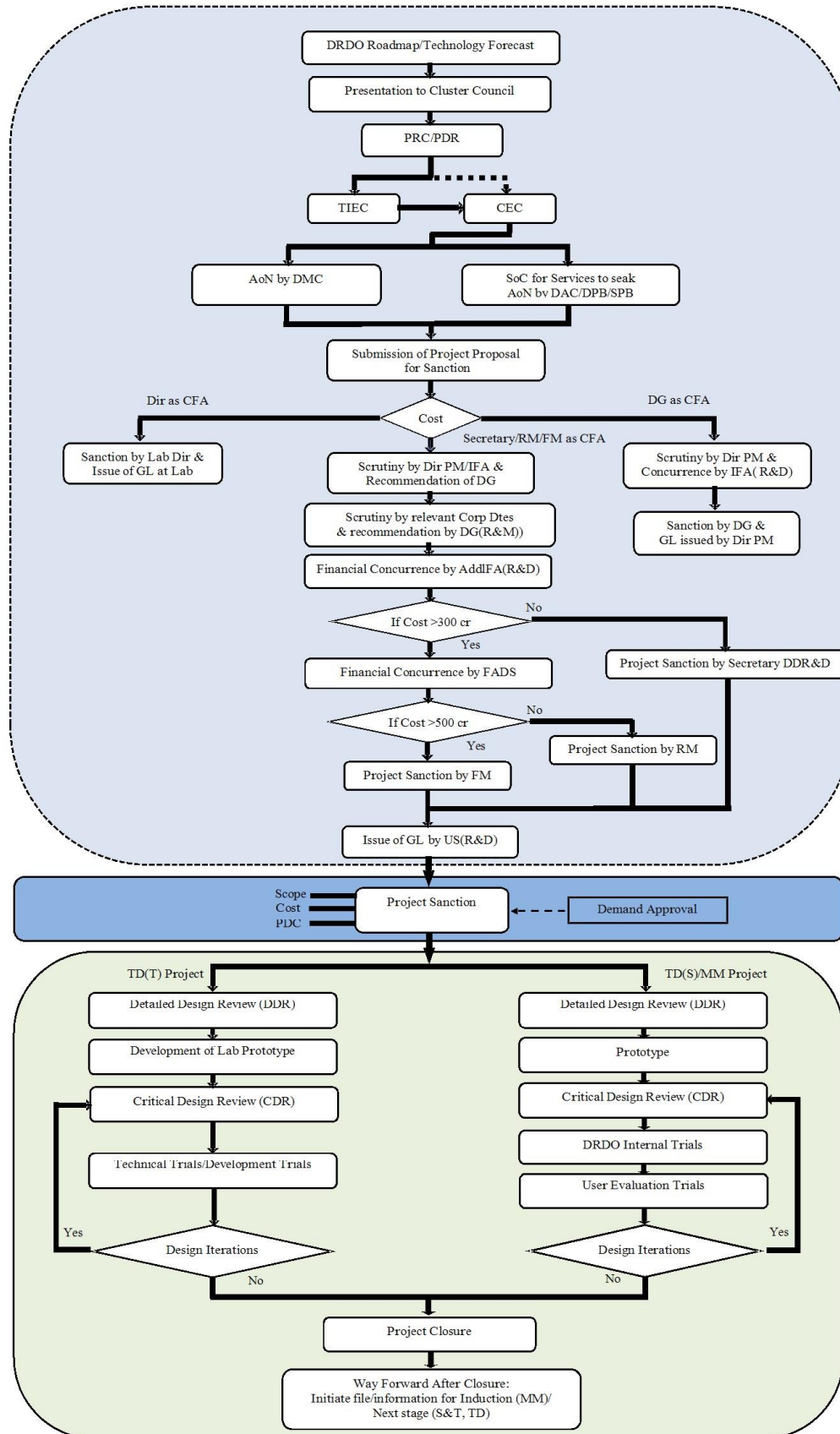
8.10.1 For **TD(S)** projects, nodal Lab should forward the proposal for AoN/Induction/Production for systems which have completed PSQR validation trials, through DG (Cluster) and DISB to SHQ. The proposal shall be moved for finalization of PSQR and AoN for TD(S) category of projects, if the same were not provided earlier. A follow-on project, if required, and the category of the follow-on project shall be decided based on agreed way ahead with the SHQ.

8.10.2 **TD (T) Projects:** If the technology is matured for Mission Mode, file should be moved to Services HQ based on the output of TD(T) to intimate about the advanced technology maturity achieved through the project and to enable Services to formulate draft QRs for developing product based on their requirement. Developed technologies be informed to other DRDO Labs/Concerned agencies for use in future projects/related activities. These technologies shall also be entered in central 'Database of Systems & Technologies'.

8.10.3 In case of **S&T projects**, probable use should be identified. A patent may be filed or paper(s) published in reputed conference(s) and journal(s) or details of the research may be communicated to all Labs through Technology Focus/DRDO Newsletter. Any technology developed in the project is to be informed to all Labs by entering the details in the central 'Database of Systems & Technologies'. Based on the outcome from the project, DRDO Labs may attempt a product or take a TD project. A next level/next generation of the S&T project may also be undertaken.

8.10.4 **IF Projects** should be put to maximum utilization for in-house requirements within DRDO. The facilities may also be extended for utilization by other Government & private agencies and industry, through provisions under Deposit Works.

PROCESS FLOW DIAGRAM



FACTORS FOR DELAYS

Challenges					
Sl. No.	Factors	Sl. No.	Factors	Sl. No.	Factors
1	Improper Project Planning	11	Limited upskilling programmes	21	Gaps in assessment of scope
2	Incorrect Time Estimates due to Wrong Assumptions	12	Delays in Obtaining Clearance from Quality/ Certification Agencies	22	Lack of accountability
3	Unforeseen technical challenges in Design (First Time Development)	13	Inadequate Manpower	23	Delayed decision making
4	Gaps in risk identification	14	Gaps in workforce utilization	24	Non-adherence to sanctioned budget
5	Lack of easy access to basic technologies	15	Delays due to Design / Algorithm Improvements for Performance / Safety	25	Non-adherence to project extension & reallocation guidelines
6	Delays in Procurement	16	Delay due to sub-Project/ CARS	26	Inadequate monitoring mechanism
7	Delay in completion of Civil Works	17	Scope Change (Design related & other reasons)	27	Limited knowledge transfer
8	Lack of framework for prioritizing projects	18	Design changes due to failure (Quality/ design flaws/ Workmanship/ etc.)	28	Unable to attract and retain talent
9	Non-adherence to procurement process	19	Failure of components	29	DRDO Internal Policies
10	Unclear roles & responsibilities	20	Limited documentation of changes	30	Suboptimal utilization of PMIS

External Factors			
Sl. No.	Factors	Sl. No.	Factors
1	Delay due to placement of order by Services	8	Delay in ToT
2	Non – availability of Components/ Sub Systems due to Export Control Regimes	9	Technology Obsolescence
3	Lack of industry partner Ecosystem	10	Delay in approval/ permission by concerned govt. authority

4	Gaps in Vendor capability and capacity	11	Gaps in contracting
5	Non-availability of Test Facilities	12	Delay due to Govt. Policies (GTE)
6	Delays in Land Acquisition / Clearances from constructions (especially for Infrastructure Project)	13	FMC: Force Majeure Clause
7	Delay by Industry/ DCP/PA in manufacturing or supply chain	14	FMC: Unforeseen Pandemic restrictions

User Factors			
Sl. No.	Factors	Sl. No.	Factors
1	Limited collaboration with User	4	Change in QRs
2	Misaligned success criteria & functional requirements	5	Scope Change
3	Non – availability of Platform	6	Change in stakeholder/users

GENERAL ADVISORIES

- **New Projects will be sanctioned to Lab based on performance during Lab Review.**
- **While preparing draft proposal/SoC for TD(S) projects, life expectancy for the system for which it is being developed should be considered so that developed product is relevant to the Services by the time it is ready for user trials.**
- **Labs should not work to develop technologies merely for the societal benefits instead work on its mandate. Technologies for societal applications can come as spin-off benefits.**
- **Labs shall allocate more resources to the futuristic or disruptive technologies. Balance shall be maintained between conventional domains and space, cyber & other futuristic technologies in prioritizing the projects.**
- **Reconciliation of Project/Programme data (Name, Type, Cost & PDC - ongoing projects till administrative closure) to be done mandatorily by Director PM O/o DG (Clusters) every month. Any mismatch with PMIS database may be intimated/corrected at the cluster level to comply with Audit authority directives. 'Sub-projects' are not to be reported as 'projects'.**
- **Any deviation from the suggested composition of Committees should be vetted by Approving Authority of the Committee based on the justification provided.**
- **Each project may engage a Project Manager who may be a Senior Technical Officer or a retired Scientist or a Consultant with experience in Project Management processes.**
- **The project files should be prepared at Lab by project team containing all reports duly signed by all concerned in bound form, detailing page numbers in a particular report. Separate file should not be created at DG (Cluster) office at DRDO HQ.**

- While sending any proposal to DRDO HQ for approval of CFA, viz. PDC Extension, Revision in Project Cost/Scope or Administrative Project Closure etc. all volumes of project file, with volume number, date of opening etc. mentioned on each file, need to be sent to HQ instead of a single part case.
- Labs are advised to follow ‘DRDO Software Asset Management Policy’ based on scope and its applicability issued by DQR&S.
- Labs are advised to follow ‘Artificial Intelligence Framework and Guidelines for Projects and Programmes’ based on scope and its applicability issued by CAIR.
- Labs should seek information on foreign funded projects (ongoing and completed) by Principal Investigator(s) of academic institutes with DRDO.
- Mission mode projects should have an element from DIPAS for Human Factor Engineering (HFE) and Human Reliability Analysis (HRA).
- Additional regular manpower in DRDS/DRTC cadre may be sought in programmes being sanctioned by CCS while preparing Draft Cabinet Note (DCN).
- Project proposals and Project Reports should be free from any plagiarised and AI generated content.
- Officials working in projects will adhere to directives promulgated in Official Secret Act while interacting in public and digital platforms.
- During review of projects, non-performance of Project Director will lead to redesignation of DPD as PD. Continuation of Lab Director may also be discussed during Lab Review. These reviews may be held on the directions of highest review committee or Committee of DGs. Based on the recommendation of Lab review, continuation of Project Director or Lab Director will be decided.

- **Timely achievement of Milestones in a project will be linked with performance appraisal of the team members including support groups. Detailed guidelines will be issued separately.**

REFERRED & RELATED DOCUMENTS

S No	Name of document	Type	Publisher	Year
1.	Business Rules: (i) Allocation of Business Rules – 1961 (ii) Transaction of Business Rules, 1961	Business Rules	Cabinet Secretariat	2025
2	Delegation of Financial Powers	Schedule	DF&MM, DRDO HQ	2019
3	DRDO Procurement Manual	Procedure	DF&MM, DRDO HQ	2025
4	Defence Acquisition Procedure (DAP)	Procedure	Ministry of Defence	2020
5	Stores Management Guidelines	Guidelines	DF&MM, DRDO HQ	2023
6	System Analysis Guidelines	Guidelines	DFTM, DRDO HQ	2021
7	Artificial Intelligence Framework & Guidelines	Framework & Guidelines	DFTM, DRDO HQ	2021
8	System Requirement Specifications Template	Template	DQR&S, DRDO HQ	2018
9	DRDO Software IV & V Policy	Policy	DQR&S, DRDO HQ	2022
10	DRDO Guidelines for Software Development (DGSD)	Guidelines	DQR&S, DRDO HQ	2023
11	Human factor Integration & Human Reliability Analysis Guidelines	Guidelines	DQR&S, DRDO HQ	2022
12	DRDO Quality & Reliability Policy Guidelines (Q & R Policy)	Policy Guidelines	DQR&S, DRDO HQ	2024
13	Policy Guidelines for AHSP Transfer of DRDO Systems to Government QA Organizations/Users	Policy Guidelines	DQR&S, DRDO HQ	2024

14	Evaluating Trustworthy AI Framework & Guidelines (ETAI)	Framework & Guidelines	SAG	2024
15	Guidelines for Selection of DcPP & Nomination of PA for DRDO Designed & Developed Systems	Guidelines	DISB, DRDO HQ	2019
16	SOP: PSQR Validation Trials of Equipment for The Indian Army	SOP	DISB, DRDO HQ	2022
17	SOP for Processing Statement of Case (Soc) Before Fielding in DPB Meeting	SOP	DISB, DRDO HQ	2024
18	SOP for Conduct of JPMT	SOP	DISB, DRDO HQ	2024
19	Cost vetting of system for AON	SOP	DISB, DRDO HQ	2024
20	SOP on Conduct of DRDO Internal Trials	SOP	DISB, DRDO HQ	2024
21	DRDO Policy & Procedures for Transfer of Technology	Policy	DIITM, DRDO HQ	2019
22	DRDO Guidelines for Production Coordination	Guidelines	DIITM, DRDO HQ	2023
23	DRDO Policy for Deposit Work	Policy	DF&MM, DRDO HQ	2024
24	SOP on DRDO Funding for Extramural Research, Research Boards & Centre's of Excellence	SOP	ER & IPR, DRDO HQ	2018
25	DRDO SOP for Exports of Military Equipment	SOP	DIITM, DRDO HQ	2019
26	Mechanism for Induction of DRDO Developed Products in CAPFs and Placement of Order by User CAPFs with DcPP/PA/ToT Partner	Mechanism	DLIC, DRDO HQ	2023
27	Guidelines for Cyber Secure Embedded Systems	Guidelines	DFTM, DRDO HQ	2024

28	DRDO Policy for Engagement of NRIs, OCIs & Foreign Nationals in R&D Activities	Policy	DFTM, DRDO HQ	2024
29	SOP for Implementing TDF Scheme	SOP	DTDF, DRDO HQ	-
30	Writing and Assigning Uniform Numbers to Technical Reports in DRDO	Policy / Guidelines	DESIDOC, Delhi	-

Note: - Current versions of referred & related documents as listed above may be used.

ABBREVIATIONS**LAB/CENTRE NAME
FOR PROJECT NO. ALLOTMENT**

(Full abbreviation of Lab/ Centre name will be used for allotment of Project No.)

ADE	Aeronautical Development Establishment
ADRDE	Aerial Delivery Research and Development Establishment
ARDE	Armament Research & Development Establishment
ASL	Advanced Systems Laboratory
CABS	Centre for Air Borne System
CAIR	Centre for Artificial Intelligence & Robotics
CFEES	Centre for Fire, Explosive and Environment Safety
CHESS	Center for High Energy Systems and Sciences
CVRDE	Combat Vehicles Research & Development Establishment
DEAL	Defence Electronics Application Laboratory
DEBEL	Defence Bioengineering and Electro-medical Laboratory
DGRE	Defence Geo-Informatics Laboratory
DIBT	Defence Institute of Biodefence Technologies
DIHAR	Defence Institute of High-Altitude Research
DIPAS	Defence Institute of Physiology & Allied Science
DIPR	Defence Institute of Psychological Research
DLJ	Defence Laboratory Jodhpur
DLRL	Defence Electronics Research Laboratory
DMRL	Defence Metallurgical Research Laboratory

DMSRDE	Defence Materials & Stores Res. & Dev. Establishment
DRDE	Defence Research & Development Establishment
DRDL	Defence Research & Development Laboratory
DRL	Defence Research Laboratory
GTRE	Gas Turbine Research Establishment
HEMRL	High Energy Materials Research Laboratory
INMAS	Institute of Nuclear Medicine & Allied Sciences
IRDE	Instruments Research & Development Establishment
ITM	Institute of Technology Management
ITR	Integrated Test Range
JCB	Joint Cipher Bureau
LRDE	Electronics & Radar Development Establishment
NMRL	Naval Materials Research Laboratory
NPOL	Naval Physical Oceanographic Laboratory
NSTL	Naval Science & Technological Laboratory
PXE	Proof & Experimental Establishment
R&DE(E)	Research & Development Establishment (Engineers)
RCI	Research Centre Imarat
SAG	Scientific Analysis Group
SSPL	Solid State Physics Laboratory
TBRL	Terminal Ballistics Research Laboratory
VRDE	Vehicles Research and Development Establishment

Note: In case of any ambiguity arising due to similar or same abbreviation for two or more Labs, DP&C will clarify and assign proper abbreviation to each Lab for allotment of Project No.

ABBREVIATIONS GENERAL TERMINOLOGY

ACAS	Assistant Chief of Air Staff
ACNS	Assistant Chief of Naval Staff
ACoAS	Assistant Chief of Army Staff
AHSP	Authorised Holder of Sealed Particulars
AO (R&D)	Accounts Office (R&D)
AoN	Acceptance of Necessity
ASQR	Air Staff Qualitative Requirement
ATP	Acceptance Test Procedure
BE	Budget Estimate
BTS	Build to Specification
CAPSI	Contract for Acquisition of Professional Services
CARS	Contract for Acquiring Research Services
CCM	Cluster Council Meeting
CCS	Cabinet Committee on Security
CDA	Controller of Defence Accounts
CDR	Critical Design Review
CEC	Cost Estimation Committee
CFA	Competent Financial Authority
CGDA	Controller General of Defence Accounts
CISC	Chief of Integrated Defence Staff
CPM	Critical Path Method
DAE	Department of Atomic Energy
DAP	Defence Acquisition Procedure

DcPP	Development cum Production Partner
DDR&D	Department of Defence Research & Development
DDR	Detailed Design Review
DFMM	Dte. of Finance and Material Management
DFP	Delegation of Financial Powers
DFTM	Directorate of Futuristic Technology Management
DG	Director General
DG (PC&SI)	Director General (Production Coordination & Service Interaction)
DG(R&M)	Director General (Resource and Management)
DGL	Draft Govt. Letter
DHRD	Directorate of Human Resource Development
DIITM	Directorate of Industry Interface & Technology Management
DISB	Directorate of Interaction with Services for Business
DMC	Deptt. of Defence R&D Management Council
DoS	Date of Sanction
DP	Development Partner
DP&C	Directorate of Planning & Coordination
DPFM	Directives for Project Formulation and Management
DPP	Draft Project Proposal
DPSU	Defence Public Sector Undertaking
DQRS	Directorate of Quality, Reliability & Safety
D-RdMAP	Roadmap of DRDO
DRDO	Defence Research & Development Organisation
DRDS	Defence Research Development Service

DRONA	DRDO Rapid Online Network Access
DRTC	Defence Research Technical Cadre
DST	Department of Science & Technology
EB	Executive Board
EDC	Expected Date of Completion
FA(DS)	Financial Advisor (Defence Services)
FE	Foreign Exchange
FERV	Foreign Exchange Rate Variation
FIM	Free Issue Material
FM	Finance Minister
FMEA	Failure Mode Effect Analysis
FMECA	Failure Mode Effect Critical Analysis
FOL	Fuel, Oil & Lubricants
FY	Financial Year
GoI	Govt. of India
GSQR	General Staff Qualitative Requirement
HQ	Head Quarters
HR	Human Resource
IC	International Cooperation
ICDP	Integrated Capability Development Plan
IDDM	Indigenous Design Development and Manufacturing
IDS	Integrated Defence Staff
IDST	Institute of Defence Scientists and Technologists
IF	Infrastructure & Facilities

IGA	Inter Government Agreement
ISRO	Indian Space Research Organisation
IV&V	Independent Verification and Validation
JPMT	Joint Project Monitoring Team
JSQR	Joint Staff Qualitative Requirement
LAO	Local Audit Officer
LATOT	License Agreement for Transfer of Technology
LP	Local Purchase
LPP	Last Purchase Price
MM	Mission Mode
MoD	Ministry of Defence
MoM	Minutes of Meeting
MoU	Memorandum of Undertaking
MPP	Manufacturing Process Plan
MSQAA	Missile System Quality Assurance Agency
NSQR	Naval Staff Qualitative Requirements
ORSA	Operation Research & System Analysis
PA	Production Agency
PBS	Product Breakdown Structure
PCDA	Principal Controller of Defence Accounts
PDC	Probable Date of Completion
PDD	Preliminary Design Document
PDR	Preliminary Design Review
PEP	Project Execution Plan

PERT	Project Evaluation and Review Technique
PJMB	Project Management Board
PM	Project Monitoring
PMB	Programme Management Board
PMIS	Project Management Information System
PMRB	Project Monitoring and Review Board
PMT	Project Monitoring Team
PRC	Peer Review Committee
PRI	Project Readiness Index
PSQR	Preliminary Staff Qualitative Requirements
PSU	Public Sector Undertaking
QA	Quality Assurance
QAP	Quality Assurance Plan
QR	Qualitative Requirement
QR&S	Quality Reliability & Safety
R&D	Research & Development
R&QA	Reliability & Quality Assurance
RDR	Revenue Debt Remittance
RE	Revised Estimate
RFP	Request For Proposal
RM	Raksha Mantri
S&T	Science & Technology
SDD	System Definition Document
SDLC	Software Development Life Cycle

SHQ	Services Head Quarters
SoC	Statement of Case
SOP	Standard Operating Procedure
SoW	Scope of Work
SWOT	Strength Weakness Opportunity & Threat
TD	Technology Demonstration
TD (S)	Technology Demonstration (System)
TD (T)	Technology Demonstration (Technology)
TEC	Technical Evaluation Committee
ToT	Transfer of Technology
TRA	Technology Readiness Assessment
TRL	Technology Readiness Level
UATT	User Assisted Technical Trials
US(R&D)	Under Secretary (Research & Development)
USC	Unique Sanction Code
UT	User Trials
WBS	Work Breakdown Structure

**HANDBOOK OF
FORMATS & CHECKLISTS OF
DIRECTIVES FOR PROJECT FORMULATION AND
MANAGEMENT IN DRDO
(DPFM 2026)
(PART-2)**



March 2026

**Directorate of Planning & Coordination
Defence Research and Development Organisation
New Delhi-110011**

INTRODUCTION

The ‘Handbook of Formats & Checklists to DPFM-2026’ contains Formats, Checklists & Abbreviations which have to be referred in conjunction with the ‘Directives for Project Formulation and Management in DRDO (DPFM-2026)’.

Contents are denoted as DRDO.DPFM.FT.SN for Formats; DRDO.DPFM.CL.SN for Checklists and DRDO.DPFM.AL.SN for Abbreviation Lists. ‘SN’ stands for corresponding serial number.

Contents of this segment of document are indicative and amendments, if required shall be promulgated by Directorate of Planning and Coordination (DP&C) with the approval of DG (R&M) based on interpretations, amendments and valuable inputs from the stakeholders.

In case of any conflict between these Formats and Checklists (as amended) and the Directives (as amended), the said provisions of the Directives will prevail.

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FORMATS

FORMAT FOR COST ESTIMATION MODULE

Lab Name:

Project Name:

Cost ₹ in Cr / Lakh

Major Head	Sub Head	Items Des/ Service	LPP/SO/ BE/OWN COST/ POV	Qty/ Service Details	Cost Per Unit/ Service rate	Currency Rate	Year of Ref	Year of Realisation	Total Without Taxes	Cost Esc % Amount (₹ in lakhs)	Import Duty % Amount (₹ in lakhs)	GST % Amount (₹ in lakhs)	Total Cost (₹ in lakhs)	Copy of Enclosure

Total Capital..... ₹ in Cr (FE_____)

Total Revenue ₹ in Cr (FE_____)

Grand Total.....₹ in Cr (FE_____)

Note: For details please refer Cost Estimation Module (CEM) on DRONA Services

FORMAT FOR TECHNOLOGY READINESS ASSESSMENT (TRA) AND PROJECT READINESS INDEX (PRI)

Activity/Parameters	Verifying Authority Remarks
System/Work breakdown chart with levels submitted	Yes/No
All sub-systems, sub-sub-systems etc. TRLs indicated	Yes/No
Supporting documents justifying current TRL	Yes/No
TRA/PRI roadmap provided for Project duration	Yes/No
Initial PRI at project sanction stage	
Expected Final PRI at project completion stage	
Suggestive project duration (Realistic & Pessimistic)	
Suggested category of Project (as per Table 2)	

Prepared

(Project Director)

Reviewed & Recommended

(Chairman, Lab Technology Council)

Date: _____

Place: _____

DRDO.DPFM.FT.03A

DMC PRESENTATION FORMAT (NEW PROJECT)

Presentation Format for seeking Acceptance of Necessity (AoN) for Project Proposals by DMC is given below:

Format 'FT.03A' for New Project Proposal (≤10 slides) for Accord of AoN**1. Health of Lab:****a) Details of projects completed in last 10 years**

S.No.	Name of Project	Date of Sanction	Category of Project	Initial PDC	Final PDC	Cost of Project	Expenditure	Is PDC extended (Yes/No)

b) No. of projects pending for technical or administrative closure by the Lab

S.No.	Name of Project	Date of Sanction	Category of Project	Initial PDC	Final PDC	No. of PDC Extn.	Date of Technical Completion (if done)

2. Brief/Overview of proposed Project

- Title of the project
- Cost (inclusive of all taxes) ₹ in Cr.
- Duration / PDC (in months):

PDC	Realistic	Pessimistic
Estimation by TiEC Committee		
Estimation by Standard / or In-house developed Software tool.		
Finally Proposed Realistic Time (to be considered as PDC)		
PDC for Sunset Clause		

Pessimistic PDC	
------------------------	--

- Category: S&T / TD(T) / TD(S) / MM / IF
 - Security classification of project (Secret/Confidential/Restricted/ Unclassified).
3. Alignment with ‘Roadmap of DRDO: 2025-2034’: Yes/No
(If no, justification may be provided for taking up the project and to ensure its inclusion during update of D-RdMAP in the next upcoming cycle)
 4. Comparison w.r.t global scenario & contemporary systems/ technologies and time taken to develop for TD(S) & TD(T) projects. Inputs from Technology Advisor/Counsellors posted in Indian embassies in foreign countries may be sought and reported.
 5. PRC or PDR Recommendations: PRC or PDR recommendations that are not agreed to or agreed with qualification (if any) have to be highlighted during presentation to DMC.
 6. Sunset Milestone & PDC (based on this milestone a decision will be taken whether to continue with the project or cancel it).
 7. Participating Labs/agencies with work share: Summary of work share assigned to participating Labs be brought out in tabular form with funds and time required for the same.
 8. Scope of the project: Scope of the project should be clearly defined in quantifiable and measurable terms.
 9. Infrastructure planned in the project.
 10. Deliverables/Output: Likely deliverables/output be mentioned after thorough discussion/consultation with User (If user involved).
 11. TRL of various sub-systems and PRI of Project.
 12. Brief of earlier work done: Similar work undertaken in the past by the project team having direct bearing on the present project may be brought out.
 13. Technology challenges/issues foreseen: Technology challenges (grey areas) likely to crop up during execution of project be highlighted adequately and possible way out to overcome the same.
 14. Cost breakdown.
 15. Details of Project Team (Documentation/AI/Test & Trial Group etc.).
 16. Demand approval and other approvals which will be sought along with Project sanction.
 17. Plan for DcPP involvement.
 18. Details of Interaction/Recommendation of Services for MM/TD(S) projects.

19. Assurance from Lab that similar technology is not available with industry & being attempted first time. Proof/justification for the same be given in DMC.

20. Realistic PERT chart with projected six-monthly milestones.

21. Ongoing commitment of Lab in terms of number of ongoing projects and availability of manpower along with the detail of documentation group manpower.

S.No.	Name of Project	Date of Sanction	Category of Project	Initial PDC	Final PDC	Cost of Project	Expenditure	Manpower Details

22. Recommendation / Approval sought from DMC

Note: AoN of DMC will be valid for 6, 4 & 3 months respectively for Secretary DDR&D, DG (Cluster) and Lab Director as CFA for Projects.

DMC PRESENTATION FORMAT (PDC REVISION)

1. Health of Lab:

a) Details of projects completed in last 10 years

S.No.	Name of Project	Date of Sanction	Category of Project	Initial PDC	Final PDC	Cost of Project	Expenditure	Is PDC extended (Yes/No)

b) No. of projects pending for technical or administrative closure by the Lab

S.No.	Name of Project	Date of Sanction	Category of Project	Initial PDC	Final PDC	No. of PDC Extn.	Date of Technical Completion (if done)

c) Details of ongoing projects and availability of manpower

S.No.	Name of Project	Date of Sanction	Category of Project	Initial PDC	Final PDC	Cost of Project	Expenditure	Manpower Details

2. Brief/Overview of Project

- Title of the project
- Cost (inclusive of all taxes) ₹ in Cr.
- Duration/PDC:- a) At sanction stage (in months): _____
b) Pessimistic PDC recorded during sanction (in months): _____
c) Proposed for revision
- Planned & Actual details of reviews held (Planned Vs Actual)
- Category: MM/TD(S)/TD(T)/S&T/IF
- Accountability, other than Project & Lab Director may be brought out.
- Any prior commitment by DG, Lab Director, PD or instruction of DAC/DPB/SPB or DMC about no further PDC extension.

3. Change in Scope, if any

4. Present Status of Projects with details of major sub-systems, reasons for delay, achievement till date/infrastructure established.
5. Bring out necessity of the project extension.
6. Additional funds required/Re-allocation of funds with justification.
7. Revised six-monthly milestones for the entire duration (including extended PDC period) with financial out lay.
8. Activities to be carried out/additional infrastructure needed to be established/revised PERT chart with milestones.
9. Name and Duration of Project Team Managers

From (Date of Sanction)	To	Name of DG	Name of Director	Name of PD	Name of DPD

10. Composition of committee and its recommendations for extension in case of MM projects.
11. Date and number of EB when the delay was first reported.
12. History of PDC revision along with DMC directive during each previous PDC revision to be brought in **present format**.
13. Decision required from DMC: PDC Revision sought (in Months)

Note: Minutes of DMC (Recommendations/Directions) shall be forwarded to the DG (Cluster). Director PM, O/o DG (Cluster) shall forward the relevant extract to the nodal Lab. Extract of DMC minutes, quoting DMC minutes No. & date should be placed in case file while forwarding the project proposal for sanction by CFA. DMC minutes should be referred in noting sheet of case file and directions/recommendations of DMC must be addressed adequately. Director (PM) of the Cluster will ensure the compliance of DMC directives before forwarding the case to CFA for sanction.

STATEMENT OF CASE FOR SANCTION OF PROJECT

1. **Name of Lab:** _____
2. **Title of the Project:** _____
3. **Category of Project** (Please tick in appropriate box):
 - ☐ Mission Mode (MM)
 - ☐ Technology Demonstration- Systems {TD(S)}
 - ☐ Technology Demonstration- Technology {TD(T)}
 - ☐ Science & Technology (S&T)
 - ☐ Infrastructure & Facilities (IF)
4. **Whether DMC has earmarked project as High Risk*:** (Yes/No)

** At any given time, total 'High-Risk' projects will not be more than 15% of total ongoing projects.*
5. **Security classification of Project/Programme**
 - ☐ Secret
 - ☐ Confidential
 - ☐ Restricted
 - ☐ Unclassified
6. **Details of Qualitative Requirements** (if any): _____ *(for TD(S) Projects)*
DAC/DPB/SPB Reference No. & Date: _____ *(for MM Projects)*
7. **Trial Directive Reference No.** *(if available):* _____
8. **Cost** (₹ in Cr): (inclusive of all taxes) _____ **Total** (FE)
(cost for High Risk Project with 20% escalation)
9. **PDC approved by DMC** (in months): _____

PDC	Realistic	Pessimistic
Estimation by TiEC Committee		
Estimation by Standard/or In- house developed Software tool.		
Proposed Realistic Time (PDC) by Lab		
Approved PDC/Remarks by DMC		

10. **Alignment with the ‘Roadmap of DRDO: 2025-34’?** Yes / No
- a) **If yes, please give Reference** _____
(Please quote Page No., Year of ‘Roadmap of DRDO document’)
- b) **If no, please give justification for undertaking the project.**
11. **Percentage of Indigenous Content (IC) of prototype/deliverables {MANDATORY for MM and TD(S) Category of projects}:** _____
- a) Dependency on the Foreign Countries (Countries, Items/Components etc.)
- b) In case mandatory criteria of 50% of IC as per IDDM Category not meeting, description of (roadmap/way forward) indigenisation efforts.
- c) By number of sub-systems (as per WBS Level-1)
12. **International Collaborations envisaged** (provide details and duration)

13. **Potential Intellectual Property Rights to be generated** (provide details of Patents, Designs and Copyrights)

14. **Export Potential, if any** _____
15. **Is ORSA Study Report (ISSA) enclosed?** Yes / No

16. **Recommendation of Cluster Council?** Yes / No
(please attach the Minutes)
17. **Recommendation of Peer Review Committee [for TD(T) & S&T]** Yes / No
(please attach the Minutes)
18. **Recommendation of PDR Committee [for MM & TD(S) Projects]** Yes / No
(please attach the Minutes)
19. **Recommendation of DMC-AoN** Yes / No
(please attach the minutes)
20. **TRL Analysis Report enclosed?** Yes / No
(Not mandatory for IF Category)
21. **Infrastructure Review Committee (IRC) Report enclosed?** Yes / No
(Mandatory only for IF Category)
22. **Project Execution Plan attached?** Yes / No
23. **Project Deliverables/Output:** () Prototype
(Please tick in appropriate box) () Technology
() Assembly or sub-assembly
() Process
() Others
24. **Name of the Programme Director, Project Director, Dy. Project Director (for approval of Competent Authority) :** _____

[Lab to issue the letter as per Format DRDO.DPFM.FT.08, after approval of Competent Authority]

Macro and Micro Details of Project / Programme

CONTENTS

Part-I (Macro Details)

Serial Number	Description	Page No.
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Part-II (Micro Details)

Serial Number	Description	Page No.
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Part – I

Macro Details of Project

1. a. **Title of the Project** _____
- b. **Short Name or Acronym** _____

2. **Scope**
 - a) **Objective**
 - b) **Proposed project deliverables:**
 - (i) No. of prototypes for testing _____
 - (ii) No. of (type approved/qualified) deliverables _____

3. **Is it a Multi-Lab Project?** Yes / No
 (If yes, please specify the participating Labs)

4. **Specify the User(s)** : (Army/Air Force/Navy/
Inter-Services/DRDO)
 (Please attach the GSQR/NSQR/ASQR/SQR
for MM/TD(S) Projects)

5. Specify the proposed DcPP or selection methodology

6. Breakup of Cost (₹ in Cr.):

(a) For MM, TD(S), TD(T) & IF projects:

(₹ in Cr.)

Minor Head	Major Head 4076 – Capital Sub Major Head – 05	Nodal Lab	Participating Lab, if any	Total (FE)
	Heads of Expenditure	Total (FE)	Total (FE)	
052 (Code Head - 929/25*)	Transportation (Movement of Stores)			
	Equipment/Stores			
	CARS/CAPSI			
	Consultancy Contracts			
	Job Work/Contracts/Hiring of Technical Services/Work Package			
	Hiring of Transport			
	FOL for Project Vehicles			
	Contingency & Miscellaneous			
	Plant & Machinery			
	Project related Vehicles			
111	Works			
TOTAL				

* Total project expenditure related to MM, TD(S), TD(T) and IF projects including equipment, hardware, consultancy, project related contingency, purchase/hiring of transport, freight, contracts for “Acquisition of Research Services (CARS)” under the project etc. will be compiled to this head.

(b) For S&T projects :

(₹ in Cr.)

Minor Head	Major Head 2080 - Revenue	Nodal Lab	Participating Lab, if any	Total (FE)
	Heads of Expenditure	Total (FE)	Total (FE)	
105	Transportation (Movement of Stores)			
110 (Code Head-856/01)**	Equipment/Stores			
	CARS/ CAPSI			
	Consultancy Contracts			
	Job Work/Contracts/Hiring of Technical Services/Work Package			
	Hiring of Transport			
	FOL for Project Vehicles			
	Contingency & Miscellaneous			
111	Works			
TOTAL (REVENUE)				
	Major Head 4076 - Capital			
052 (Code Head-929/24)	Plant & Machinery			
	Project related Vehicles			
	Works			
TOTAL (CAPITAL)				
GRAND TOTAL(REVENUE & CAPITAL)				

**Booking of Expenditure for Science & Technology (S&T) projects will be as per PM-2025 (Para 2.5.2) as amended from time to time.

Note:

- (i) In case, project is jointly funded by DRDO, Services, PSUs or any other agencies, details of funds along with relevant budget heads of Services, PSUs or any other agencies be given in tabular form and grand total be mentioned accordingly.

(ii) The details of the equipment, machinery and store costing more than ₹50 lakh or 10% of project cost- whichever is less to be provided as Annexure, giving cost, FE (if any), and the month/year of procurement during implementation of the project.

(iii) Project sanction file will not move for vehicle sanction to DHRD. However, the case will be processed separately only after project approval is accorded.

7. Additional requirement of mechanical transport vehicles (type of vehicle to be mentioned) specific to the project, for equipment/developed systems and stores (with justifications). While proposing additional requirement of mechanical transport vehicles in the project the procedures/SOP for purchase/hiring of vehicles in projects issued by DHRD may be adhered.

8. Proposed Six monthly milestones along with financial outlay (₹ in Cr):

S. No.	Time (Months)	Six Monthly Technical Milestone	Financial Outlay (₹ in Cr)
		sunset milestone(s)	
Total			

9. Procurement Plan

An overall procurement plan should be prepared adequately linked with Work Breakdown Structure, detailing all items / services products, components to be procured with their estimated timelines for different procedural actions, cost and source of supply.

Sl. No	Name of the item/ Service	Brief description / Purpose	Source of supply	Mode of tendering	Expected cost	Expected Date of Tendering	Expected Date of Delivery

10. Details of Demand approvals required

- Demand approval can be sought along with the processing for project sanction as per Procurement Manual-2025 as amended from time to time.
- Labs may initiate procurement actions for projects being submitted for approvals/PDC extension/cost enhancement as per Procurement Manual-2025 as amended from time to time.
- Demand approval along with project sanction is recommended in MM projects as part of the main sanction file. For other categories of projects, the file for Demand approval may move in parallel with the project sanction file and demand can be approved once project is sanctioned.

11. List of enclosures: CCM minutes, PRC, PDR, TiEC Report, CEC Report, DMC AoN, PERT Chart, Bio-Data of PgD, PD, Dy. PD etc.

Part – II

Micro Details of Project

1. **Brief technical appreciation** (To include scope, relevance of the project to Lab Charter and Mission in maximum 1 or 2 pages only).
 - ❖ Justification (need) for undertaking the project along with the recommendation of the Cluster Council and DMC.
 - ❖ What will be achieved by taking this project?
 - ❖ Competence level and preliminary work done to acquire the same.
 - ❖ DRDO will work only on advanced and critical technologies where industry does not have the capability. Where industry has the capability, such projects will not be undertaken by the Labs. All the systems/components which can be done by industry will be procured/taken from industry on bought out/BTS.
 - ❖ Brief of TRL analysis. Initial TRL indices at the time of undertaking the project and likely TRL levels at the time of closure of the project as accepted by PRC/PDR.
 - ❖ Peer Review Committee [For TD(T) & S&T project]/Infrastructure Review Committee (for IF project) recommendations (please enclose the copy of approved Minutes of the Peer Review Committee/Infrastructure Review Committee meeting).
 - ❖ For MM and TD(S) project, PDR should be conducted in lieu of PRC. Enclose the copy of PDR and major recommendations in case of MM and TD(S) projects.
 - ❖ Action Plan for prototype development.
 - ❖ The proposal should contain the number of design iterations required at the component/sub-system level in a TD(T) project and number of design iterations required at the overall system/project level in TD(S) and MM projects. Time penalty for each design iteration should be given.
 - ❖ Realisation Plan
 - ❖ Testing Plan – Should include details on tests required, tests equipment/facilities, tests methodology to meet the performance as envisaged in the project. Details of test infrastructure, its location and availability be clearly brought out. In case of test facility in other Lab/Institute/Country, a letter be attached from the Lab/Institute that it has been made aware of the requirement and probable time.
 - ❖ Six monthly milestones linking with financial outlay.
 - ❖ Critical factors/technology involved.

- ❖ Development risk areas and remedial actions proposed.
- ❖ Responsibility and Accountability Matrix (RACI Matrix)
- ❖ Development cum Production Partners (DcPP)
- ❖ Costs benefit analysis/spin-off benefits.
- ❖ Project management and monitoring structure proposed.
- ❖ PERT Chart.
- ❖ Time Estimation Module outputs and findings

2. If the project is to be executed by multiple Labs, please outline the agreed work-packages among the various Labs. A letter from participating Lab may be attached to indicate that participating Lab is aware and **willing to take up** the activity assigned as part of the main project.

S. No.	Title of Sub-Project	Scope of Work	Lab	Cost (₹ in Cr)	PDC (in months)

3. List of major additional facilities (Capital) required for the project

(Please highlight status of similar facilities existing within the Lab as well as other DRDO Labs and why they cannot be utilized)

- (a) _____
- (b) _____

4. Major technical trainings/skills identified for Project Manpower (specific for executing the project)

S. No.	Discipline/area for training	Agency Proposed	No. of Personnel proposed to be trained	Duration	Funding Agency (DHRD/Lab)	Remarks

Note: Correspondence established with the agencies contacted for imparting the training and their willingness should be enclosed.

5. Details of Work Packages

S. No.	Name of Govt agencies	Work Package	Scope of Work	Cost (₹ in Cr)	PDC (in months)

6. Details of CARS

S. No.	Name of Institute/ Agency	Name of the identified Professor	Area where R&D is required	Cost	PDC

Note: Correspondence established with academic institutes contacted for undertaking the research activity and their willingness should be enclosed.

7. Details of CAPSI**8. Details of Consultancy requirements**

S. No.	Discipline/ Area	Agency	Name of person / expert	Cost	Process that will be followed

Note: Correspondence established with the agency contacted for offering consultancy and their willingness should be enclosed.

Details of proposed project team

S. No.	Name	Designation	Role	Period of Association whether full time/ partial (project stage)	Remarks (Source- within Lab/ Outside Lab)
			PD	Full Time	

Note: Number of minimum DRDS/DRTC working on project may be mentioned in the proposal. Variation from number mentioned in the 'Roadmap of DRDO' be brought out with justification.

Project based manpower requirement

S. No.	Designation/ Rank	Discipline & Role	Number(s)*	Period of Association	Remarks (Scientific/ Technical/ Support)

* Additional manpower requirement should be based on 'Policy for Project Based Manpower' under process by DHRD.

10. Details of additional building space requirement

- Funds for construction of building should be booked under Major Head-4076 (Capital)/Sub Head-111(Works) for MM, TD(S/T) & IF Projects and under Major Head-4076 (Capital)/Sub Head-052 for S&T projects.
- Approximate Estimates (based on recommendation of 'Infrastructure Review Committee') should be obtained from concerned CCE/DCW&E and appended with the proposal.

11. Health of Lab:

A (Projects/Programmes completed in last 10 years):

S. No.	Title of Project	Category of project	Date of Sanction	Initial PDC	Final PDC	No. of PDC extensions	Final Cost (₹ in Cr)	Outcome of project

B (Ongoing Projects/Programmes):

Sl. No	Title of Project	Category of project	Date of Sanction	Initial PDC	Final PDC	No. of PDC extensions	Final Cost (₹ in Cr)	Name of Project Director	Status (On time/ will seek PDC extension)

12. It is certified that:-

- (a) SoC of the Project completely covers the Capital Acquisition Proposals of the Services for which AoN has been accorded by DAC/DPB/SPB (for MM projects only)
- (b) This technology is not available in India
- (c) Selected DcPP/Industry is not in negative list of vendors
- (d) This project cannot be done by Industry and Academia

13. Comments of Project Director with signature and date

14. Following details need to be certified by Lab Director

- (a) Manpower of Lab: DRDS - DRTC -
- (b) Availability of manpower for this project

Recommendations of Lab Director with signature and date.

Note: Lab Director will provide his recommendations on the proposal submitted by Project Director. Lab Director will also certify that manpower requirement for the proposed project as projected in Para-9 above. No further requirement shall be raised during currency of the project which attributes to project delay. Availability of building space, if required, for housing the new equipment likely to be purchased under the project will be made available from the existing building space. In case additional manpower/building space is needed, the details may be given as per S.No. 9 & 10.

15. Recommendations of DG (Cluster) with signature and date.

Note: DG (Cluster) will offer his comments/recommendations taking a holistic view on necessity, availability of technology/expertise, manpower, resource availability and overall possibility of successful execution of the Project/Programme within timelines.

CCS FORMAT

(Specimen Format of the Main Note)

SECRET

Copy No. _____

File No. _____

MINISTRY OF DEFENCE

DEPARTMENT OF DEFENCE R&D

New Delhi, the (date of the note)

NOTE FOR THE CABINET COMMITTEE ON SECURITY

Subject: _____

1. INTRODUCTION

A snapshot of the proposal in 3 or 4 sentences.

2. BACKGROUND

A brief background of the proposal to understand its genesis. This would include consideration of the matter earlier by Cabinet/Cabinet Committees/EGoM/GoM or Commissions/High Level Committees etc. if relevant and other pertinent details.

3. PROPOSAL

The proposal may be stated with clarity and precision so that there is no ambiguity in what the Ministry/Department aims to achieve by implementing the proposal. The time-lines for completion of different stages of the projects/scheme/plan etc, where relevant, need to be clearly spelt out.

4. JUSTIFICATION

Rationale of the proposal may be brought out in this part of the note. The Ministries/Departments are also required to bring out clearly in this para, while proposing a new legislation, the essentiality of the legislation being proposed and the reasons why the purpose cannot be served through an existing legislation or by framing subordinate legislation or through executive orders etc.

5. INTER-MINISTERIAL CONSULTATIONS

Details of all inter-ministerial consultations and their views/comments as elaborated in the consolidated instructions should be brought out in this section. This should also give details of the appraisal of the proposal by any appraisals bodies or financial institutions.

6. FINANCIAL IMPLICATIONS

The financial implications of the proposal may be worked out as accurately as possible and should be detailed in this section. Further, the manner in which the expenditure is proposed to be borne may also be clearly indicated.

7. EMPLOYMENT GENERATION POTENTIAL

The Employment Generation Potential of the proposal should be worked out and be reflected clearly in this Paragraph.

8. ATMANIRBHAR BHARAT

The potential of the proposal in realizing the goal of Atmanirbhar Bharat may be reflected in this paragraph.

9. TECHNOLOGY IMPLICATIONS

10. APPROVAL SOUGHT

The approval paragraph is the most crucial paragraph containing the proposal on which consideration and approval of the Cabinet/ Committee of the Cabinet is solicited. It should be a self-contained paragraph and drafted with clarity and precision leaving no scope for ambiguity or differing interpretations.

11. STATEMENT OF IMPLEMENTATION SCHEDULE (APPENDIX-I)

As per prescribed format and conforming to the approval paragraph.

12. STATEMENT OF EQUITY, INNOVATION AND PUBLIC ACCOUNTABILITY (APPENDIX-II)

As per prescribed format. Brief details of how the proposal will subserve the three criteria to be included in the main body of the note.

13. STATEMENT ON MAJOR MILESTONES AND CORRESPONDING TARGET DATES (APPENDIX-III)

In the format prescribed for the purpose (This is required only for infrastructure project related proposals).

14. APPROVAL OF THE MINISTER-IN-CHARGE

The last paragraph should indicate about the approval of the Minister-in-Charge to the proposal(s) contained in the note.

Signature_____

Name_____

(Secretary DDR&D, Ministry of Defence)

Telephone No._____

Note:-

Kindly refer Handbook on writing Cabinet Notes available at <https://cabsec.gov.in>

APPENDIX-I

SECRET

File No. -----

Ministry of Defence

Department of Defence R&D

STATEMENT OF IMPLEMENTATION SCHEDULE

Subject: _____

Gist of decision requited	Project benefits/results	Time-frame and manner of implementation/reporting to Cabinet Secretariat.

Signature_____

Name_____

(Secretary DDR&D, Ministry of Defence)

Telephone No._____

APPENDIX-II

SECRET

File No. _____

MINISTRY OF DEFENCE

DEPARTMENT OF DEFENCE R&D

STATEMENT OF EQUITY, INNOVATION AND PUBLIC ACCOUNTABILITY

Subject: _____

S. No.	The required goal	How does the proposal advance this goal?
1.	Equity or Inclusiveness	
2.	Innovation	
3.	Public Accountability	

APPENDIX-III

SECRET

File No. _____

MINISTRY OF DEFENCE

DEPARTMENT OF DEFENCE R&D

STATEMENT ON MAJOR MILESTONES AND TARGET DATES

Subject: _____

S. No.	Major milestones	Time frame for completion/target date	Financial Outflow (₹ in Cr)
1.			
2.			
3.			

Signature_____

Name_____

Designation_____

(Secretary DDR&D, Ministry of Defence)

Telephone No._____

Annexure of CCS format

SPECIMEN FORMAT FOR THE PRESS BRIEF

Ministry of Defence

Date of decision

Subject of the Note for the Cabinet/Cabinet Committees

1. Decision
2. Point-wise details
3. Background
4. Implementation strategy and targets
5. Major impact
6. Expenditure involved
7. No. of beneficiaries
8. States/districts covered
9. Details and progress of scheme if already running

SPECIMEN FORMAT FOR PROJECT SANCTION LETTER

No. _____/_____/D(R&D)/20XX

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग

नई दिल्ली-110011

दिनांक: _____, 202X

सेवा में,

अध्यक्ष,

रक्षा अनुसंधान एवं विकास संगठन,
रक्षा मंत्रालय, डीआरडीओ भवन,
नई दिल्ली-110011

*(For projects, where CFA is Secretary DDR&D and above addressee shall be Chairman DRDO;
For projects, where CFA is DG (Cluster) addressee shall be concerned Lab Director;
For projects, where CFA is Lab Director addressee shall be Project Director)*

विषय: Sanction of Project " _____ "

महोदय,

I am hereby directed to convey the sanction of the President of India for undertaking the Project as per following details.

1. Title of the project : _____
2. Nodal Lab : _____
3. Other Participating Labs, if any : i) _____
(with name & title of sub-project) ii) _____
4. Project No. : _____
(i) For Nodal Lab: _____
(ii) For Participating Lab 1: _____
(iii) For Participating Lab 2: _____ (if more than 2, add accordingly)
5. Project Linkage with DRDO Roadmap/
AoN by DAC/DPB/SPB or Apex Council of DRDO-MHA: _____
6. **Whether DMC has earmarked project as High Risk:** (Yes/No)

7. Total estimated cost : Total _____ ₹ in Cr (FE: _____ ₹ in Cr)
(Inclusive of all taxes)
(Rupees in words)

Break-up of Share in case of Jointly Funded Projects

8. Start Date : 01st Jan/ 01st Apr/ 01st Jul / 01st Oct of the year: YYYY
(Please select 1st date of upcoming Quarter)
9. PDC (Months & Date) : _____
10. Scope
- a) Objective : _____
- b) Deliverables : _____
11. Details of funds (₹ in Cr) : Break-up of Estimated Funds

- a) For MM, TD(S), TD(T) and IF projects :

(₹ in Cr)

Minor Head	Major Head 4076 – Capital Sub Major Head – 05	Nodal Lab	Participating Lab, if any	Total (FE)
	Heads of Expenditure	Total (FE)	Total (FE)	
052 (Code Head - 929/25*)	Transportation (Movement of Stores)			
	Equipment/Stores			
	CARS/CAPSI			
	Consultancy Contracts			
	Job Work/Contracts/Hiring of Technical Services/Work Package			
	Hiring of Transport			
	FOL for Project Vehicles			
	Contingency & Miscellaneous			
	Plant & Machinery			
	Project related Vehicles			
111	Works			
TOTAL				

* Total project expenditure related to MM, TD(S), TD(T) and IF projects including equipment, hardware, consultancy, project related contingency, purchase/hiring of transport, freight, contracts for “Acquisition of Research Services (CARS)” under the project etc. will be compiled to this head.

b) For S&T Projects:

(₹ in Cr)

Minor Head	Major Head 2080 - Revenue	Nodal Lab	Participating Lab, if any	Total (FE)
	Heads of Expenditure	Total (FE)	Total (FE)	
105	Transportation (Movement of Stores)			
110 (Code Head- 856/01)**	Equipment/Stores			
	CARS/ CAPSI			
	Consultancy Contracts			
	Job Work/Contracts/Hiring of Technical Services/Work Package			
	Hiring of Transport			
	FOL for Project Vehicles			
	Contingency & Miscellaneous			
111	Works			
TOTAL (REVENUE)				
	Major Head 4076 - Capital			
052 (Code Head- 929/24)	Plant & Machinery			
	Project related Vehicles			
	Works			
TOTAL (CAPITAL)				
GRAND TOTAL (REVENUE & CAPITAL)				

**Booking of Expenditure for Science & Technology (S&T) projects will be as per PM-2025 (Para 2.5.2) as amended from time to time.

Note: In case of project jointly funded by DRDO, Services, PSUs etc., details of funds of Services, PSUs etc. along with budget heads be given and grand total be mentioned accordingly.

12. Add para **If Required/Applicable**. If Demand approval is accorded along with project sanction, additional information viz. name of equipment(s), estimated cost, delivery period, probable source of supply, mode of tendering, wherever applicable be mentioned. *(Demand Approval at sanction stage will be given by concerned CFA as per DFP)*.

13. Add para **(If Required/Applicable)** The works pertaining to Minor Head 111 will be executed by CCE (R&D) concerned at a cost of _____ Cr as per RDCE WP 2024 and as amended. This letter shall be considered as admin approval for civil works under Minor Head 111 and expenditure will be debitable to Major Head 4076 Capital Sub Major Head 05, Minor Head 111 (Works) against Project No. _____ within Project PDC of ____ months from date of issue of Govt letter. The civil work will be executed in three packages as mentioned below: -

- (a) Provision of Composite Civil Work for Project _____
- (b) Provision of Principal Consultancy Services for Project _____
- (c) Provision of Project Management Consultancy (including contingency).

14. Director, Nodal Lab is authorized to allot sub-projects to other Defence R&D Labs as per DPFM-2026 after approval of concerned CFA, for development/manufacture of sub-systems / sub-assemblies required for the project within the sanctioned funds under specific budget heads and likely date of completion. *(Not applicable for CCS approved Programmes)*

15. Director(s), PD(s) of Nodal Lab and Participating Labs (if applicable) are also authorized to place supply orders for development, consultancy, or research contracts with other Govt./public/private sector organizations, academic institutions etc. (as applicable) restricting it within the total sanctioned funds (within the budget sanctioned under the relevant heads) subject to financial powers authorized vide Government of India, Min of Defence, Deptt. of Defence Research & Development, Letter No. DRDO/DFMM/PL/83226/M/01/1976/ D(R&D) dated 18th Dec 2019, as amended from time to time.

16. Procurement of stores will be made in accordance with rules and procedures for procurement of Equipment/Stores and services within the available powers and in terms of DRDO 'Procurement Manual-2025' issued vide Govt. of India, Ministry of Defence, Deptt. of Defence Research & Development, Letter No. DRDO/DFMM/85002/PM-25/A/M/I/434/D(R&D) dated 01st May 2025 as amended from time to time.

17. Authority letter for appointment of Programme Director and Project Director will be issued separately to concerned CDA (R&D)/Agency by the Competent Authority as per DPFM 2026.

18. Director (Nodal Lab) is authorized to change the FE element as given in the Govt. of India letter which can be converted to IC, subject to the overall sanctioned cost of the project not being exceeded.

19. The Project Monitoring & Review Mechanism will be followed as per DPFM-2026 and monitoring and review structure is enclosed as Annexure-'A' to this Govt. Sanction Letter.

20. Six-Monthly milestones and PERT Chart as per Statement of Case of project proposal are enclosed to this Govt. Sanction Letter as Annexure-'B' and Annexure-'C' respectively. However, any deviations w.r.t. completion date of milestones will not lead to change in the Govt. Sanction Letter, if these are achieved within the current PDC of the project.

21. Para on Special Powers (as applicable), directions and other specific issues, if any: should be mentioned (if approved on file).

22. Expenditure will be debited to the relevant Minor Heads, under Major Head 2080 “Revenue” and Major Head 4076 “Capital” (as the case may be), of Defence Services Estimates, Research & Development.

23. **Add para, If Required/Applicable:** Para on Debit of Expenditure/Flow of funds in case of Joint Projects with Services/PSUs and other agencies.

24. Unique Sanction Code (USC) : _____

(Please refer guidelines issued vide Letter No.- DBFA/FA/83301/M/01 dated 31st Mar 2014, available on DRONA portal of DF&MM, DRDO HQ)

25. This issues with the concurrence of IFA (R&D) or Ministry of Defence (Finance/R&D) (as the case may be) vide their Dy. No _____/MoD/Fin/R&D dated _____.

भवदीय,

()

अवर सचिव, भारत सरकार

Note: The circulation of the sanction letter of the project will be as per the table given below:

Ink Signed Copy	Copy to
• DG (Cluster) • Nodal Lab Director • Participating Lab(s) Director • DP&C • CDA (R&D) of concerned Lab • CDA (R&D) of participating Lab • PCDA (R&D), New Delhi • CDA (Services) – for jointly funded projects • Controller General of Defence Accounts (CGDA) • D(R&D), Ministry of Defence • File Copy	• Principal Director of Audit, Defence Services (PDADS), Pune • Addl. FA (R&D) & JS, New Delhi • IFA (Cluster) • DF&MM • DCW&E – if there is provision for Works in Rev/Cap • User (Services), if any

FORMAT FOR APPOINTMENT OF PgD, PD, DPD
(For new Programmes/Projects/Sub-Projects)

It is certified that: -

- 1) Sh./Ms./Dr. _____, _____ (Designation) may be appointed as Programme Director/Project Director/Dy. Project Director for the Programme/Project/Sub-Project _____ (Name, Project No.) sanctioned to _____ (Lab) vide Govt. of India Letter No. _____ dated _____.
- 2) Sh./Ms./Dr. _____ is presently not holding the Post of Programme Director/Project Director/Dy. Project Director of any other Programme/Project/Sub-Project.
- 3) Age of the Scientist as on date (____) is ____ (Yrs./Months), his/her remaining service period left is ____ (Yrs./Months) and PDC of the Programme / Project/ Sub-Project is upto _____ (Date).
- 4) Whether nominated Scientist has undergone mandatory 'Two-week of Project Management trainings course' organized or co-ordinated by ITM/DIAT (Yes / No).
- 5) eCOP PIN No. of the Scientist is _____.
- 6) DRONA Mail ID _____ Internet Mail ID _____.

(Signature of Lab Director)

{Signature of DG (Cluster)}

Date: _____

Place: _____

FORMAT FOR APPOINTMENT OF PgD, PD, DPD
(For already sanctioned Programmes/Projects/Sub-Projects)

It is certified that: -

- 1) Sh./Ms./Dr. _____, _____ (Designation) presently the Programme Director/Project Director/Dy. Project Director of Programme/Project/Sub-Project _____ (Name, Project No.) sanctioned to _____ (Lab) vide Govt. of India Letter No. _____ dated _____ is transferred/ moved or superannuating (*as the case may be*) on _____.
- 2) Sh./Ms./Dr. _____, _____ (Designation) may be appointed as Programme Director / Project Director/ Dy. Project Director.
- 3) Sh./Ms./Dr. _____ is presently not holding the Post of Programme Director/Project Director/Dy. Project Director of any other Programme/Project/Sub-Project.
- 4) Age of the Scientist as on date (____) is ____ (Yrs./Months), his/her remaining service period left is ____ (Yrs./Months) and PDC of the Programme/Project/Sub-Project is upto _____ (Date).
- 5) Whether nominated Scientist has undergone mandatory 'Two-week of Project Management trainings course' organized or co-ordinated by ITM/DIAT (Yes / No).
- 6) eCOP PIN No. of the Scientist is _____.
- 7) DRONA Mail ID _____ Internet Mail ID _____.

(Signature of Lab Director)

{Signature of DG (Cluster)}

Date: _____

Place: _____

**FORMAT FOR INTIMATION TO PCDA FOR APPOINTMENT OF
PROGRAMME / PROJECT DIRECTOR**

No. _____/_____/_____
Government of India
Ministry of Defence
Deptt. of Defence Research & Development
Lab (Name)

Date ____ Month, Year

To,

LAO/AO(R&D)/CDA(R&D)/PCDA(R&D)

Sub: Appointment of Programme Director (PgD) / Project Director (PD)

1. Reference
 - (a) Project Sanction Letter
 - (b) PgD/PD Appointment Letter
 - (c) Prevailing Delegation of Financial Powers as amended from time to time
2. Sh./Ms./Dr. _____ is appointed as Programme Director (PgD)
/ Project Director (PD for the following Programme / Project:

Programme / Project Name: _____
Programme / Project Number: _____
3. DRONA Mail ID _____ Internet Mail ID _____.
4. His / her signature is attested below:

Attested by Lab Director

Copy To,

**Director PM, O/o DG (Cluster)
Programme Director /Project Director,
DFMM, DRDO HQ
DP&C, DRDO HQ**

ANNUAL AUDITED STATEMENT OF EXPENDITURE

As a project progresses, expenditure record is maintained by the Lab BFA section and also by AO(R&D)/CDA/PCDA. Interim progressive expenditure figures/monthly expenditure reports for each project are being promulgated by the Labs. These progressive expenditures have to be rolled up each Financial Year (FY) and an Annual Audited Statement of Expenditure of each project under approved budget heads has to be sought by the Nodal/Participating Labs from respective AO(R&D)s/CDAs/PCDAs. The statement will be vetted by the Accounts Officer. A format is given below. The audited statement of expenditure has to be sought for each financial year right from year of sanction till formal completion. Participating Labs will provide the respective sub-project expenditure statement duly audited and vetted by respective AO(R&D)s/CDAs/PCDAs. At the project closure stage all the vetted statements should be enclosed with the Administrative Closure Report in support of total expenditure incurred under the project. This will obviate the tedious exercise of complete project expenditure audit at project closure stage.

ANNUAL AUDITED EXPENDITURE STATEMENT

(a) Programme/ Project / Sub-Project Name & No.:

(b) Date of Sanction: DD/MM/YYYY

(c) Latest PDC: DD/MM/YYYY

(d) Project Cost (Original & Revised): Rs. Total (FE)

(e) Unit Code:

(f) Programme /Project/ Sub-Project annual expenditure details:

For Programme (PG), MM, TD(S), TD(T) & IF projects

(₹ in Cr)

Minor Head	Major Head 4076-Capital	Expenditure during FY, YYYY-YY	
	Sub Major Head-05	Funds Allocated	Expenditure Incurred
	Heads of Expenditure	Total (FE)	Total (FE)
052 (Code Head-929/25)*	Transportation (Movement of Stores)		
	Equipment/Stores		
	CARS/CAPSI		
	Consultancy Contracts		
	Job Work/Contracts/ Technical Services/Work Package		

	Hiring of Transport		
	FOL for Project Vehicles		
	Contingency & Miscellaneous		
	Plant & Machinery		
	Project related Vehicles		
111	Works		
TOTAL			

* Total project expenditure related to MM, TD(S), TD(T) & IF projects including equipment, hardware, consultancy, project related contingency, purchase/hiring of transport, freight, contracts for “Acquisition of Research Services (CARS)” under the project etc. will be compiled to this head.

For S&T projects

(₹ in Cr)

Minor Head	Major Head 2080 - Revenue	Expenditure during FY, YYYY-YY	
		Funds Allocated	Expenditure Incurred
	Heads of Expenditure	Total (FE)	Total (FE)
105	Transportation (Movement of Stores)		
110 (Code Head- 856/01)**	Equipment/Stores		
	CARS/ CAPSI		
	Consultancy Contracts		
	Job Work/Contracts/ Hiring of Technical Services/Work Package		
	Hiring of Transport		
	FOL for Project Vehicles		
	Contingency & Miscellaneous		
111	Works		
TOTAL (REVENUE)			

	Major Head 4076 - Capital		
052 (Code Head- 929/24)	Plant & Machinery		
	Project related Vehicles		
	Works		
TOTAL (CAPITAL)			
GRAND TOTAL (REVENUE & CAPITAL)			

**Booking of Expenditure for Science & Technology (S&T) projects will be as per PM-2025 (Para 2.5.2) as amended from time to time.

Audited Statement of Expenditure

It is certified that an expenditure of ₹ _____ (in words) including FE ₹ _____ (in words) was incurred under the programme/project titled “ _____ (No. _____)” against an allocation of ₹ _____ including FE ₹ _____ during the FY, YYYY – YY.

(Accounts Officer)
with stamp

Date: _____

Place: _____

To – Lab Director

Copy to – DF&MM/ DRDO HQ

- DG (Cluster)

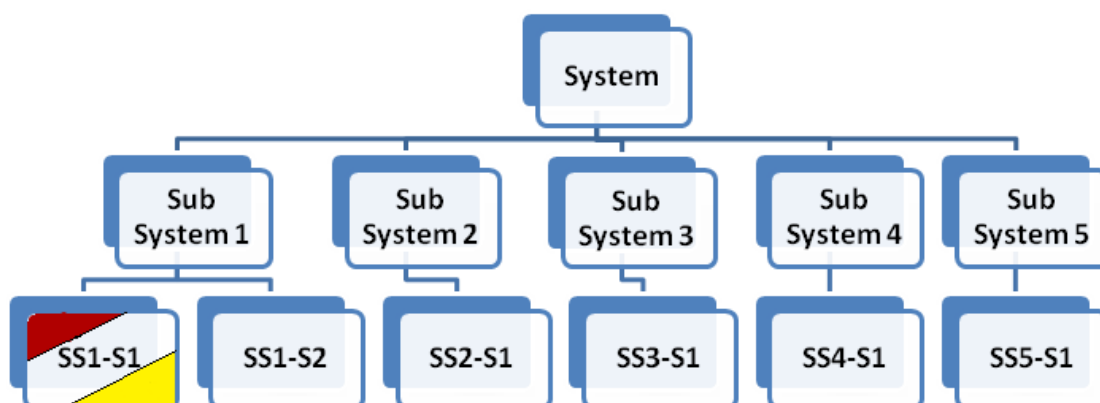
FORMAT FOR BRIEFING PAPERS OF APEX, PMB/EB & PJMB/PMRB MEETINGS

1. Project Attributes:

a.	Project Title				
b.	Category				
c.	Date of Sanction				
d.	Nodal & Participating Labs				
e.	Scope (including Objective & Deliverables)				
f.	PDC	Original	Rev.1	Rev.2....	Current
g.	Cost break-up (including FE)	Original	Rev.1	Rev.2....	Current
h.	No. of ABs, PMBs/EBs and PJMBs/ PMRBs held	AB (Planned/Held)	PMB/EB (Planned/Held)	PMRB (Planned/Held)	
		- / -	- / -	- / -	
i.	Current Status of Project	Design	Realisation	Testing & Evaluation	Ready for Closure

2. Schematic configuration:

3. Overall WBS (Minimum 4 Level for Programmes, MM & TD(S) and 3 Level for others):



Please colour code each module/subsystem/algorithm/process as per scheme given below.

Stage (Upper corner)	Design	Realisation	Testing & Evaluation	Ready for Closure
Module (Lower corner)	In-House development	BTS	BTP	COTS

4. Particulars of Meetings:

- Ratification of **recommendations** of last PMB/EB & PMRB review.
- Last AB/PMB/EB action points with Expected Date of Completion (EDC) & current status.
- Technical / User Reviews (if any).

S. No.	Action point	Expected Date of Completion	Actual date of completion	Responsible agency/person	Status/Remarks
				PD/MMG/Finance/O/o DG/HQ	

- Milestones achieved prior to this AB/PMB/EB period. Assessment of delay for each milestone should be projected w.r.t. overall impact on PDC of the project (Revised PERT diagram and milestone chart to be provided).
- Review of Sunset milestone.
- Details of work and current status of sub system with major milestones (since last AB/PMB/EB)
 - Work carried, achievements, test results etc.
 - TRL table with TRL at sanction stage and current stage indicating overall PRI.

S. No	TRL		
Sub System (SS)	Sanction	Last AB/PMB/EB	Current
SS-1			
SS-2 etc			

- Risk Matrix/Mitigation Plan/Status: Assessment of risks shall be done at every review and accordingly Risk Management Plan incorporating “5-step risk mitigation and response mechanism: Escalate, Avoid, Transfer, Mitigate, Accept” may be modified for each milestone.

8. Details of Procurement plan (Major Items):

S. No	Name of item	Cost	Type of tender	Demand Initiation date	Approval date	RFP floated on	TEC conducted	TPC conducted	Supply order placed	Delivery/ Inspection (ATP)	Current status / Remarks

Note: Any other stages of procurement may be chosen for better appreciation.

9. Overall Financial status:

For MM, TD(S), TD(T) & IF projects

(₹ in Cr)

Minor Head	Major Head 4076 – Capital	Sanctioned	Expenditure	Commitment	In-Pipeline	Balance
	Sub Major Head – 05	(₹ in Cr)				
	Heads of Expenditure	Total (FE)	Total (FE)	Total (FE)	Total (FE)	Total (FE)
052 (Code Head – 929/25)*	Transportation (Movement of Stores)					
	Equipment/Stores					
	CARS/CAPSI					
	Consultancy Contracts					
	Job Work/Contracts/ Technical Services/Work Package					
	Hiring of Transport					
	FOL for Project Vehicles					
	Contingency & Miscellaneous					
	Plant & Machinery					
	Project related Vehicles					
111	Works					
TOTAL						

* Total project expenditure related to MM, TD(S), TD(T) & IF projects including equipment, hardware, consultancy, project related contingency, purchase/hiring of transport, freight, contracts for “Acquisition of Research Services (CARS)” under the project etc. will be compiled to this head.

For S&T projects

(₹ in Cr)

Minor Head	Major Head 2080 - Revenue	Sanctioned	Expenditure	Commitment	In-Pipeline	Balance
	(₹ in Cr)					
	Heads of Expenditure	Total (FE)	Total (FE)	Total (FE)	Total (FE)	Total (FE)
105	Transportation (Movement of Stores)					
110 (Code Head-929/25)**	Equipment/Stores					
	CARS/CAPSI					
	Consultancy Contracts					
	Job Work/Contracts/ Technical Services/Work Package					
	Hiring of Transport					
	FOL for Project Vehicles					
	Contingency & Miscellaneous					
111	Works					
TOTAL (REVENUE)						
	Major Head 4076 - Capital					
052 (Code Head-929/24)	Plant & Machinery					
	Project related Vehicles					
	Works					
TOTAL (CAPITAL)						
GRAND TOTAL (REVENUE & CAPITAL)						

**Booking of Expenditure for Science & Technology (S&T) projects will be as per PM-2025 (Para 2.5.2) as amended from time to time.

10. Status update and Action Plan of six-monthly milestones with financial outlay:

S. No.	As per Project Proposal / SoC			Actual Progress as on Date		Future Action Plan		Impact on Project PDC	
	Technical Milestone	Milestone Completion Date	Financial Outlay (₹ in Cr.)	Percentage Completion of Technical Milestone	Financial Outlay (₹ in Cr.)	Expected Date of Completion	Action Plan	Yes/No	Overall Delay in Months
1	A1	MM-YYYY	20	100	20	-	-	No	-
2	A2	MM1-YYYY1	50	70	20	MM2-YYYY2	ABC	No	Nil

3	A3	MM3- YYYY3	80	50	30	MM4- YYYY4	DEF	Yes	NN
---	----	---------------	----	----	----	---------------	-----	-----	----

11. Data related to 'Valuation of Technologies' (to be updated cumulatively in every review): -

Manpower Utilisation in days	Man-days utilised				(Cumulative Past Years)	(Cumulative Till Date)
	(1 st Quarter)	(2 nd Quarter)	(3 rd Quarter)	(4 th Quarter)		
Scientist						
Technical						
Admin & Allied						

a) Manpower Utilisation in days

b) Utilisation of Established Infrastructure/ Facilities of the Lab/ Sister Lab

(1 st Quarter)		(2 nd Quarter)		(3 rd Quarter)		(4 th Quarter)		(Cumulative Past Years)	(Cumulative Till Date)
Name of Infra/ Facility	Days Utilized	Name of Infra/ Facility	Days Utilized	Name of Infra/ Facility	Days Utilized	Name of Infra/ Facility	Days Utilized		

c) Training

(1 st Quarter)		(2 nd Quarter)		(3 rd Quarter)		(4 th Quarter)		(Cumulative Past Years)	(Cumulative Till Date)
Name of Training	Cost	Name of Training	Cost	Name of Training	Cost	Name of Training	Cost		

12. Data related to 'Economic Impact' of Project (to be updated cumulatively in every review):-

- a) Percentage Indigenous Content, Dependent Foreign Countries, Items imported and Indigenization efforts
- b) International Collaborations executed (provide details and duration)
- c) Intellectual Property Rights generated (provide details of Patents, Designs and Copyrights)
- d) Export Potential, if any
- e) Infrastructure created, if any

13. PERT chart of overall project schedule (Original & Current):-

14. Issues:

15. Other Relevant points (if any).

16. Decision/Recommendations sought from AB/ EB:

Note:

- *Agenda mentioned in Chapter 6 on 'Project Monitoring and Review' be referred while making briefing papers.*
- *Action plan and Impact on Overall PDC (if any), as mentioned at S. No. 9 should mandatorily form part of AB/ PMB/ EB minutes which should be released within 10 days of meeting. If the minutes of meeting to be vetted by outside offices cut off dates should be given beyond which minutes would be assumed to be approved.*
- *Format for Apex Board, Programme Management Board or Executive Board may be chosen based on the review being scheduled for a project.*
- *Detailed technical discussions on each sub systems to be deliberated and recorded during PMB/ EB / PMRB. Ratification points from the higher review committee to be clearly mentioned in the minutes.*
- *For PDC extension cases, the defensible reason why PDC could not be adhered & remedial steps to be taken to avoid further PDC extension may also be presented as per the table given below & recorded in minutes.*

<i>S. No.</i>	<i>Reason for time over run</i>	<i>Responsible agency</i>	<i>Remedial Steps / Remarks</i>
-	-	-	-

Slides for presentation to be numbered 1 of n.

FORMAT FOR RECORDING MINUTES OF MEETING OF REVIEW (AB, PMB/EB & PJMB/PMRB)

Record/ File No. _____ dated _____

Minutes of Apex Board, PMB/EB and PJMB/PMRB Meeting for Project titled
“ _____ ” held on DD-MM-YYYY at “Venue’

1. Project Details:

a.	Project Title				
b.	Category				
c.	Date of Sanction				
d.	Nodal & Participating Labs				
g.	PDC	Original	Rev.1	Rev.2....	Current
h.	Cost	Original	Rev.1	Rev.2....	Current
i.	No. of ABs, PMBs, EBs and PMRBs held	AB (Planned/Held) - / -	PMB/EB (Planned/Held) - / -	PMRB (Planned/Held) - / -	
j.	Current Status of Project	Design	Realisation	Testing & Evaluation	Ready for Closure

2. Following members were present during the meeting :

Name	Designation	Lab/Agency

3. Opening/ Introductory remarks (If any).

4. Record of discussions and action points of Previous Review -

Item Codes:- **A:** Action, **C:** Comment, **D:-** Decision, **R:-** Recommendation

Sl. No	Description	Item Code	Expected Date of Completion	Actual Date of Completion	Responsibility (Agency/ Individual Name)

5. Achievement / Status of major systems/ sub systems and sub projects (if any)

6. Details of procurements items envisaged in the projects along with status.

7. Financial Status presented during the review

Major Head						(₹ in Cr.) (FE in brackets)
Minor Heads	Budget Head Description	Sanctioned Funds		Expenditure		Balance Funds
			Spent	Commitments	In Pipeline	
Total ₹ in Cr (FE ₹ in Cr)						

8. Status update and Action Plan of Technical milestones.

S. No	As per Project Proposal / SoC		Actual Progress as on Date	Future Action Plan		Impact on Project PDC
	Technical Milestone	Milestone Completion Date	Percentage Completion of Technical Milestone	Expected Date of Completion (if delayed)	Action Plan (if delayed)	Overall Delay in Months
1	A1	MM-YYYY	100	-	-	-

2	A2	MM1- YYYY1	70	MM2- YYYY2	ABC	Nil
3	A3	MM3- YYYY3	50	MM4- YYYY4	DEF	NN

9. Action plan for the next 06 months.

Item Codes:- **A:** Action, **C:** Comment, **D:-** Decision, **R:-** Recommendation

S. No	Description	Item Code	Expected Date of Completion	Actual Date of Completion	Responsibility (Agency/ Individual Name)

10. Revised PERT Chart, if applicable

11. Other relevant points

12. Recommendations

13. Concluding remarks (If any)

14. These minutes are being issued with the approval of Chairman AB /PMB/ EB/ PMRB.

(Member Secretary)

Addressees

STATEMENT OF CASE (SoC) FOR PDC EXTENSION & CHANGE IN SCOPE¹

1. Title of the Project/Programme : _____
2. Project/Programme No. : _____
3. Name of Lab : _____
4. Sanction Cost (₹ in Cr.) : Total _____ (FE _____)
5. Date of Sanction : _____
6. Amount spent so far (₹ in Cr.) : Spent - Total _____ (FE _____)
: Committed - Total _____ (FE _____)
: In Pipeline - Total _____ (FE _____)
7. PDC (at sanction stage) : (i) (DD:MM:YYYY) & Months _____
(revised PDC, if any) : (ii) (DD:MM:YYYY) & Months _____
- 8a) Pessimistic PDC presented to DMC at sanction stage: _____
- 8b) Revised PDC proposed with : (DD:MM:YYYY) & Months _____
financial or any other implications
9. Details of Additional funds required: _____
(if any)
10. Original Scope (if change is needed)
a) Objective : _____
b) Deliverables : _____
11. Revised Scope (if change is needed)
a) Objective : _____
b) Deliverables : _____
12. Reasons for seeking change in Scope (if change is needed): _____
13. Minutes of the project review
meetings held so far and recommendations
of the highest review committee

¹ Above format may be customized for Change of Category and Nodal Lab to Other Lab etc.

for PDC extension of the project or Change in Scope (whichever is applicable):

14. Recommendations of Peer Review Committee constituted by Secretary DDR&D for PDC Extension: _____
15. Recommendations & Minutes of DMC meeting for PDC extension of the project : _____
16. Reasons (point wise) why PDC could not be adhered (details may be attached) : _____
17. Work done so far with milestones achieved (details may be attached) : _____
18. Six monthly milestones along with financial outlay and timelines (**INITIAL proposed timelines vis-a-vis REVISED timelines in Tabular Format as illustrated below**):
- | S.No | Six-Monthly Milestone | Initial Timeline (Months) | Revised Timeline (Months) | Initial Financial Outlay (Rs. In Cr) | Revised Financial Outlay (Rs. In Cr) |
|------|-----------------------|---------------------------|---------------------------|--------------------------------------|--------------------------------------|
| 1. | Milestone-1 | T0 + 4 (July 2026) | T0 + 6 (Sep 2026) | 20 | 30 |
| 2. | Milestone-2 | T0 + 8 (Nov 2026) | T0 + 9 (Dec 2026) | 25 | 40 |
| 3. | ----- | ----- | ----- | ----- | ----- |
19. Revised **Comparative PERT Chart (INITIALLY Proposed vis-a-vis REVISED Version)**: _____
20. Check list for submission of case for PDC Extension duly signed by competent authority (enclosed / not enclosed) : _____

Note: SoC should be prepared by the Nodal Lab based on the inputs from the Participating Lab(s)

Programme / Project Director

**Recommended
(Lab Director)**

Date: _____

**Approved
(Director General)**

Date: _____

STATEMENT OF CASE (SoC) FOR COST REVISION AND RE-ALLOCATION OF FUNDS IN PROJECT

1. Name of Lab : _____
2. Project/Programme No. : _____
3. Title of the Project : _____
4. Sanction Cost (₹ in Cr) : Total _____ (FE _____)
5. Date of Sanction : _____
6. Amount spent so far (₹ in Cr) : Spent - Total _____ (FE _____)
: Committed - Total _____ (FE _____)
: In Pipeline - Total _____ (FE _____)
7. PDC original given and Subsequent amendments, if any: : (i) DD:MM:YYYY) & Months _____
(ii) DD:MM:YYYY) & Months _____
8. Revised six monthly milestones along with financial outlay (Initially Proposed Vs Revised Version)
9. Work done so far with milestones achieved : _____
10. Re-allocation of funds:
 - i) Details of head-wise funds to be re-allocated
 - ii) Reasons/Justification for re-allocation of funds
(Point-wise justifiable and defensible reasons may be highlighted)
 - iii) Recommendations of highest monitoring committee for re-allocation of funds
 - iv) Details of project monitoring committees meetings planned as per the Govt. letter vis-a-vis actually held.
11. Cost revision
 - i) Proposed project cost (₹ in Cr)
 - ii) Head-wise details of revised / proposed project cost (₹ in Cr)
 - iii) Reasons/justification for cost revision

(Point-wise justifiable and defensible reasons may be highlighted)

- iv) Recommendations of highest monitoring committee for cost revision
(please append the minutes)
- v) Supporting documents (budgetary quotes, e-mail quotes etc) for cost revision
- vi) Details of project monitoring committees meetings planned vis-à-vis held as per the
Govt. letter (please append the minutes)

12. General Assessment of Status (health) of Project
(views of the Project Director may be brought out)

13. Check list for submission of case for
Re-allocation of funds / Cost Revision
Duly signed by Competent Authority
(enclosed / not enclosed) : _____

14. Certification from Project Director on availability of funds

Note: SoC should be prepared by the Nodal Lab based on the inputs from the Participating Lab(s)

Programme / Project Director

**Recommended
(Lab Director)**

Date: _____

**Approved
(Director General)**

Date: _____

STATEMENT OF CASE FOR TECHNICAL COMPLETION OF PROJECT (FOR ALL TYPE OF COMPLETION/CLOSURE CASES)

1. Name of Lab : _____
2. Title of the Project : _____
3. Project No. : _____
4. Category of Project : _____
5. User : _____
(Army/Navy/Airforce & MHA)
6. Date of Sanction : _____
7. PDC original given and : (DD:MM:YYYY) & Months _____
Subsequent amendments, if any : (DD:MM:YYYY) & Months _____
8. Sanctioned Cost (₹ in Cr) : Total _____ (FE _____)
DRDO & User Share

Subsequent amendments, if any : Total _____ (FE _____)
DRDO & User Share
9. Statement of Accounts (₹ in Cr) : Expenditure (incurred, committed, in-pipeline and balance)
: Total _____ (FE _____)
10. Economic Impact of Project (pertaining to):
 - a) Percentage Indigenous Content, Dependent Foreign Countries, Items imported and Indigenization efforts
 - b) International Collaborations executed (provide details and duration)
 - c) Intellectual Property Rights generated (provide details of Patents, Designs and Copyrights)
 - d) Export Potential, if any
 - e) Infrastructure created, if any
11. Data pertaining to 'Valuation of Technology':

Man-days Utilized in Project			Infrastructure/ Facilities of the Lab/ Sister Lab Utilized in Project		Training Undertaken in the Project	
Scientist	Technical	Admin & Allied	Name of Infra/ Facility	Days Utilized	Name of Training	Cost

12. Present Status : _____

(Give details of work done so far and stage up to which project has been progressed).

13. Detailed reasons/considerations for project completed with Partial Success/Stage Closed and Cancellation (*if applicable*): _____

14. Recommendation of PSQR validation trial outcome report (for AoN accorded MM projects by DAC/DPB/SPB): _____

15. Recommendation of 'Independent DRDO Technical Committee for Evaluation of Programme/Project Success' constituted by DG (Cluster) [for projects other than MM]:

16. Minutes of Monitoring Committee Meetings held so far and recommendations of the highest monitoring committee for closure of the project:

17. Direction of DMC (for projects with partial success, stage closed & cancelled)

18. Lessons Learnt (including risk management) : _____

19. Other relevant details.

Programme / Project Director

Note: SoC should be prepared by the Nodal Lab based on the inputs from the Participating Lab(s)

**Recommended
(Nodal Lab Director)
(Approved / Not Approved)**

Date: _____

Approval of DG (Cluster) (Nodal Lab)

Date: _____

ADMINISTRATIVE CLOSURE OF PROJECT / PROGRAMME AND AUDITED STATEMENT OF EXPENDITURE

(FOR ALL TYPE OF COMPLETION/CLOSURE CASES)

Part - I

1. Name of the Lab : _____
2. Title of the Project/Programme : _____
3. Date of Sanction : _____
4. Category of the Project : _____
5. Cost (₹ in Cr): Total (FE) (original & revised): _____

Cost	Original	Revised
RE		
FE		
Total (FE)		

6. PDC of the Project (months & date) : _____

No. of Revisions	Original (DD:MM:YYYY) & Months	Revised (DD:MM:YYYY) & Months

7. Expenditure : Total ₹ _____ (FE _____)
8. Scope
 - a) Objective : _____
 - b) Deliverables : _____

9. Status of Deliverables as brought out in Govt. Letter: _____

10. List of sub-projects

S. No.	Sub Projects, Name & No.	Executing Lab	Cost	Current Status of Scope

11. List of CARS & CAPSI

S. No.	CARS & CAPSI Name & No.	Institute	Cost	Current Status

12. List of Consultancies

S. No.	Aim	Agency	Amount	Date

13. Details of Facilities created (as proposed in the project/programme, if applicable)

14. Brief of Trial Results (if applicable): _____

15. Achievements (based on Scope):

S. No.	Targets as Envisaged	Targets as Achieved	Remark
	Technology/ Deliverables/Products/ Learnings/ Knowledge as envisaged in the project proposal	Technology/ Deliverables/Products/ Learnings/ Knowledge etc. as achieved against each objective	

16. Recommendation of 'Independent DRDO Technical Committee for Evaluation of Project Success' constituted by DG (Cluster) {for S&T, TD(T), TD(S) and IF}: _____

17. Recommendation of PSQR validation trial outcome report (for AoN accorded MM projects by DAC/DPB/SPB): _____
18. Minutes of Monitoring Committee Meetings held so far and recommendations of the highest monitoring committee for completion of the project/programme
19. Certificate from Lab MMG/Store Section stating no live supply order or contracts is pending. List of payments, to be made due to contractual obligation be enclosed.
20. Certified that Scope set for the project have been met as per Technical Completion Report No. _____.

(Programme / Project Director)

Part-II

Statement of Accounts (Expenditure)

21. It is certified that the project “_____” Project No. _____ has incurred the expenditure of ₹ _____ including FE _____ against the sanctioned cost of ₹ _____ including FE _____ as per the enclosed Audited Statement of Expenditure.

All the stores/equipment undertaken in the project has been accounted for.

Date: _____

(Lab Accounts Officer or equivalent)

(Signature with name and office seal)

Part - III

The Statement of Accounts (Expenditure) of the project has been audited and reconciled.

Date: _____

Audit Authorities

(Local Audit Officer / CDA R&D)

(Signature with name and office seal)

Date: _____

Recommendation of the Nodal Lab Director

Date: _____

(Approved / Not Approved)

Approval of DG (Cluster) (Nodal Lab)

Note:

- (i) For **multi-lab projects**, format number DRDO.DPFM.FT.15 used for 'administrative closure of projects/programmes and audited statement of expenditure' needs to be prepared by **consolidating data** from the participating Labs. Formats by participating Labs may be utilized by Nodal Lab and may not be attached while seeking sanction of Competent Authority for Administrative Closure.
- (ii) Audited Statement of expenditure duly vetted by audit authorities {Local Audit Officer/CDA (R&D)} must be appended with the proposal
- (iii) Distribution of Technical Project Completion Report and Administrative Closure Report to
(a) Participating Labs (b) Concerned DGs (if applicable) and (c) Director PM, O/o DG (Cluster)

**SPECIMEN FORMAT OF GOVT. LETTER FOR PDC EXTENSION &
CHANGE IN SCOPE**

No. _____/_____/D(R&D)/20XX

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग

नई दिल्ली-110011

दिनांक: _____, 202X

सेवा में,

अध्यक्ष,

रक्षा अनुसंधान एवं विकास संगठन,
रक्षा मंत्रालय, डीआरडीओ भवन,
नई दिल्ली-110011

*(For projects, where CFA is Secretary DDR&D and above addressee shall be Chairman DRDO;
For projects, where CFA is DG (Cluster) addressee shall be concerned Lab Director;
For projects, where CFA is Lab Director addressee shall be Project Director)*

विषय: PDC Extension / Change in Scope of Project (Name) _____(Project No.)_____

महोदय,

I am directed to refer to Govt. of India, Ministry of Defence Letter No. _____ dated _____ as amended vide Letter No. _____ dated _____ and to convey the sanction of the President of India, in respect of Project _____, Project No. _____ for PDC Extension / Change in Scope of the project by another __ months, i.e. from _____(date) (Total __ months) to _____(date) (Total __ months) with / without any financial implications.

2. All other contents of the aforesaid Government of India letter dated _____ as amended from time to time shall remain same.

3. Directions from DMC w.r.t. further PDC extension, closure beyond a date/non achievement of milestone, review of Project Director etc. may be added.

4. Unique Sanction Code: _____

5. This issues with the concurrence of Ministry of Defence (Finance) vide their Dy. No _____/MoD/Fin(R&D) dated _____.

भवदीय,

()

अवर सचिव, भारत सरकार

<u>Ink Signed Copy to :</u>	<u>Copy to</u>
DG (Cluster) Director Lab Director Participating Labs Director P&C CGDA, New Delhi PCDA (R&D), New Delhi CDA (R&D), Concerned D(R&D), Ministry of Defence File copy	Principal Director of Audit, Defence Services, Pune Addl. FA (R&D) & JS IFA (R&D), Concerned Cluster DF&MM DCW&E (if applicable)

SPECIMEN FORMAT OF GOVT. LETTER FOR COST REVISION / RE-ALLOCATION OF FUNDS

No. _____/_____/D(R&D)/20XX

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग

नई दिल्ली-110011

दिनांक: _____, 202X

सेवा में,

अध्यक्ष,

रक्षा अनुसंधान एवं विकास संगठन,
रक्षा मंत्रालय, डीआरडीओ भवन,
नई दिल्ली-110011

*(For projects, where CFA is Secretary DDR&D and above addressee shall be Chairman DRDO;**For projects, where CFA is DG (Cluster) addressee shall be concerned Lab Director;**For projects, where CFA is Lab Director addressee shall be Project Director)***विषय : Re-allocation of Funds/Cost Revision in Project (Name) ____ (Project No.) ____**

महोदय,

I am directed to refer to Govt. of India, Ministry of Defence Letter No. _____ dated _____ as amended vide Letter No. _____ dated _____ and to convey the sanction of the President of India, in respect of Project _____, Project No. _____ for Re-allocation of Funds/Cost Revision of the project as per Annexure '____' to this letter *(mention revised fund details in Annexure '____')*

2. All other contents of the aforesaid Government of India letter dated _____ as amended from time to time shall remain same.
3. Unique Sanction Code: _____
4. This issues with the concurrence of Ministry of Defence (Finance) vide their Dy. No _____/MoD/Fin(R&D) dated _____.

भवदीय,

()

अवर सचिव, भारत सरकार

<u>Ink Signed Copy to :</u>	<u>Copy to</u>
DG (Cluster)	Principal Director of Audit, Defence
Director Lab	Services, Pune
Director Participating Labs	Addl. FA (R&D) & JS
Director P&C	IFA (R&D), Concerned Cluster
CGDA, New Delhi	DF&MM
PCDA (R&D), New Delhi	DCW&E (if applicable)
CDA (R&D), Concerned	
D(R&D), Ministry of Defence	
File copy	

SPECIMEN FORMAT OF GOVT. LETTER FOR PROJECT COMPLETION

No. _____/_____/D(R&D)/20XX

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग

नई दिल्ली-110011

दिनांक: _____, 202X

सेवा में,

अध्यक्ष,रक्षा अनुसंधान एवं विकास संगठन,
रक्षा मंत्रालय, डीआरडीओ भवन,
नई दिल्ली-110011

(For projects, where CFA is Secretary DDR&D and above addressee shall be Chairman DRDO;
For projects, where CFA is DG (Cluster) addressee shall be concerned Lab Director;
For projects, where CFA is Lab Director addressee shall be Project Director)

विषय: Administrative Closure of Project “Project (Name) _____” (Project No.)

महोदय,

I am hereby directed to convey the sanction of the President of India for Administrative Closure of the Project _____ Project No. _____ on its successful completion (as the case may be) sanctioned to _____ (Lab Name) vide Govt. of India, Ministry of Defence, Letter No. _____ dated _____ as amended vide Letter No. _____ dated _____, w.e.f. _____ (last date of PDC or earlier date whichever is applicable).

2. The total audited expenditure incurred on the said project is ₹ _____ Cr (in words _____) including FE ₹ _____ Cr (in words _____) against the sanctioned cost of ₹ _____ Cr (in words _____) including FE ₹ _____ Cr (in words _____)

3. Unique Sanction Code (USC): _____

4. This issues with the concurrence of Ministry of Defence (Finance) vide their Dy. No _____/MoD/Fin(R&D) dated _____.

भवदीय,

()

अवर सचिव, भारत सरकार

Ink Signed Copy to :DG (Cluster)
Director Lab
Director Participating Labs
Director P&C
CGDA, New Delhi
PCDA (R&D), New Delhi
CDA (R&D), Concerned
D(R&D), Ministry of Defence
File copy**Copy to**Principal Director of Audit, Defence
Services, Pune
Addl. FA (R&D) & JS
IFA (R&D), Concerned Cluster
DF&MM
DCW&E (if applicable)

**SPECIMEN FORMAT FOR CHANGE OF PROJECT FROM
NODAL LAB TO OTHER LAB**

No. _____/_____/D(R&D)/20XX

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग

नई दिल्ली-110011

दिनांक: _____, 202X

सेवा में,

अध्यक्ष,

रक्षा अनुसंधान एवं विकास संगठन,
रक्षा मंत्रालय, डीआरडीओ भवन,
नई दिल्ली-110011

*(For projects, where CFA is Secretary DDR&D and above addressee shall be Chairman DRDO;
For projects, where CFA is DG (Cluster) addressee shall be concerned Lab Director;
For projects, where CFA is Lab Director addressee shall be Project Director)*

विषय: Change of Nodal Lab for Project (name)..... From (Nodal Lab name) to (Other Lab name)

Ref: Letter No.(previous Govt. of India sanctioned letter no) dated

महोदय ,

I am directed to refer to Government of India, Ministry of Defence Letter No. dated and to convey the sanction of the President of India for change of Nodal Lab for Project (name) from (Nodal Lab) to (Name of New Lab) with effect from dated The changed details are as follows:

Project No. : New Lab Project No. :
Participating Lab Sub-project No. :

2. Project No. of previous Nodal Lab: and Participating Lab (Sub-Project No).
..... shall stands **cancelled with effect from (Date)**.....

3. Nodal Lab (previous) will ensure that expenditure incurred till cut-off date
..... will be vetted by concerned CDA (R&D) and stores will be transferred to (Other Lab name).

4. All other authorization and responsibilities as in referred Govt. letters are also transferred to (Other Lab name).

5. All other contents of the original project sanction letter and subsequent revisions to the sanction if any, will remain unchanged, except for the change in name of the Lab.

6. All outstanding liabilities and observations of Audit shall stand transferred to (Lab name).... with effect from (date).....
7. Unique Sanction Code (USC):
8. This issues with the concurrence of Ministry of Defence (Finance) vide their Dy. No. _____/MoD/Fin(R&D) dated _____.

भवदीय,

(_____)
 अवर सचिव, भारत सरकार

<u>Ink Signed Copy to :</u>	<u>Copy to</u>
DG (Cluster of Nodal Lab)	Principal Director of Audit, Defence Services, Pune
DG (Cluster of New Lab)	Addl. FA (R&D) & JS
Director Nodal Lab	IFA (R&D), Cluster
Director of New Lab	DF&MM
Director of Participating Labs	DCW&E (if applicable)
Director P&C	
CGDA, New Delhi	
PCDA (R&D), New Delhi	
CDA (R&D), Concerned	
D(R&D), Ministry of Defence	
File copy	

Annexure

- Transfer of original Project Sanction File
- Statement of Expenditure signed by concerned CDA till cut-off date
- List of all procurement cases and files to new Lab.
- Finalisation of all audit observations and issues till cut-off date with status and details of replies
- Transfer of record and project inventory to new Lab along with outstanding supply orders for all procurement cases

SPECIMEN FORMAT FOR CHANGE OF PROJECT CATEGORY DURING PROJECT CURRENCY

No. _____/_____/D(R&D)/20XX

भारत सरकार, रक्षा मंत्रालय
रक्षा अनुसंधान और विकास विभाग

दिनांक: नई दिल्ली-110011
_____, 202X

सेवा में,

अध्यक्ष,

रक्षा अनुसंधान एवं विकास संगठन,
रक्षा मंत्रालय, डीआरडीओ भवन,
नई दिल्ली-110011

*(For projects, where CFA is Secretary DDR&D and above addressee shall be Chairman DRDO;
For projects, where CFA is DG (Cluster) addressee shall be concerned Lab Director;
For projects, where CFA is Lab Director addressee shall be Project Director)*

**विषय: Change of category of project (name)..... of nodal Lab (name) from (current
category) to (new category)**

Ref: Letter no.(previous Govt. of India sanctioned letter no) dated

महोदय ,

I am directed to refer to Government of India, Ministry of Defence Letter No. dated and amendments (if any) and convey the sanction of the President of India for change of category for project (name) from (current category) to (new category) with effect from dated The changed details are as follows:

Project No.: Nodal Lab new category project No.

Participating Lab new category Sub-project No.

2. Project No. of Nodal Lab: and Participating Lab (Sub-project No.).
..... shall stand **cancelled with effect from (Date).....**

3. (The Scope/Cost of project in new category to be specified as approved).

4. Nodal Lab will ensure that expenditure incurred till cut-off date (Date)..... will be vetted by concerned CDA (R&D) and stores will be transferred in new category of project of (Lab name).

5. All other contents of the project original sanction letter and subsequent revisions to the sanction, if any, will remain unchanged, except for the change in category of the project.

6. Unique Sanction Code (USC):
7. This issues with the concurrence of Ministry of Defence (Finance/R&D) vide their Dy. No. _____/MoD/Fin/R&D dated _____.

भवदीय,

(_____)
 अवर सचिव, भारत सरकार

<u>Ink Signed Copy to :</u>	<u>Copy to</u>
DG (Cluster of Nodal Lab)	Principal Director of Audit, Defence Services, Pune
DG (Cluster of Participating Lab)	
Director Nodal Lab	Addl. FA (R&D) & JS
Director Participating Labs	IFA (R&D), Cluster
Director P&C	DF&MM
CGDA, New Delhi	DCW&E (if applicable)
PCDA (R&D), New Delhi	
CDA(R&D), Concerned	
D(R&D), Ministry of Defence	
File copy	

CHECKLISTS

CHECKLIST FOR TRANSITION TO NEW TRL (INDICATIVE)

TRL-1	Basic principles observed and reported. When a technology is at TRL 1, scientific research is beginning and those results are being translated into future research and development. Questions for transition to next TRL: • Can all the relevant physical and technical parameters be enumerated? • Are there relationships between the above that can be stated quantitatively as equations? • Is it possible to relate the above equations to references in textbooks or journals?
TRL-2	Technology concept and/or application formulated. TRL 2 occurs once the basic principles have been studied and practical applications can be applied to those initial findings. TRL 2 technology is very speculative, as there is little to no experimental proof of concept for the technology. Questions for transition to next TRL: • Can the physical and technical parameters be related to applications-specific utilities or Figure of Merit (FoMs)? • Can the sensitivity of utility parameters to physical and technical parameters be estimated? • Can trade-offs and convex optimizations be identified for the utilities and FoMs?

TRL-3	Analytical and experimental critical function and/or characteristic proof-of-concept. When active research and design begin, a technology is elevated to TRL 3. Generally, both analytical and Lab studies are required at this level to see if a technology is viable and ready to proceed further the development process. Often during TRL 3, a proof-of-concept model is constructed. Questions for transition to next TRL: • Have the relations between physical and technical parameters been experimentally verified? • Have relations between physical & technical parameters and utility parameters been formulated analytically? • Have relations between physical & technical parameters and utility parameters been experimentally verified to match the analytic expectations? • Did the experimentation on a Proof of Concept (PoC) model validate the expected trade-offs and optimizations?		
TRL - 4	Technology validated in Lab environment	• Configuration finalisation for system / product specifications. • Generation of System/Product Tree. • Detailed Design Review completed. • Lab prototype realised.	• Identification of Development cum Production Partner (DcPP) for a TD (System) Project. • Detailed Design Document and Layout Drawings. • Integration Layouts.
Description	Component validation in Lab environment. Once the proof-of-concept technology is ready, the technology advances to TRL 4. During TRL 4, multiple component pieces are tested with one another. Questions for transition to next TRL: • Could all the components be integrated for an end-to-end functionality test in the Lab? • Could parameter ranges be found that allowed each component to be in an acceptable state? • Could the trade-offs and optimizations be re-established for the integrated system?		

TRL – 5	Technology validated in simulated environment	• Components / Sub-systems manufactured as per Acceptance Test Procedures (ATP). • Design validation for performance at component / sub-system level. • Components/sub-systems integrated in Lab environment for functional assessment.	• ATP document. • Approved sub-system/ component design and drawings for prototype manufacture. • Approved integration plan. • Component/Sub-system Functional Test Reports. • Quality Assurance Plan for sub-system / component.
Description	Component validation in relevant environment. TRL 5 is a continuation of TRL 4, however, a technology that is at 5 is identified as a breadboard technology and must undergo more rigorous testing than technology that is only at TRL 4. Simulation should be run in environments that are as close to realistic as possible. Questions for transition to next TRL: • Were the design margins and stability of the components characterized with realistic noise? • Was behaviour of the components studied with parameter variations and fault injection? • Were the components tested to work with each other across full range of input values?		

TRL - 6	System/ Sub-system model or prototype readiness for trials in simulated environment	• Sub-systems/Systems manufactured as per Quality Assurance Plan (QAP) & accepted as per ATP. • Completion of Qualification tests for components/sub-systems from DcPP. • Critical Design Review and implementation of recommendations. • Readiness for Prototype Trials in simulated environment.	• Qualification Test Certificates for components/ sub-systems. • Approved Interface Control Documents • Critical Design Review report of components/ sub-system. • Manufacturing Drawings of QAP and ATP cleared sub-systems/ components. • Prototype Trials Plans.
Description	System / Sub-system prototype or model validation in relevant environment. Once the testing of TRL 5 is complete, a technology may advance to TRL 6. A TRL 6 technology has a fully functional prototype or representational model. Questions for transition to next TRL: • Did the system composed of the above components function as anticipated? • Did this prototype meet the expected quantitative performance requirements? • Did this prototype meet the ancillary / non-functional requirements as required for usage?		
TRL - 7	System prototype demonstration in relevant / simulated environment	• Realisation of prototype from DcPP. • Completion of Prototype Trials. • Prototype cleared for Preliminary Staff Qualitative Requirement (PSQR) Validation Trials. • Completion of Technical Documentation.	• Prototype Trial Reports. • PSQR Validation Trial definition document at system/ sub-system level. • Approved Technical Documents. • Approval by Executive Body.
Description	System prototype demonstration in environment. TRL 7 technology requires that		

	the working model or prototype be demonstrated in realistic environment. Questions for transition to next TRL: • Were adequate numbers of prototypes produced meeting the design specifications? • Did it operate under field conditions, interworking with existing systems and infrastructure? • Did it meet its functional requirements, or tweaked to subsequently do so? • Did it impact existing infrastructure / systems, or need to be operated under restrictions?		
TRL - 8	Actual system completion and qualified through PSQR validation trials for performance requirement against Trial Directive.	• Completion of PSQR Validation Trials on systems manufactured by DcPP-based on QA/ ATP, for performance requirements. • Completion of Technical Documentation.	• PSQR Validation Trial (for performance evaluation) reports duly vetted by the Trials Team.
Description	Actual system completed and qualified through test and demonstration. TRL 8 technology has been tested and “flight qualified” and it’s ready for implementation into an already existing technology or technology system. Questions for transition to next TRL: • Did DcPP do a limited series production? • Did the system undergo successful User trials and evaluation? • Is the system cleared for acquisition and induction?		

TRL - 9	Actual System operated over the full range of operational conditions as per PSQR parameters against Trial Directive.	• Completion of PSQR validation/ User Trials / Final Acceptance Trials.	• Jointly vetted PSQR Validation Trial reports. • Approval by the highest Review Committee.
Description	Actual system proven through successful mission operations. Once a technology has been “flight proven” during a successful mission, it can be called TRL 9. Questions for transition to next TRL: • Was the system inducted for use? • Did the system undergo usage in mission (actual or exercises)? • Was post-usage feedback incorporated into a new production version?		
TRL - 10	System in production/ COTS	• System/sub-system/ components under production meeting the performance requirements.	• Production order.

CHECKLIST FOR NEW PROJECT SANCTION**(This check-list should be placed on the file on top of Project Proposal)**

1.	Project Proposal and its Executive Summary (Yes/No)
2.	TRL report prepared by PD and reviewed by Chairman, Lab Technology Council (Yes/No)
3.	SHQ commitment for MM, TD(S) projects (Yes/No)
4.	i) Recommendation of the Cluster Council for taking project (please append the minutes) along with the In-principle approval of CCM of all sub-project Lab (Yes/No) ii) Minutes of the PRC/ IRC/ PDR Committee (Yes/No) (Recommendation of PRC/ PDR on TRL & PRI indices for taking a new project shall include initial TRL indices at the time of undertaking the project and likely TRL levels at the time of closure of the project shall be a part of Minutes of Meeting) iii) Whether the recommendations of PRC/ IRC/ PDR have been implemented and proposal has been modified accordingly? (Yes/No) iv) Minutes of Time Estimation Committee (TiEC) (Yes/No) v) Recommendation of the DMC (AoN) to undertake the project (the minutes to be enclosed). In case of validity of AoN by DMC which is 2/4/6 months for Director/DG (Cluster)/Secretary DDR&D as CFA has expired, may be re-validated. (Yes/No)
5.	Recommendations of Cost Estimation Committee (Yes/No)
6.	Six monthly milestones along-with financial outlay and timelines (Yes/ No)
7.	PERT Chart (Yes/No)
8.	Review mechanism as per laid down guidelines of DPFM-2026 (Yes/No)

9.	Manpower, Building space and Vehicle requirement if any	(Yes/No)
10.	Details of sub-projects indicating scope, cost and PDC enclosed	(Yes/No)
11.	Details of CARS, CAPSI, Consultancy, Training etc. if proposed	(Yes/No)
12.	Procurement plan (Necessary documents for EPC approval enclosed)	(Yes/No)
13.	Risk Management Plan	(Yes/No)
14.	Responsibility & Accountability Matrix (RACI Matrix)	(Yes/No)
15.	Comments of Lab Director on ongoing project commitmentts	(Yes/No)
16.	Recommendation of DG (Cluster)	(Yes/No)
17.	Draft Govt. Letter (DGL) prepared and placed in File	(Yes/No)
18.	PMIS/ERP database updated on DRONA	(Yes/No)
	{Signature of Director PM O/o of DG (Cluster)}	

CHECKLIST FOR PDC EXTENSION & CHANGE IN SCOPE OF THE PROJECT

1. Expenditure Details (Yes/No)
2. Reasons for not adhering to initial PDC (Yes/No)
3. List of activities to be carried out with revised milestones for the extended period of PDC (Yes/No)
4. Revised Comparative PERT Chart (Initially Proposed Vs Revised Version) (Yes/No)
5. Six-monthly milestones along with financial outlay and timelines (**INITIAL timelines vis-a-vis REVISED proposed timelines in Tabular Format**) (Yes/No)
6. In case of MM projects, recommendation of **Peer Review** (Yes/No)
7. Recommendations of **DMC** and **Highest Review Committee** for extension of PDC (Yes/No)
8. Revised Comparative PERT Chart (**INITIALLY proposed vis-à-vis REVISED Version**) (Yes/No)
9. Original and Revised Scope (Yes/No)
10. Reasons for seeking change in Scope (Yes/No)
11. Recommendations of Highest Review Committee for change in Scope (Yes/No)
12. Details of project review committees meetings planned as per the Govt. letter vis-a-vis held with their respective minutes (Yes/No)
13. Remedial steps undertaken to avoid further PDC extension is (Yes/No)
14. General Assessment of Status (health) of Project (Yes/No)
15. DGL placed in file (Yes/No)
16. PMIS/ERP database updated on DRONA (Yes/No)

It is certified that points raised in check list has been taken into consideration and necessary information / papers have been placed in file.

(Signature of Director PM, O/o DG (Cluster))

Date: _____

Place: _____

CHECKLIST FOR PROJECT COST REVISION & RE-ALLOCATION OF FUNDS

1. Expenditure Details - Total _____ (FE____) (Yes/No)
2. **Re-allocation**
 - (i) Details of Re-allocation of funds (Yes/No)
 - (ii) Reasons (Justification) for re-allocation of funds (Yes/No)
 - (iii) Recommendations of highest monitoring committee for re-allocation of funds (Yes/No)
 - (iv) Details of project monitoring committees meetings held (Yes/No)
 - (v) Revised six monthly milestones along with financial outlay and timelines (Yes/No)
3. **Cost Revision**
 - i) Details of cost revision (Yes/No)
 - ii) Reasons for cost revision (Yes/No)
 - iii) Recommendations of highest review committee for cost revision (Yes/No)
 - iv) Supporting documents (budgetary quotes, e-mail quotes etc.) for cost revision (Yes/No)
 - v) Details of project review committees' meetings held (Yes/No)
 - vi) Revised six monthly milestones along with financial outlay and timelines (Yes/No)
4. General Assessment of Status (health) of Project (Yes/No)
5. DGL placed in File (Yes/No)
6. PMIS/ERP database updated on DRONA (Yes/No)

It is certified that points raised in check list has been taken into consideration and necessary information/papers have been placed in file.

{Signature of Director PM, O/o DG (Cluster)}

Date: _____

Place: _____

ABBREVIATIONS

**LAB/CENTRE NAME ABBREVIATIONS
FOR PROJECT NO. ALLOTMENT**

(Full abbreviation of Lab/ Centre name will be used for allotment of Project No.)

ADE	Aeronautical Development Establishment
ADRDE	Aerial Delivery Research and Development Establishment
ARDE	Armament Research & Development Establishment
ASL	Advanced Systems Laboratory
CABS	Centre for Air Borne System
CAIR	Centre for Artificial Intelligence & Robotics
CFEES	Centre for Fire, Explosive and Environment Safety
CHESS	Center for High Energy Systems and Sciences
CVRDE	Combat Vehicles Research & Development Establishment
DEAL	Defence Electronics Application Laboratory
DEBEL	Defence Bioengineering and Electro-medical Laboratory
DGRE	Defence Geo-Informatics Laboratory
DIBT	Defence Institute of Biodefence Technologies
DIHAR	Defence Institute of High-Altitude Research
DIPAS	Defence Institute of Physiology & Allied Science
DIPR	Defence Institute of Psychological Research
DLJ	Defence Laboratory Jodhpur
DLRL	Defence Electronics Research Laboratory
DMRL	Defence Metallurgical Research Laboratory
DMSRDE	Defence Materials & Stores Res. & Dev. Establishment

DRDE	Defence Research & Development Establishment
DRDL	Defence Research & Development Laboratory
DRL	Defence Research Laboratory
GTRE	Gas Turbine Research Establishment
HEMRL	High Energy Materials Research Laboratory
INMAS	Institute of Nuclear Medicine & Allied Sciences
IRDE	Instruments Research & Development Establishment
ITM	Institute of Technology Management
ITR	Integrated Test Range
JCB	Joint Cipher Bureau
LRDE	Electronics & Radar Development Establishment
NMRL	Naval Materials Research Laboratory
NPOL	Naval Physical Oceanographic Laboratory
NSTL	Naval Science & Technological Laboratory
PXE	Proof & Experimental Establishment
R&DE(E)	Research & Development Establishment (Engineers)
RCI	Research Centre Imarat
SAG	Scientific Analysis Group
SSPL	Solid State Physics Laboratory
TBRL	Terminal Ballistics Research Laboratory
VRDE	Vehicles Research and Development Establishment

Note: In case of any ambiguity arising due to similar or same abbreviation for two or more Labs, DP&C will clarify and assign proper abbreviation to each Lab for allotment of Project No.

ABBREVIATIONS GENERAL TERMINOLOGY

ACAS	Assistant Chief of Air Staff
ACNS	Assistant Chief of Naval Staff
ACoAS	Assistant Chief of Army Staff
AHSP	Authorized Holder of Sealed Particulars
AO (R&D)	Accounts Office (R&D)
AoN	Acceptance of Necessity
ASQR	Air Staff Qualitative Requirement
ATP	Acceptance Test Procedure
BE	Budget Estimate
BTS	Build to Specification
CAPSI	Contract for Acquisition of Professional Services
CARS	Contract for Acquiring Research Services
CCM	Cluster Council Meeting
CCS	Cabinet Committee on Security
CDA	Controller of Defence Accounts
CDR	Critical Design Review
CEC	Cost Estimation Committee
CFA	Competent Financial Authority
CGDA	Controller General of Defence Accounts
CISC	Chief of Integrated Defence Staff
CPM	Critical Path Method
DAE	Department of Atomic Energy

DAP	Defence Acquisition Procedure
DcPP	Development cum Production Partner
DDR&D	Department of Defence Research & Development
DDR	Detailed Design Review
DFMM	Dte. of Finance and Material Management
DFP	Delegation of Financial Powers
DFTM	Directorate of Futuristic Technology Management
DG	Director General
DG (PC&SI)	Director General (Production Coordination & Service Interaction)
DG(R&M)	Director General (Resource and Management)
DGL	Draft Govt. Letter
DHRD	Directorate of Human Resource Development
DIITM	Directorate of Industry Interface & Technology Management
DISB	Directorate of Interaction with Services for Business
DMC	Deptt. of Defence R&D Management Council
DoS	Date of Sanction
DP	Development Partner
DP&C	Directorate of Planning & Coordination
DPFM	Directives for Project Formulation and Management
DPP	Draft Project Proposal
DPSU	Defence Public Sector Undertaking
DQRS	Directorate of Quality, Reliability & Safety
D-RdMAP	Roadmap of DRDO
DRDO	Defence Research & Development Organisation

DRDS	Defence Research Development Service
DRONA	DRDO Rapid Online Network Access
DRTC	Defence Research Technical Cadre
DST	Department of Science & Technology
EB	Executive Board
EDC	Expected Date of Completion
FA(DS)	Financial Advisor (Defence Services)
FE	Foreign Exchange
FERV	Foreign Exchange Rate Variation
FIM	Free Issue Material
FM	Finance Minister
FMEA	Failure Mode Effect Analysis
FMECA	Failure Mode Effect Critical Analysis
FOL	Fuel, Oil & Lubricants
FY	Financial Year
GoI	Govt. of India
GSQR	General Staff Qualitative Requirement
HQ	Head Quarters
HR	Human Resource
IC	International Cooperation
ICDP	Integrated Capability Development Plan
IDDM	Indigenous Design Development and Manufacturing
IDS	Integrated Defence Staff
IDST	Institute of Defence Scientists and Technologists

IF	Infrastructure & Facilities
IGA	Inter Government Agreement
ISRO	Indian Space Research Organisation
IV&V	Independent Verification and Validation
JPMT	Joint Project Monitoring Team
JSQR	Joint Staff Qualitative Requirement
LAO	Local Audit Officer
LATOT	License Agreement for Transfer of Technology
LP	Local Purchase
LPP	Last Purchase Price
MM	Mission Mode
MoD	Ministry of Defence
MoM	Minutes of Meeting
MoU	Memorandum of Undertaking
MPP	Manufacturing Process Plan
MSQAA	Missile System Quality Assurance Agency
NSQR	Naval Staff Qualitative Requirements
ORSA	Operation Research & System Analysis
PA	Production Agency
PBS	Product Breakdown Structure
PCDA	Principal Controller of Defence Accounts
PDC	Probable Date of Completion
PDD	Preliminary Design Document
PDR	Preliminary Design Review

PEP	Project Execution Plan
PERT	Project Evaluation and Review Technique
PJMB	Project Management Board
PM	Project Monitoring
PMB	Programme Management Board
PMIS	Project Management Information System
PMRB	Project Monitoring and Review Board
PMT	Project Monitoring Team
PRC	Peer Review Committee
PRI	Project Readiness Index
PSQR	Preliminary Staff Qualitative Requirements
PSU	Public Sector Undertaking
QA	Quality Assurance
QAP	Quality Assurance Plan
QR	Qualitative Requirement
QR&S	Quality Reliability & Safety
R&D	Research & Development
R&QA	Reliability & Quality Assurance
RDR	Revenue Debt Remittance
RE	Revised Estimate
RFP	Request For Proposal
RM	Raksha Mantri
S&T	Science & Technology
SDD	System Definition Document

SDLC	Software Development Life Cycle
SHQ	Services Head Quarters
SoC	Statement of Case
SOP	Standard Operating Procedure
SoW	Scope of Work
SWOT	Strength Weakness Opportunity & Threat
TD	Technology Demonstration
TD (S)	Technology Demonstration (System)
TD (T)	Technology Demonstration (Technology)
TEC	Technical Evaluation Committee
ToT	Transfer of Technology
TRA	Technology Readiness Assessment
TRL	Technology Readiness Level
UATT	User Assisted Technical Trials
US(R&D)	Under Secretary (Research & Development)
USC	Unique Sanction Code
UT	User Trials
WBS	Work Breakdown Structure